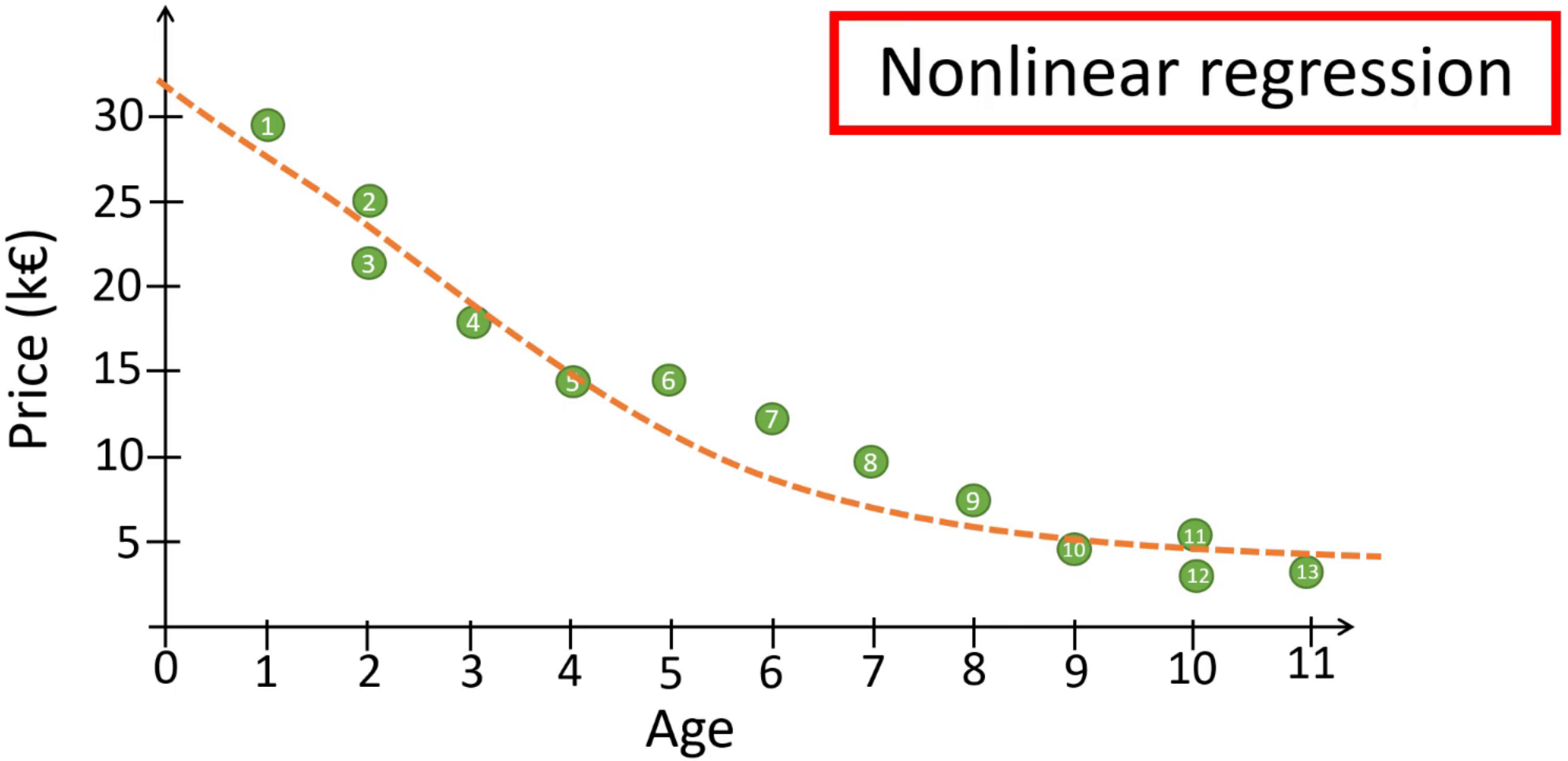
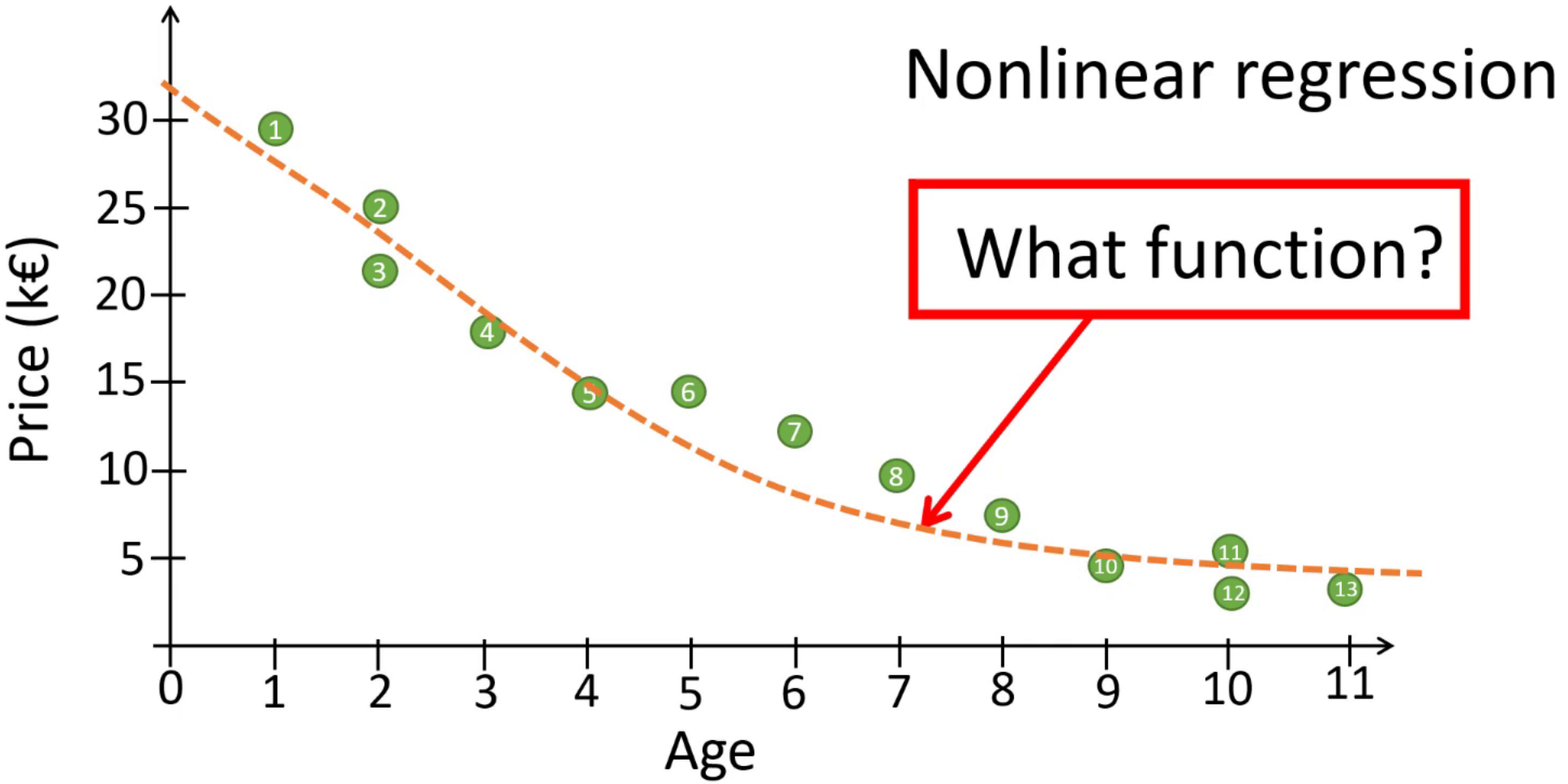
Nonlinear regression



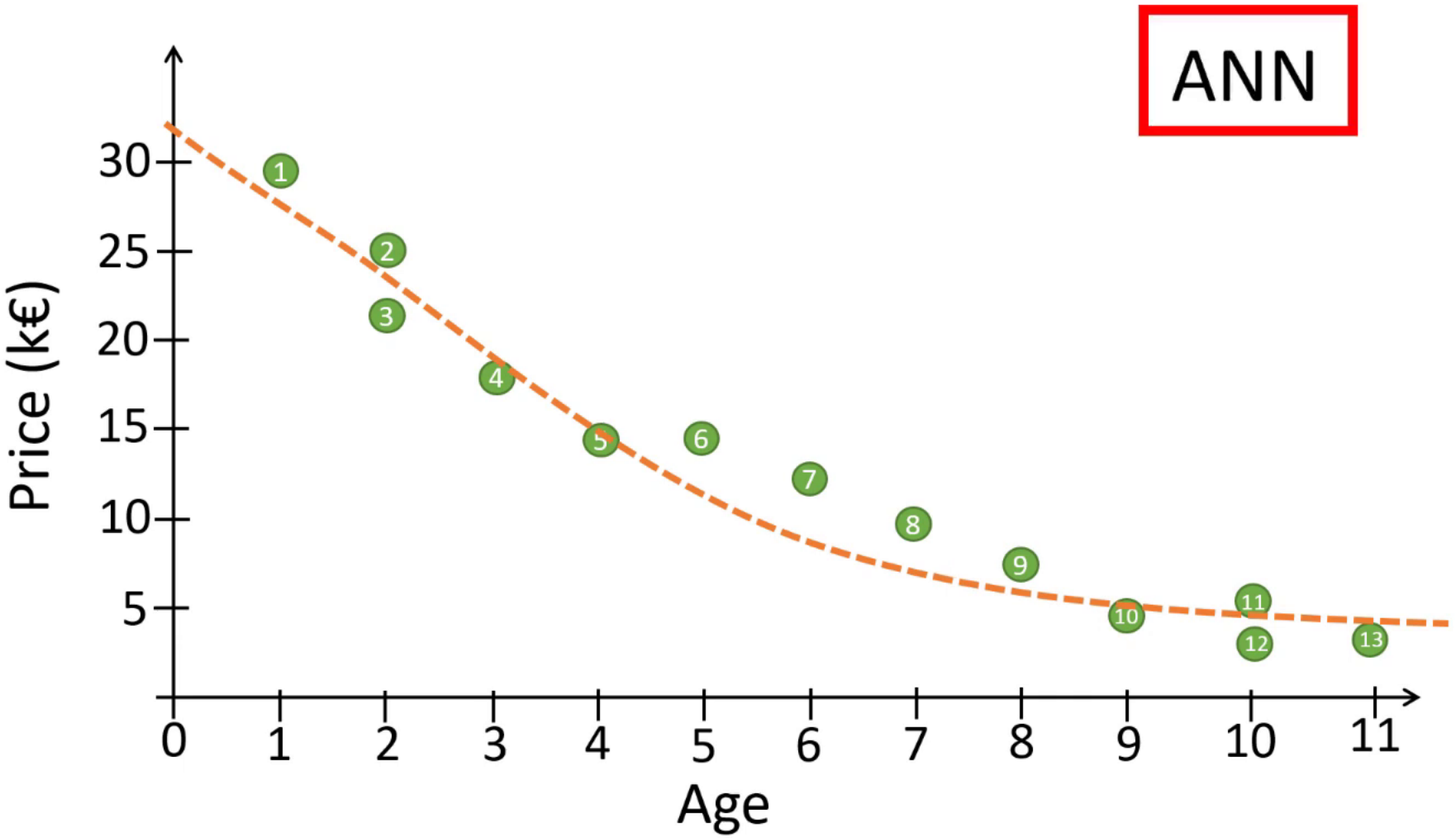
One way is to use nonlinear regression where we fit a nonlinear function to the data.

Nonlinear regression



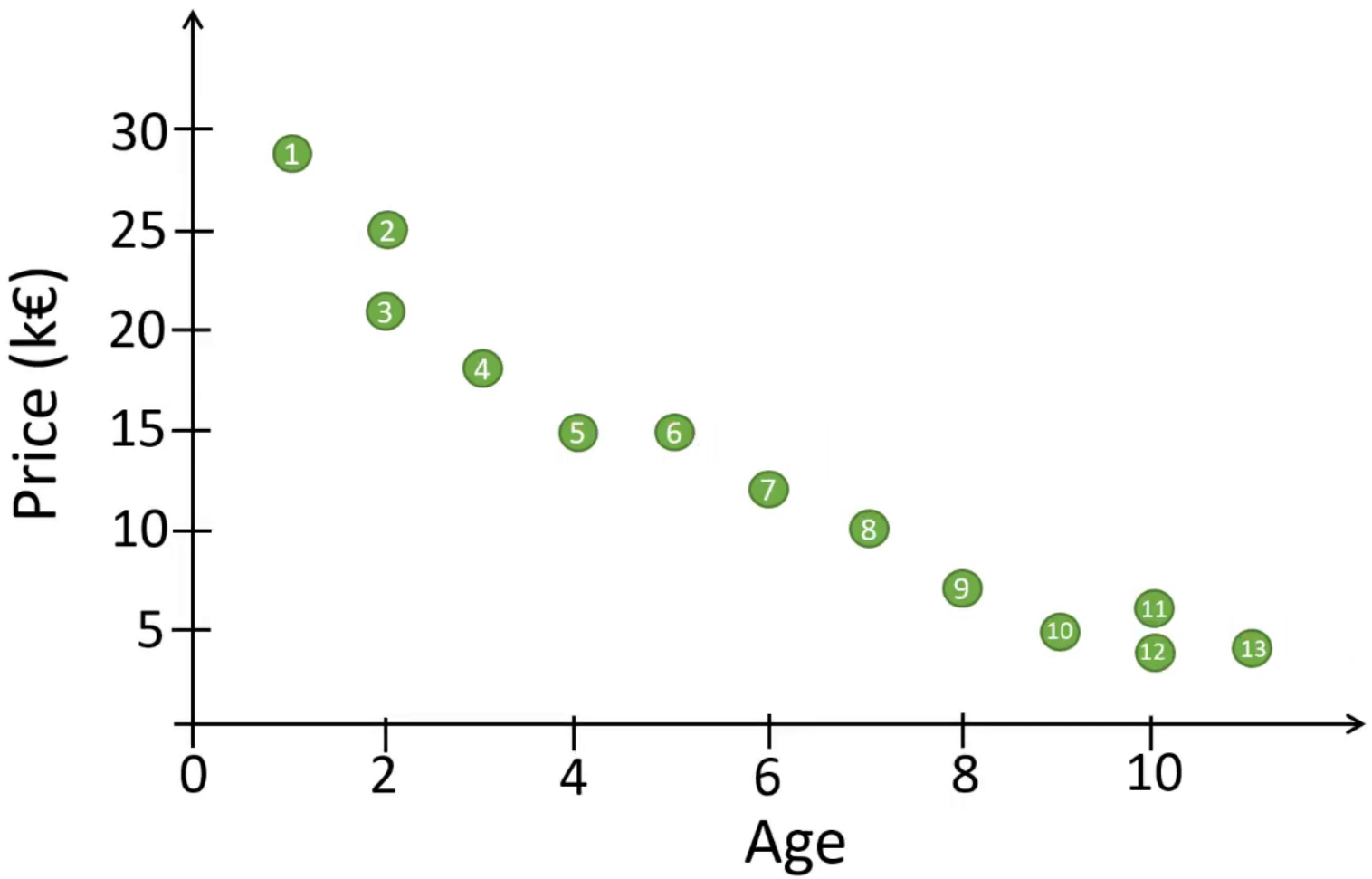
The problem with nonlinear regression is that we need to figure out an appropriate mathematical function that can be used to generate a curve that fits the data. Finding such a function might be tricky when we have several input variables.

ANN



This is where artificial neural networks become handy because they can help us to find nonlinear functions that can be fitted to the data.

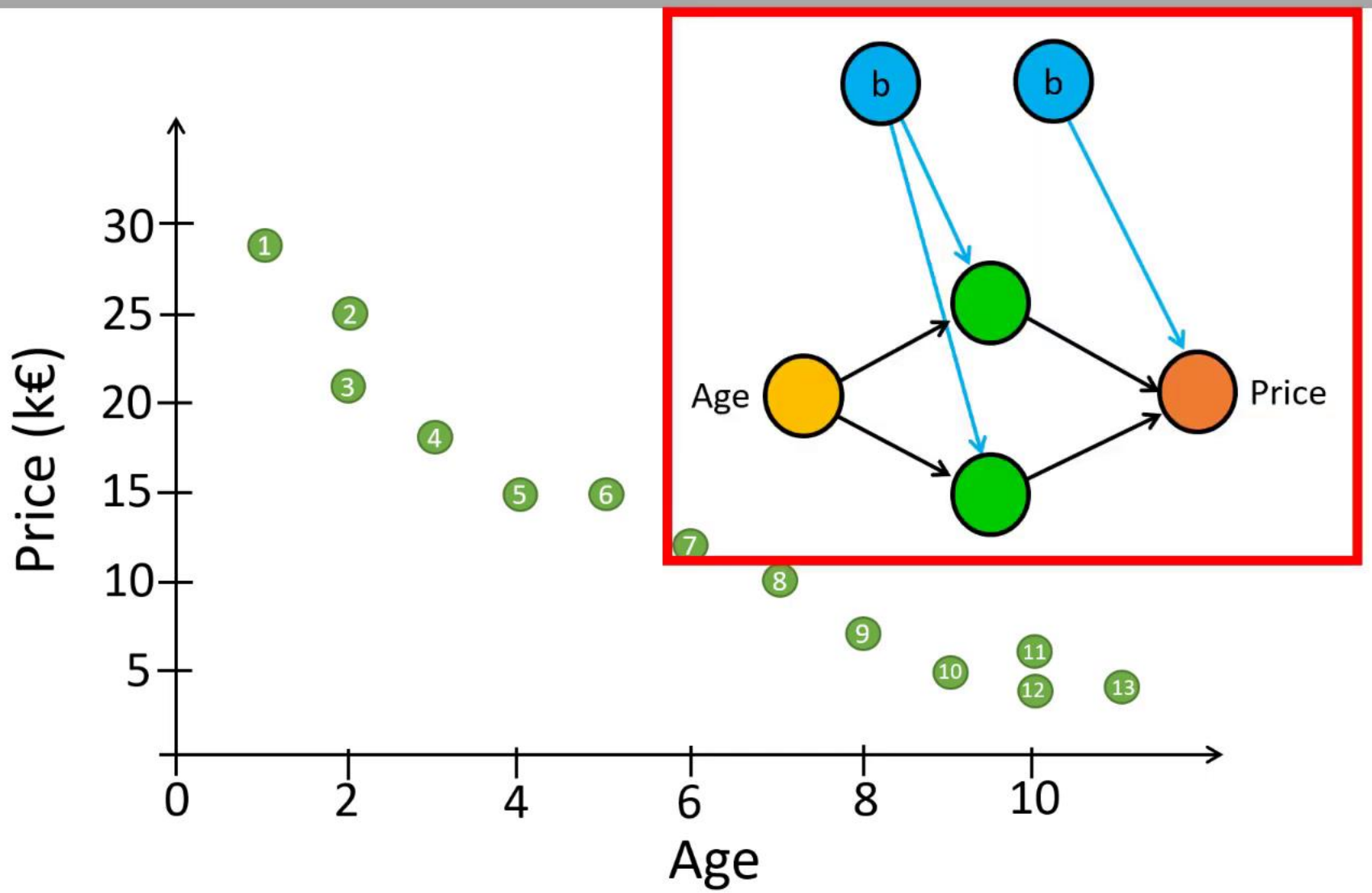
ANN



Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5
7	12	6
8	10	7
9	7	8
10	5	9
11	6	10
12	4	10
13	4	11

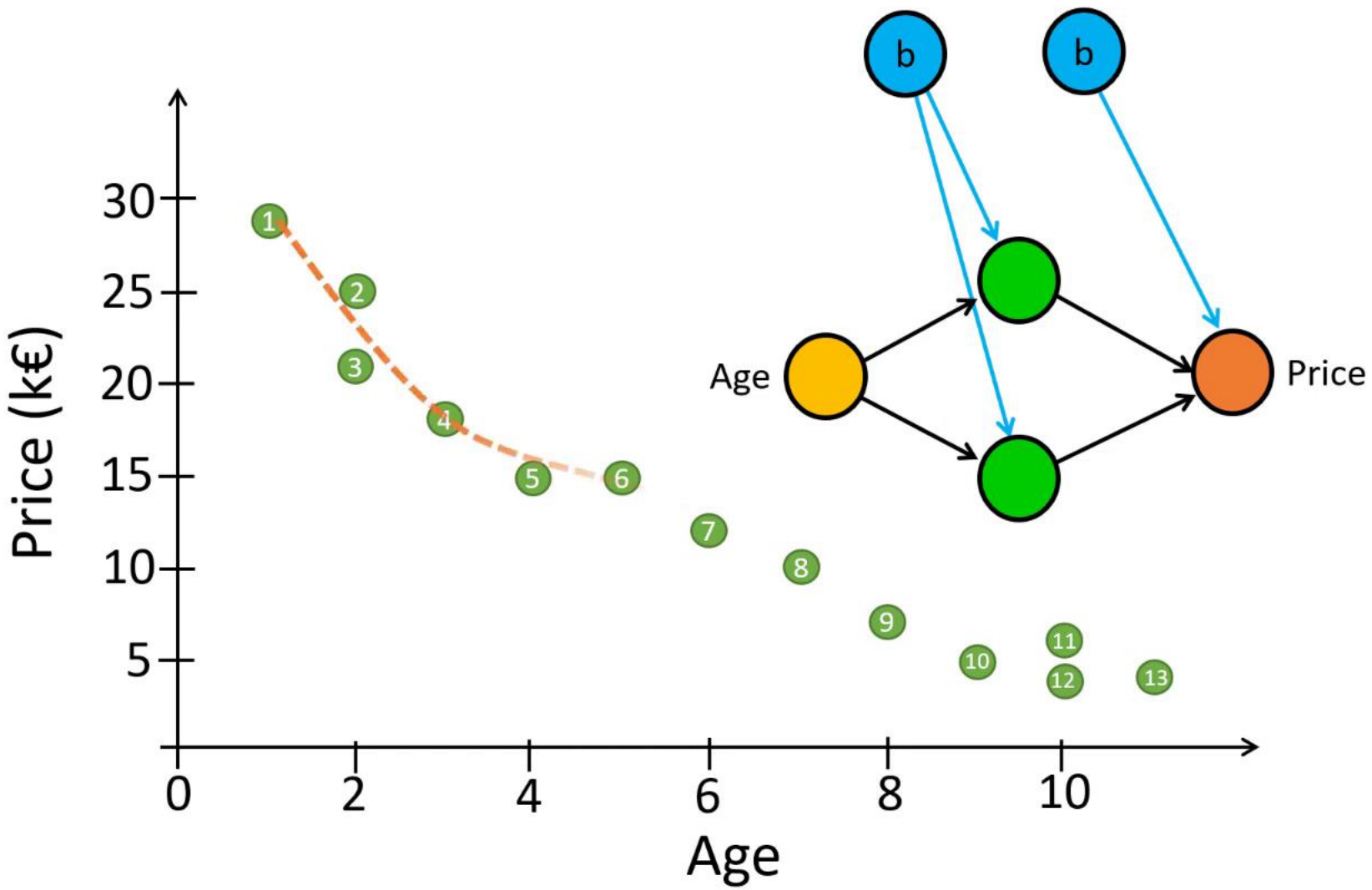
Suppose that we would train a neural network on the following data.

ANN



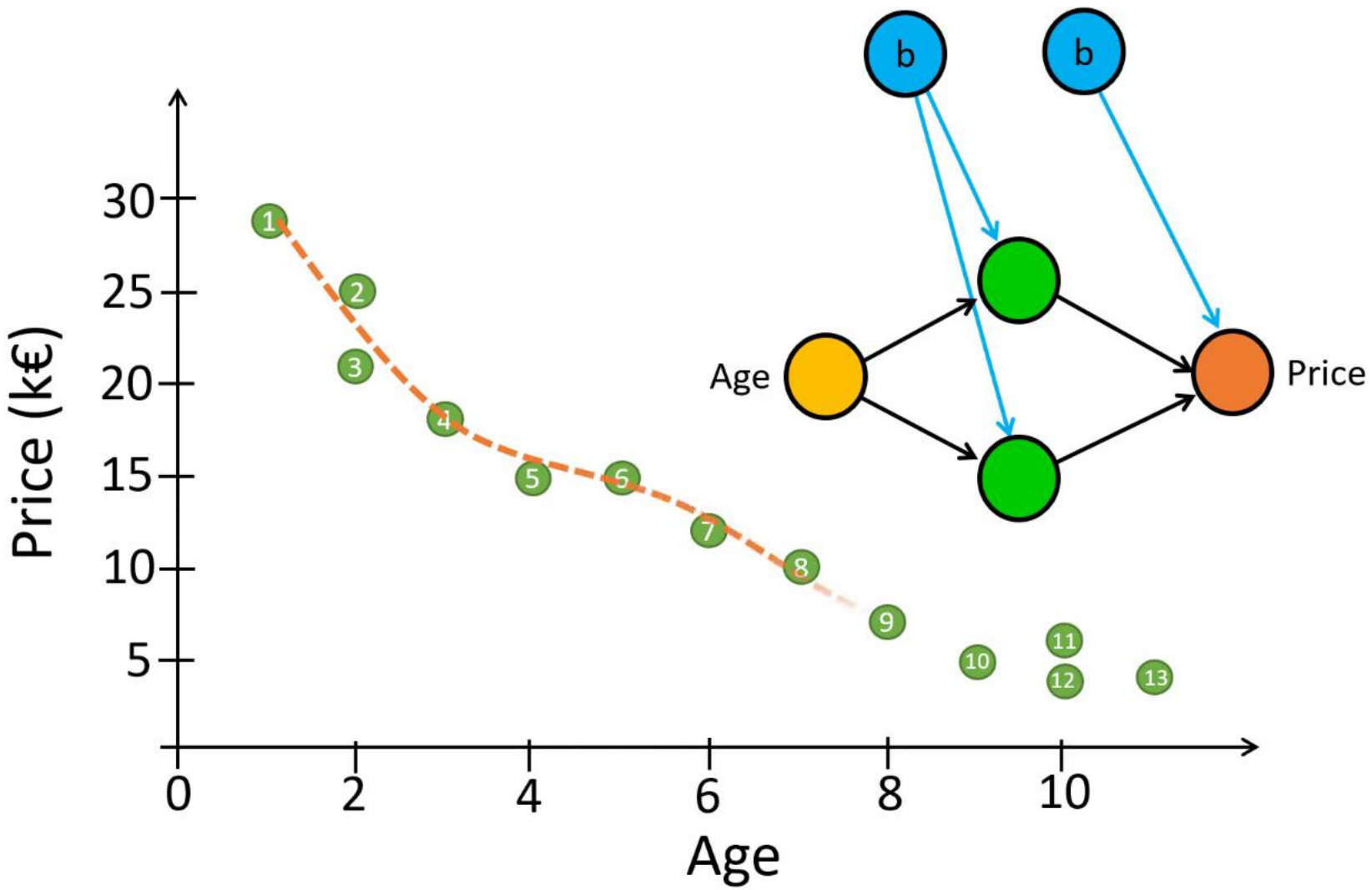
If we use the following network,

ANN



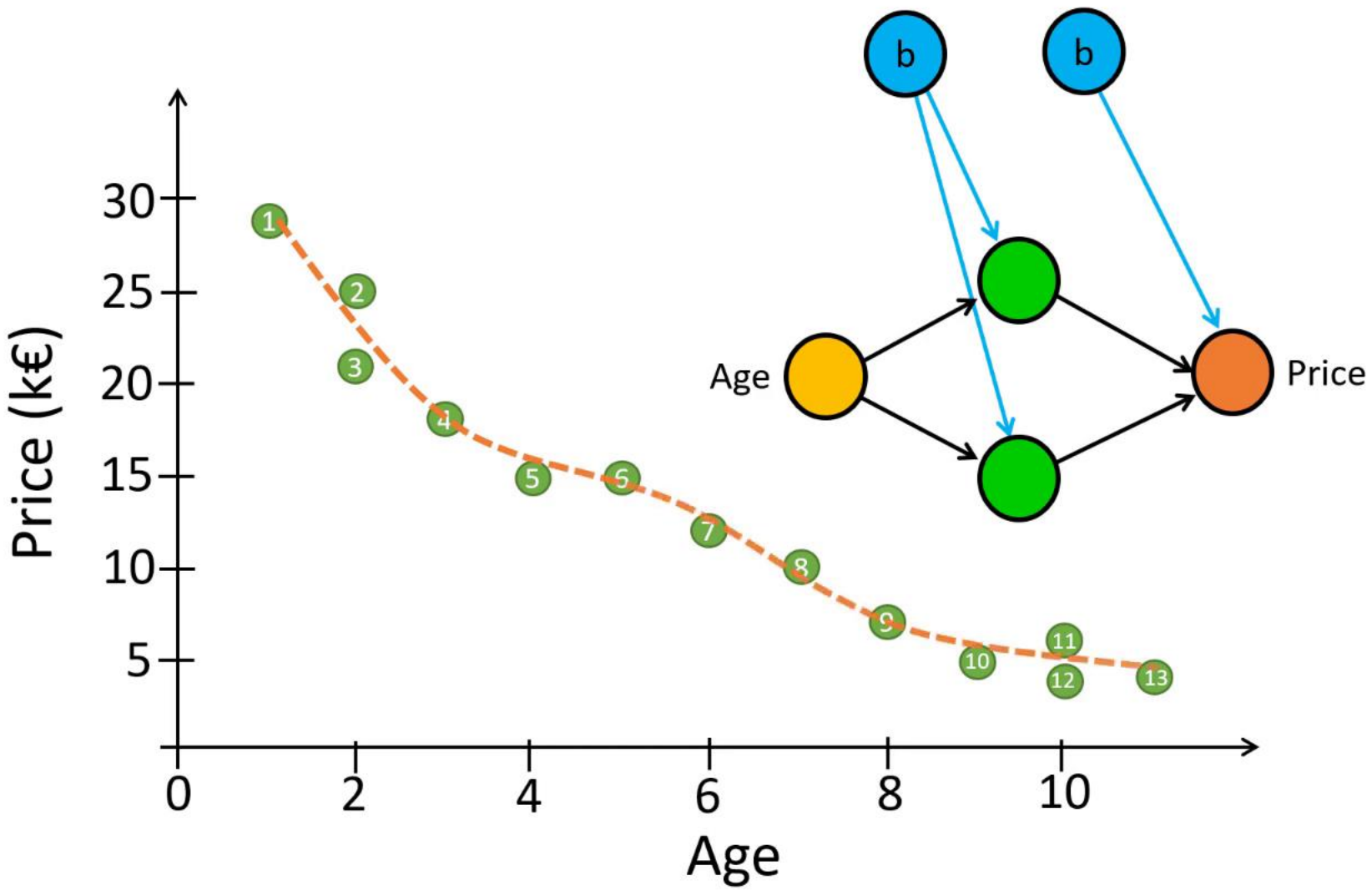
then we will be able to generate the following curve that fits the data.

ANN



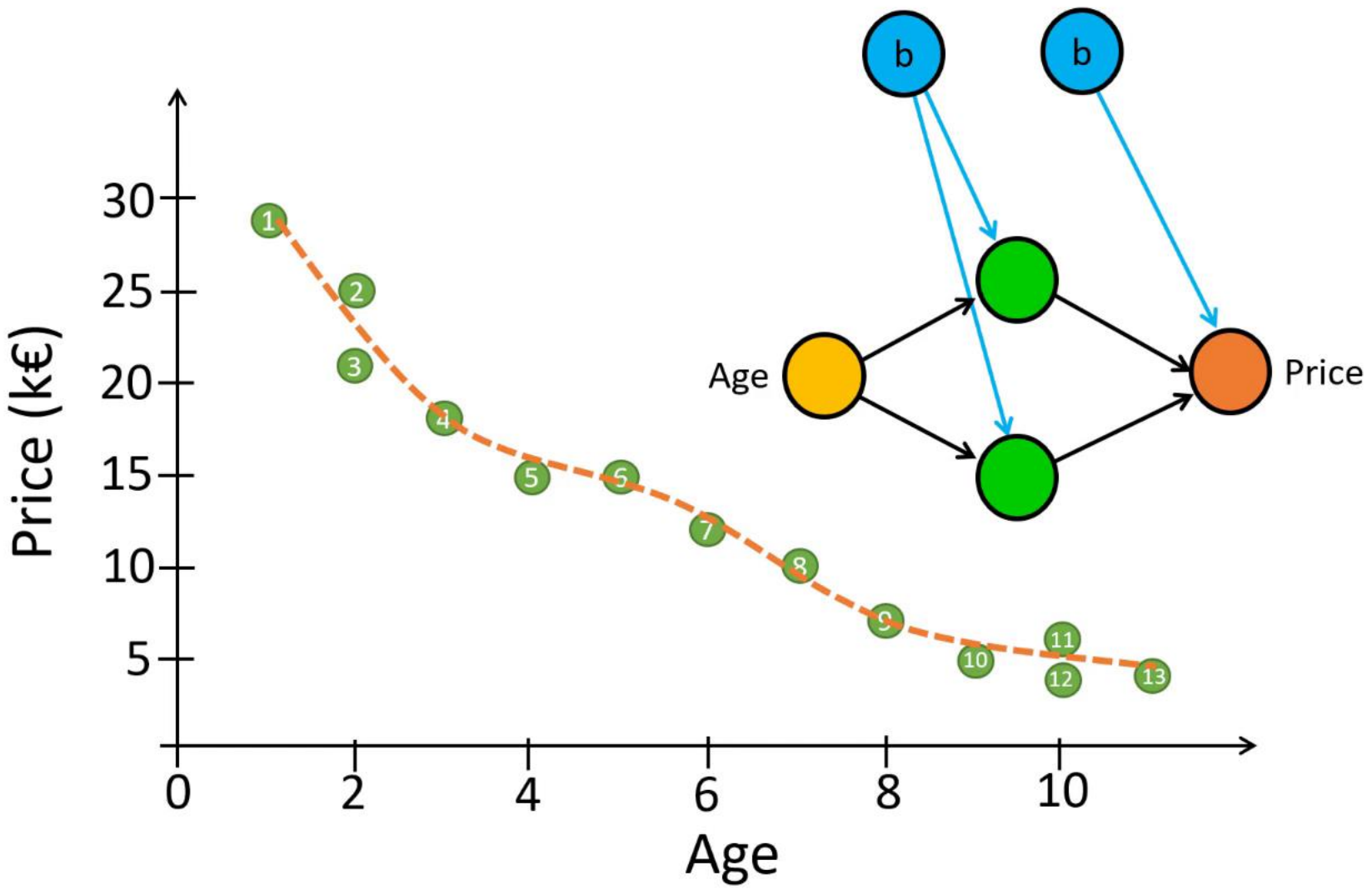
then we will be able to generate the following curve that fits the data.

ANN



then we will be able to generate the following curve that fits the data.

ANN

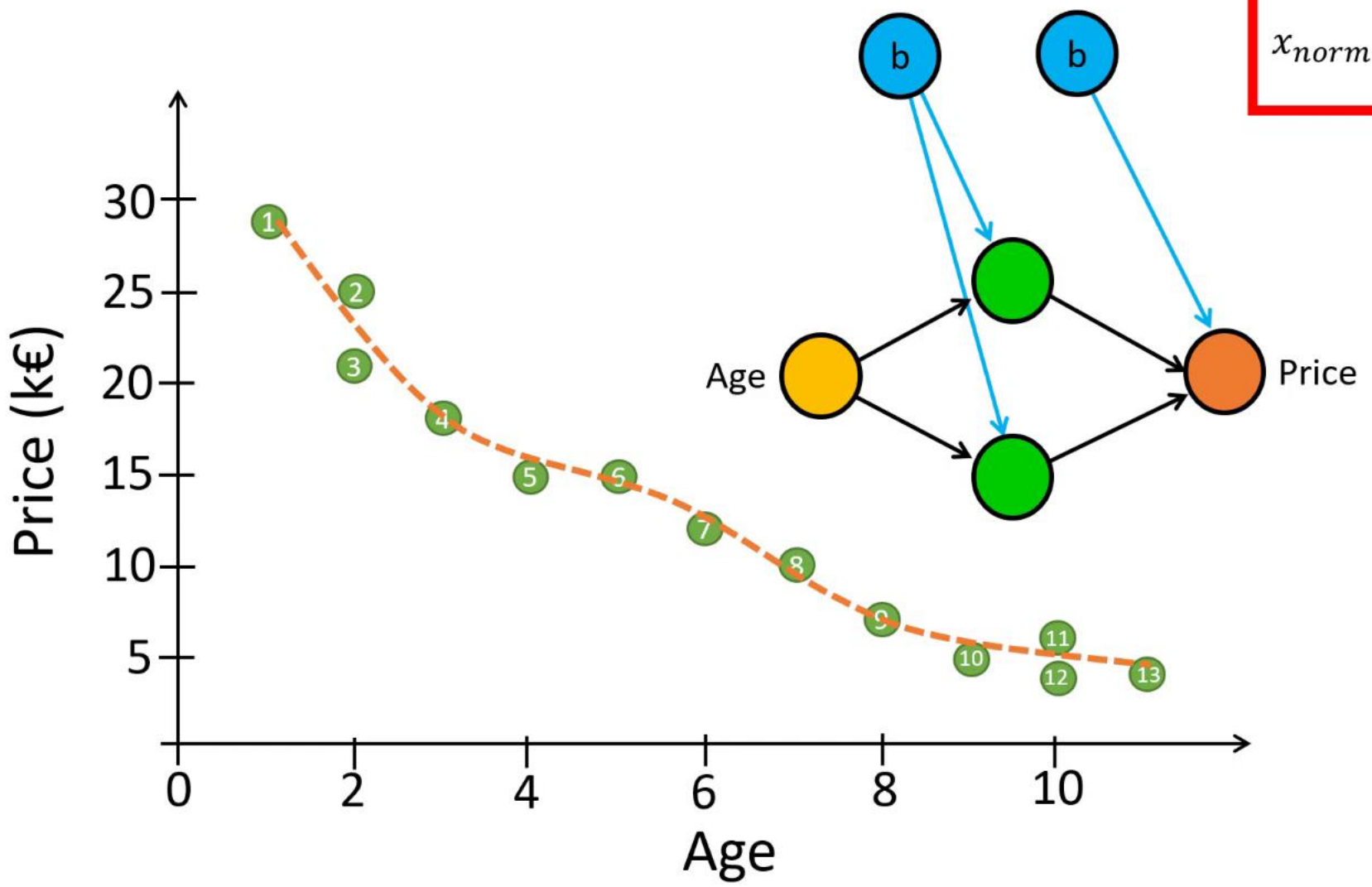


Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5
7	12	6
8	10	7
9	7	8
10	5	9
11	6	10
12	4	10
13	4	11

When we train a neural network on some data, it is recommended that we first somehow normalize the data because neural nets work best on such type of data.

ANN

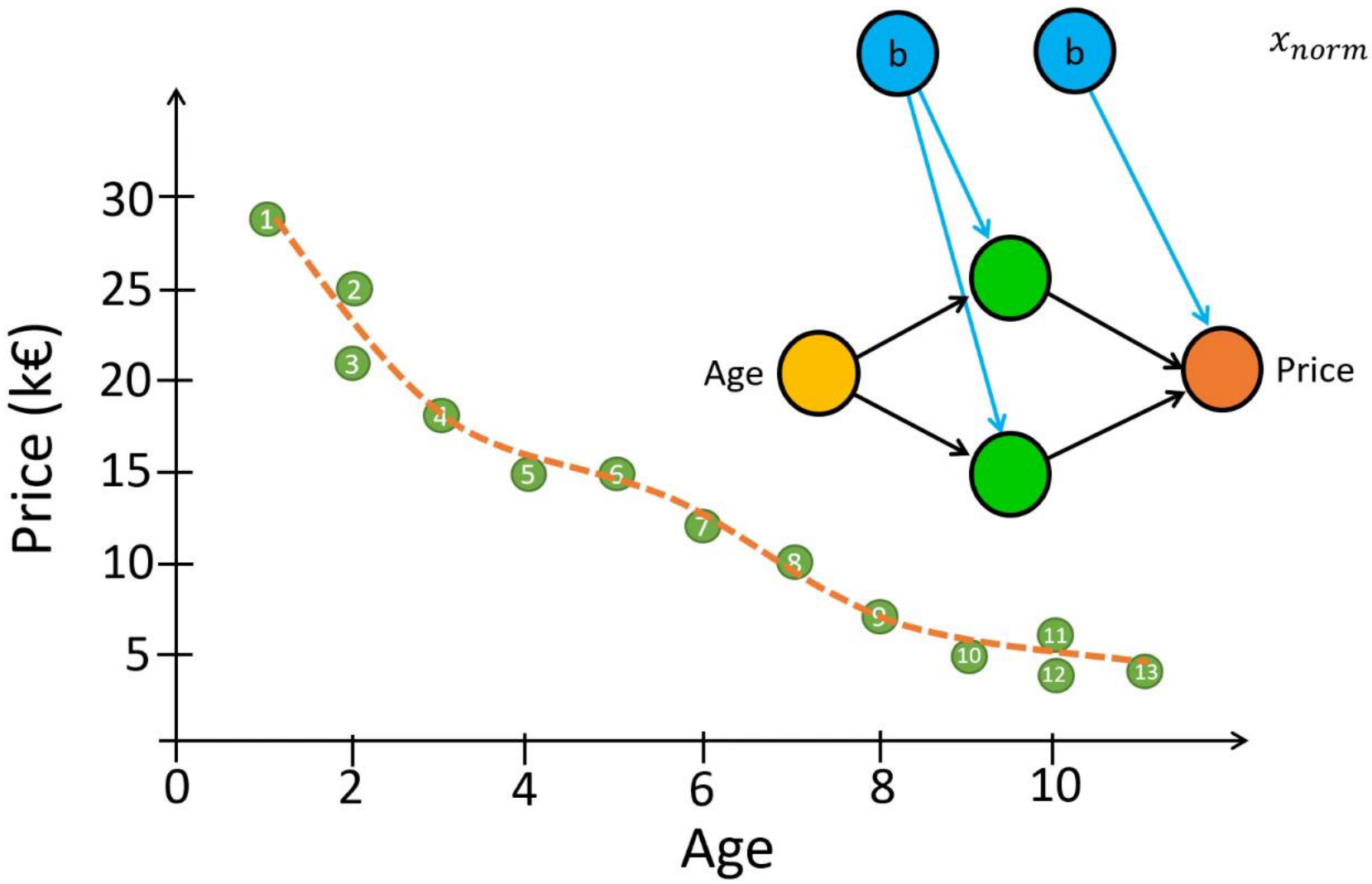
$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)}$$



Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5
7	12	6
8	10	7
9	7	8
10	5	9
11	6	10
12	4	10
13	4	11

One way to normalize the data is to use the min-max normalization, where each variable is scaled so that it has a range between zero and one.

ANN

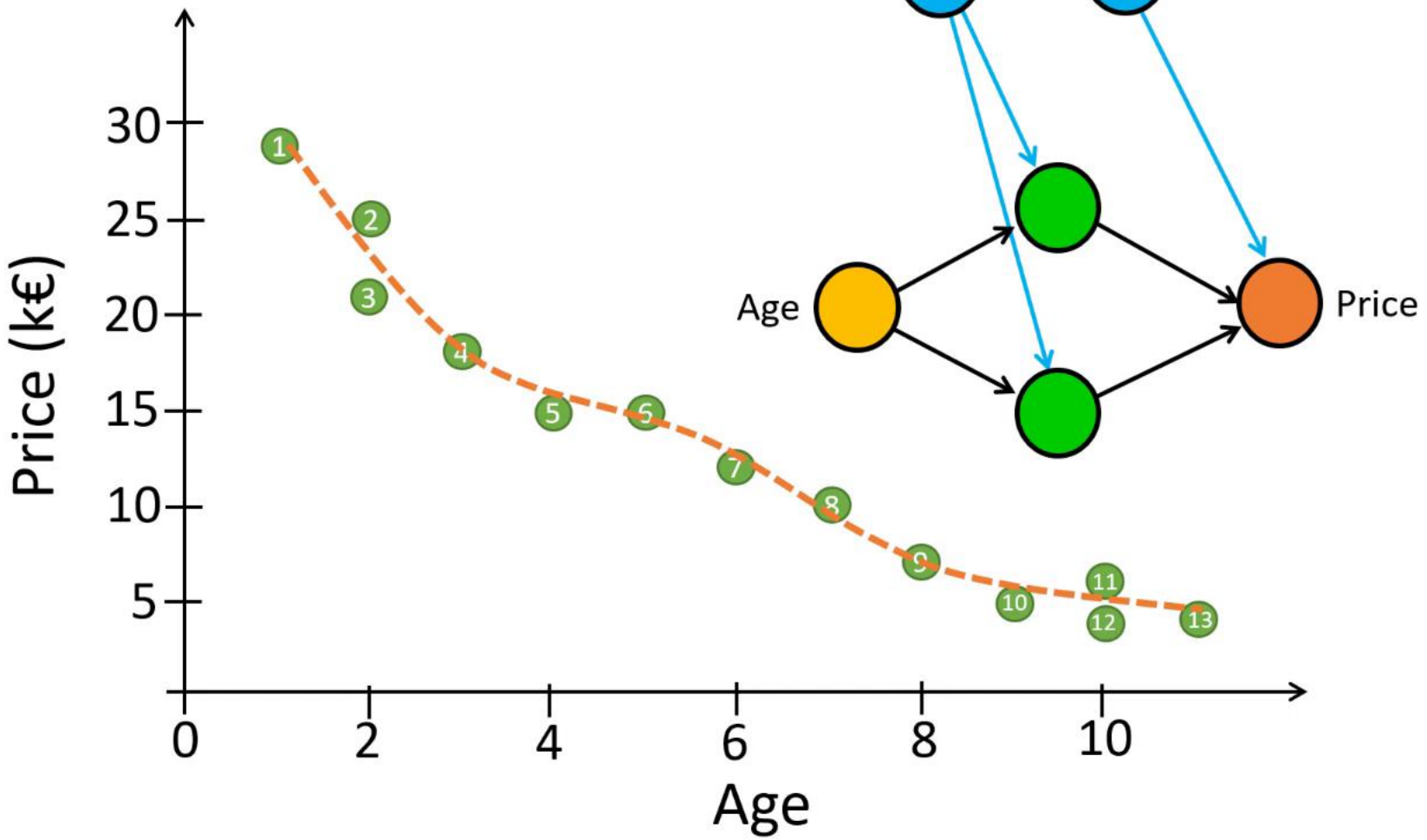


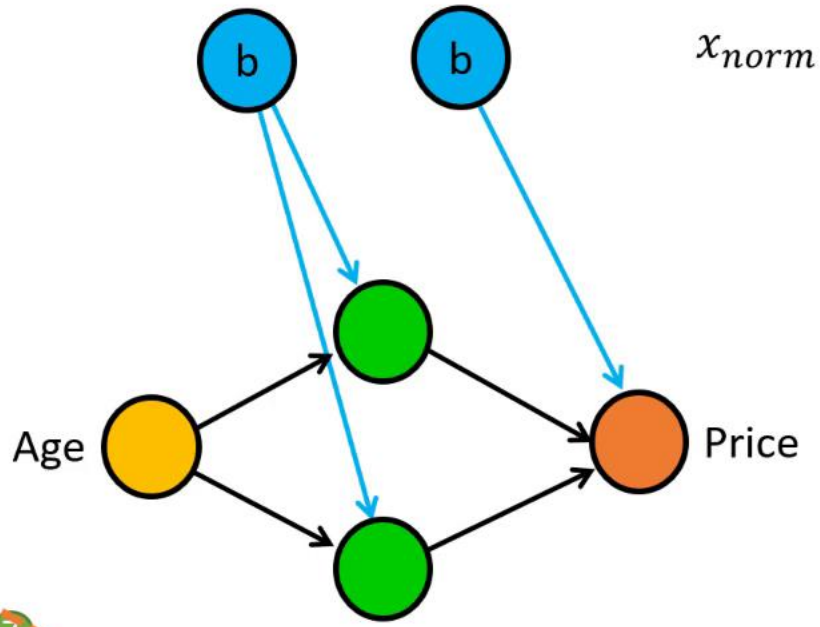
$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5
7	12	6
8	10	7
9	7	8
10	5	9
11	6	10
12	4	10
13	4	11

For example, suppose that we would normalize the variable price,

ANN



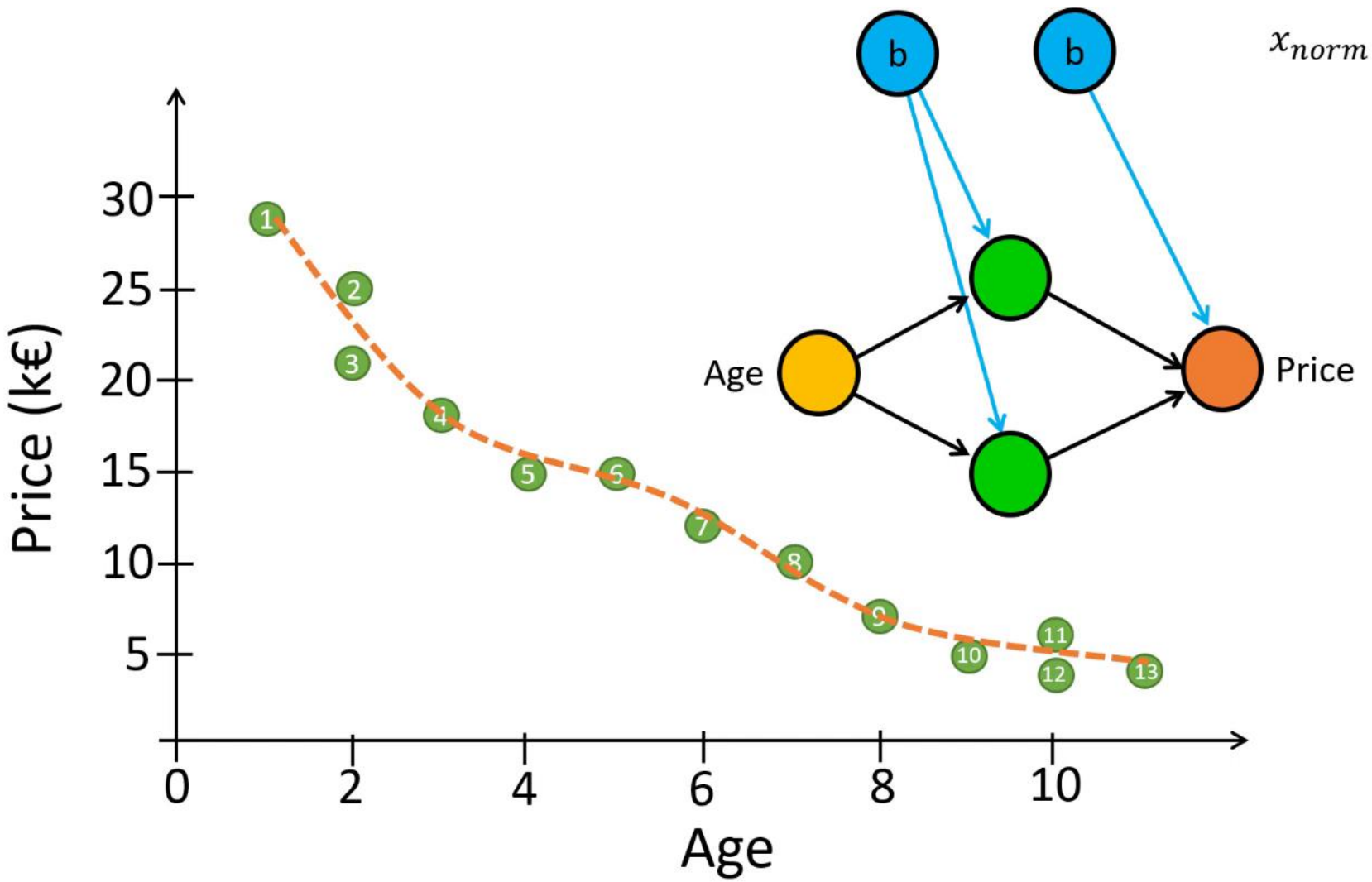


$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5
7	12	6
8	10	7
9	7	8
10	5	9
11	6	10
12	4	10
13	4	11

we then plug in the minimum value here,

ANN

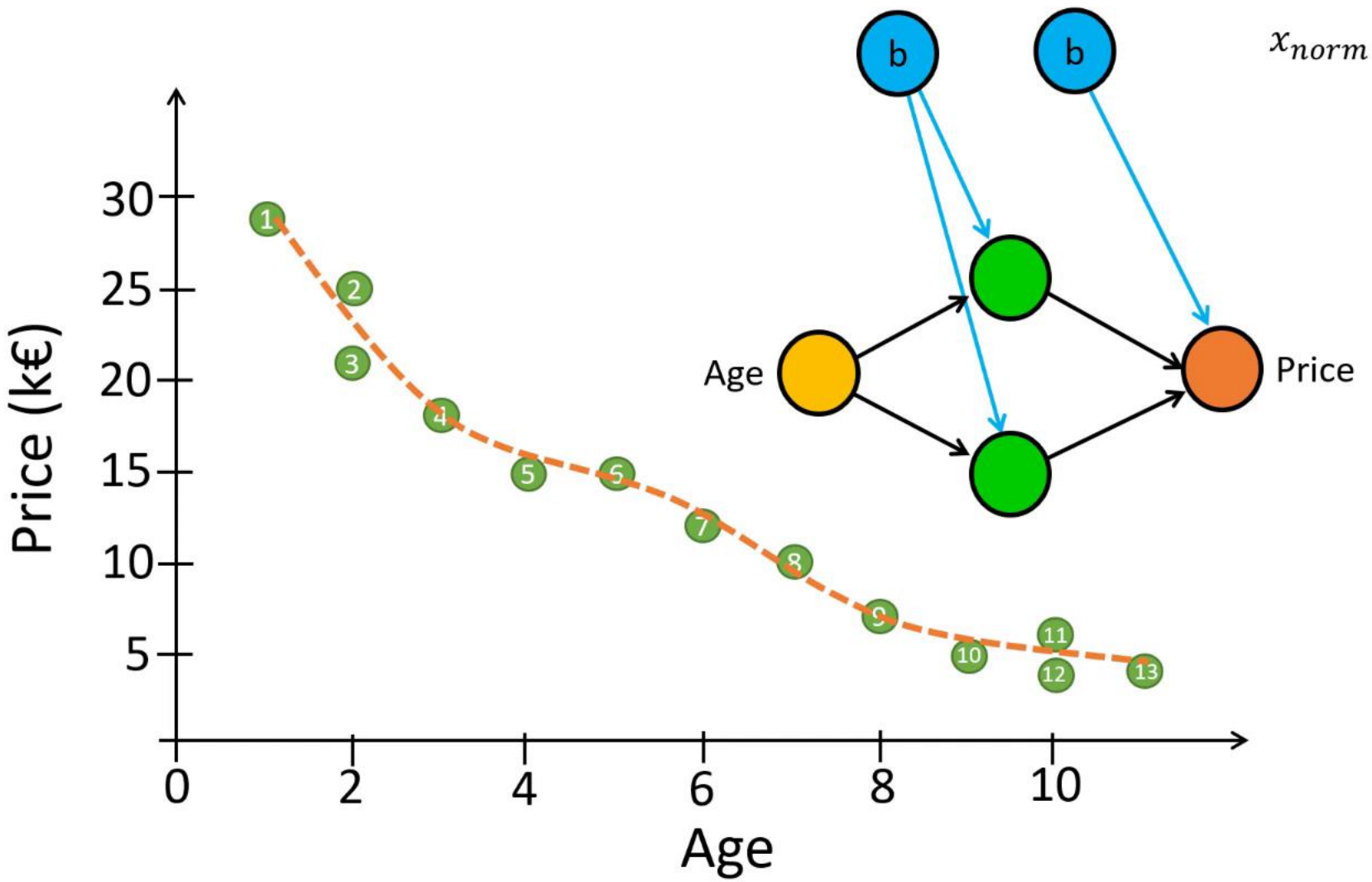


$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5
7	12	6
8	10	7
9	7	8
10	5	9
11	6	10
12	4	10
13	4	11

and the maximum value here.

ANN

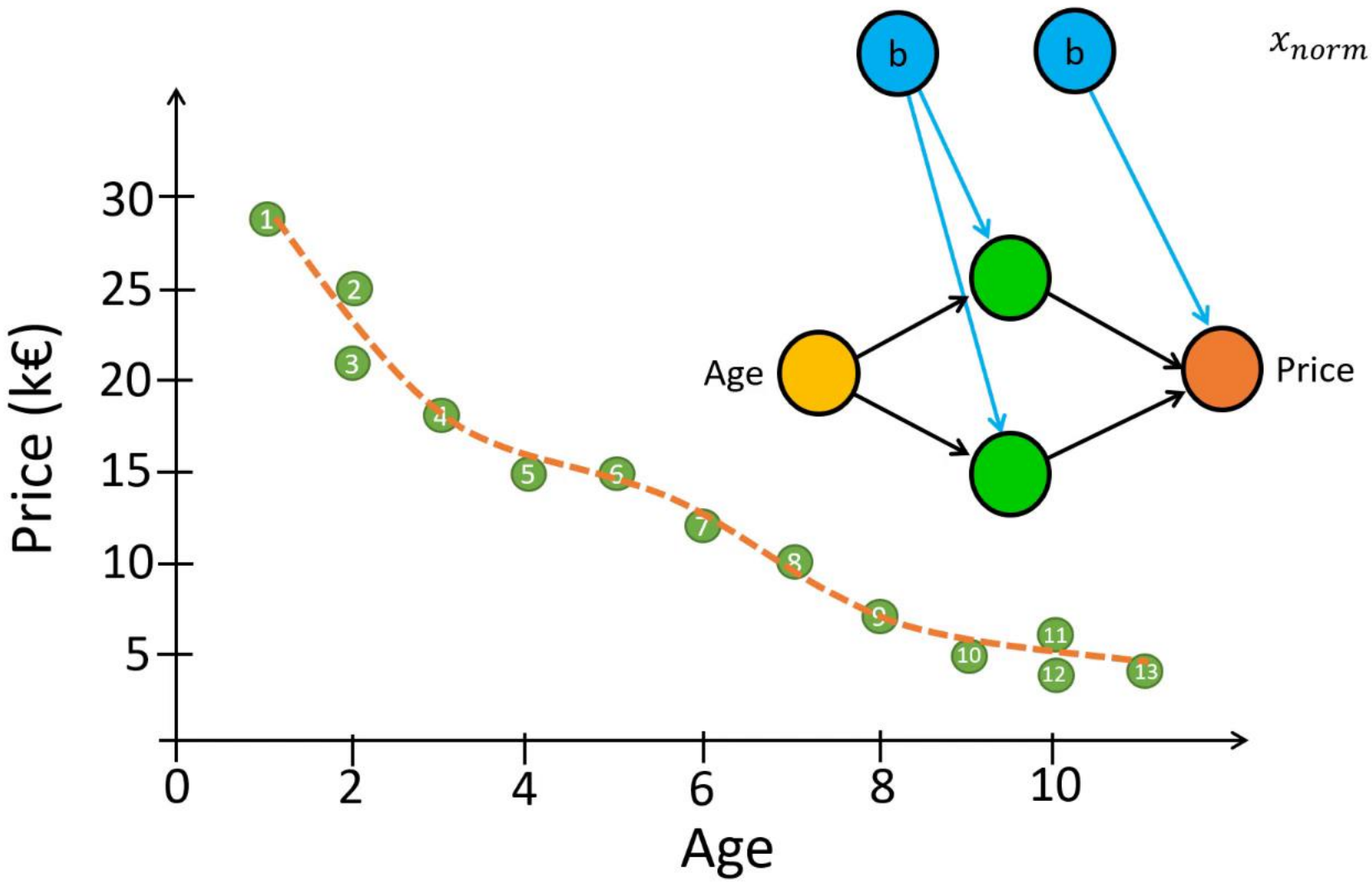


$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)} = \frac{x - 4}{29 - 4}$$

Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5
7	12	6
8	10	7
9	7	8
10	5	9
11	6	10
12	4	10
13	4	11

For example, the normalized value of the price of car number two,

ANN

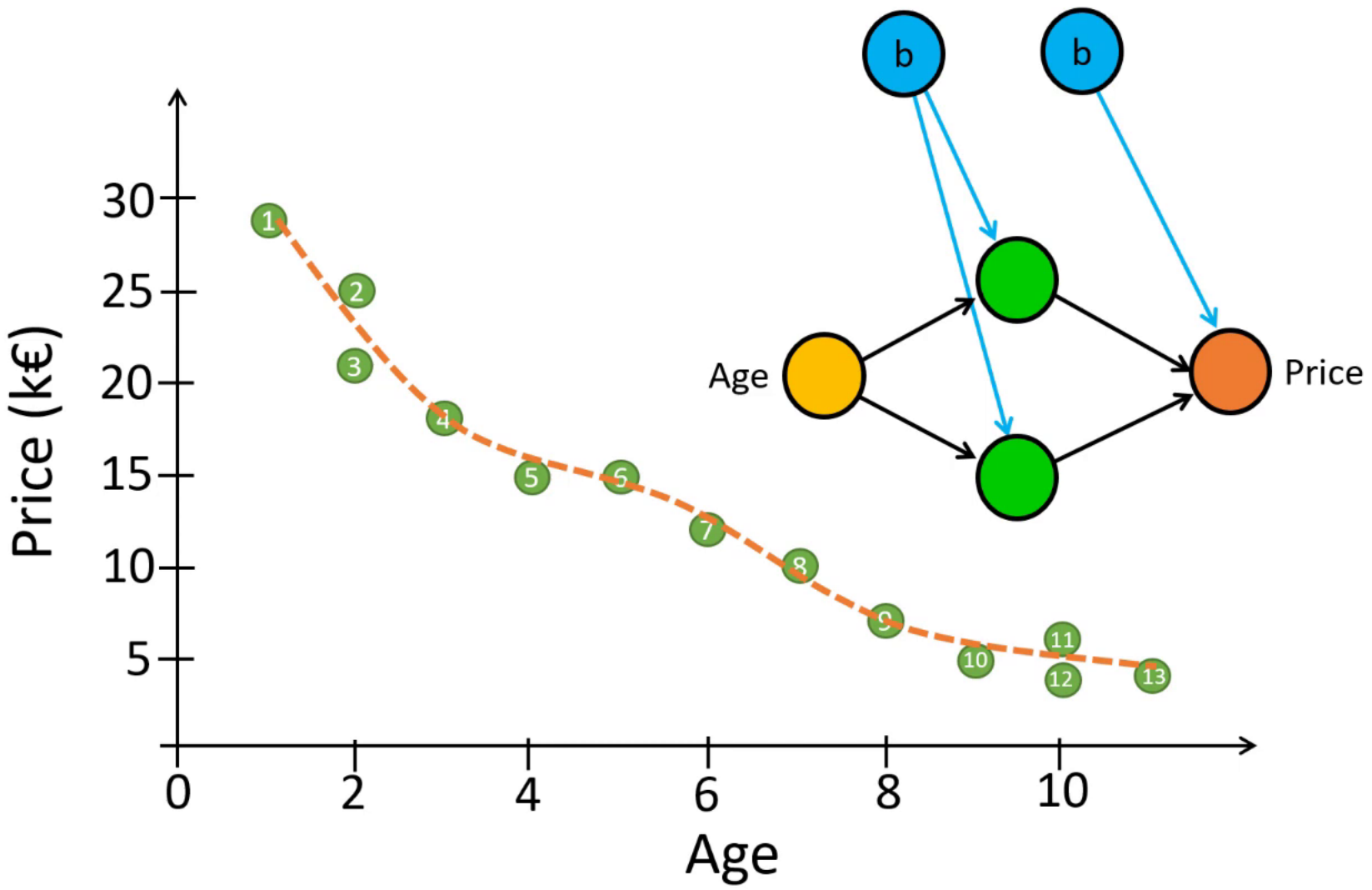


$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)} = \frac{x - 4}{29 - 4} = 0.84$$

Car	Price (k€)	Age (years)
1	29	1
2	25	2
3	21	2
4	18	3
5	15	4
6	15	5
7	12	6
8	10	7
9	7	8
10	5	9
11	6	10
12	4	10
13	4	11

is 0.84.

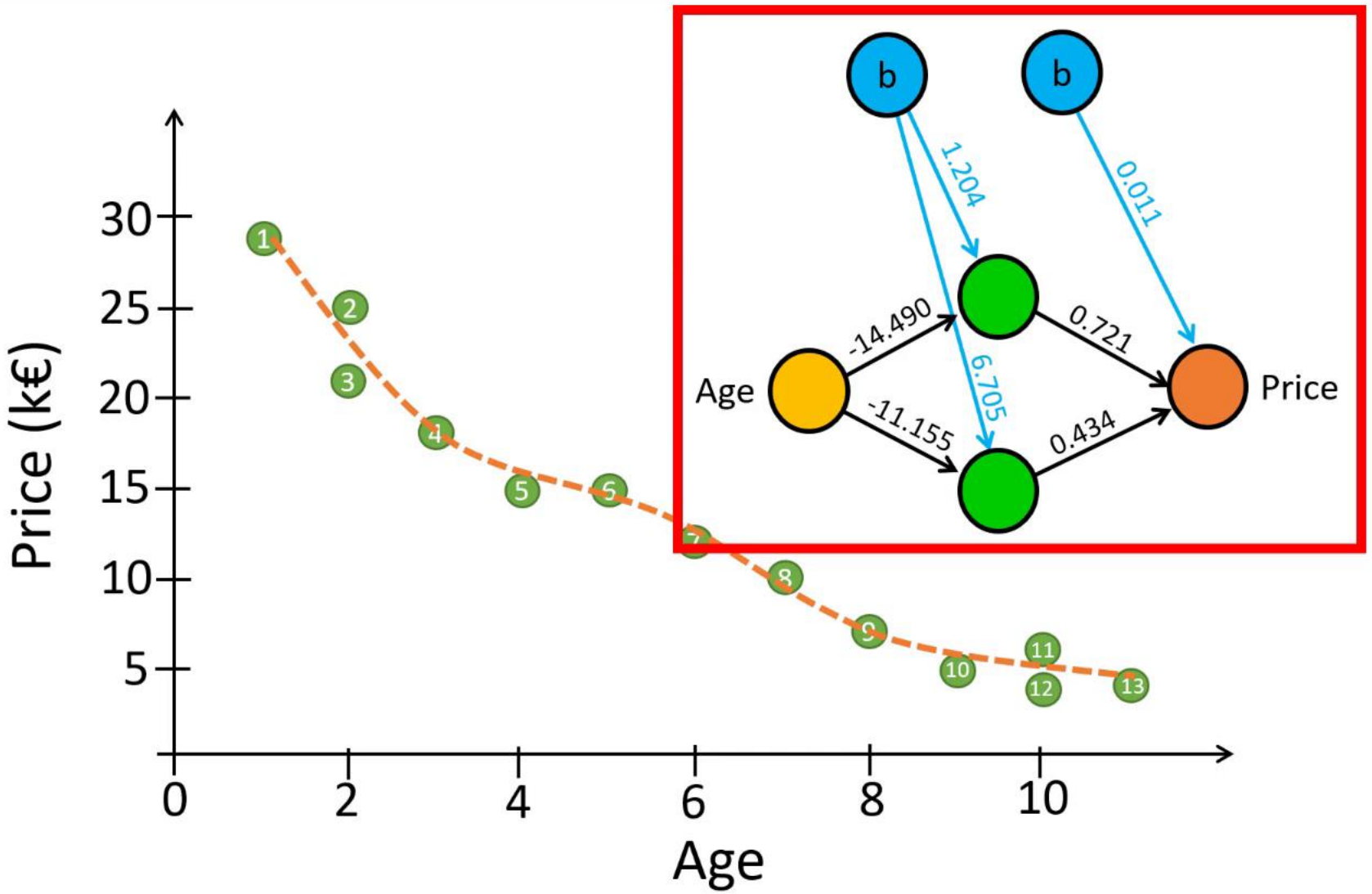
ANN



Car	Price (k€)	Age (years)	Norm Price	Norm Age
1	29	1	1.00	0.0
2	25	2	0.84	0.1
3	21	2	0.68	0.1
4	18	3	0.56	0.2
5	15	4	0.44	0.3
6	15	5	0.44	0.4
7	12	6	0.32	0.5
8	10	7	0.24	0.6
9	7	8	0.12	0.7
10	5	9	0.04	0.8
11	6	10	0.08	0.9
12	4	10	0.0	0.9
13	4	11	0.0	1.0

These are the normalized values of the two features, or variables.

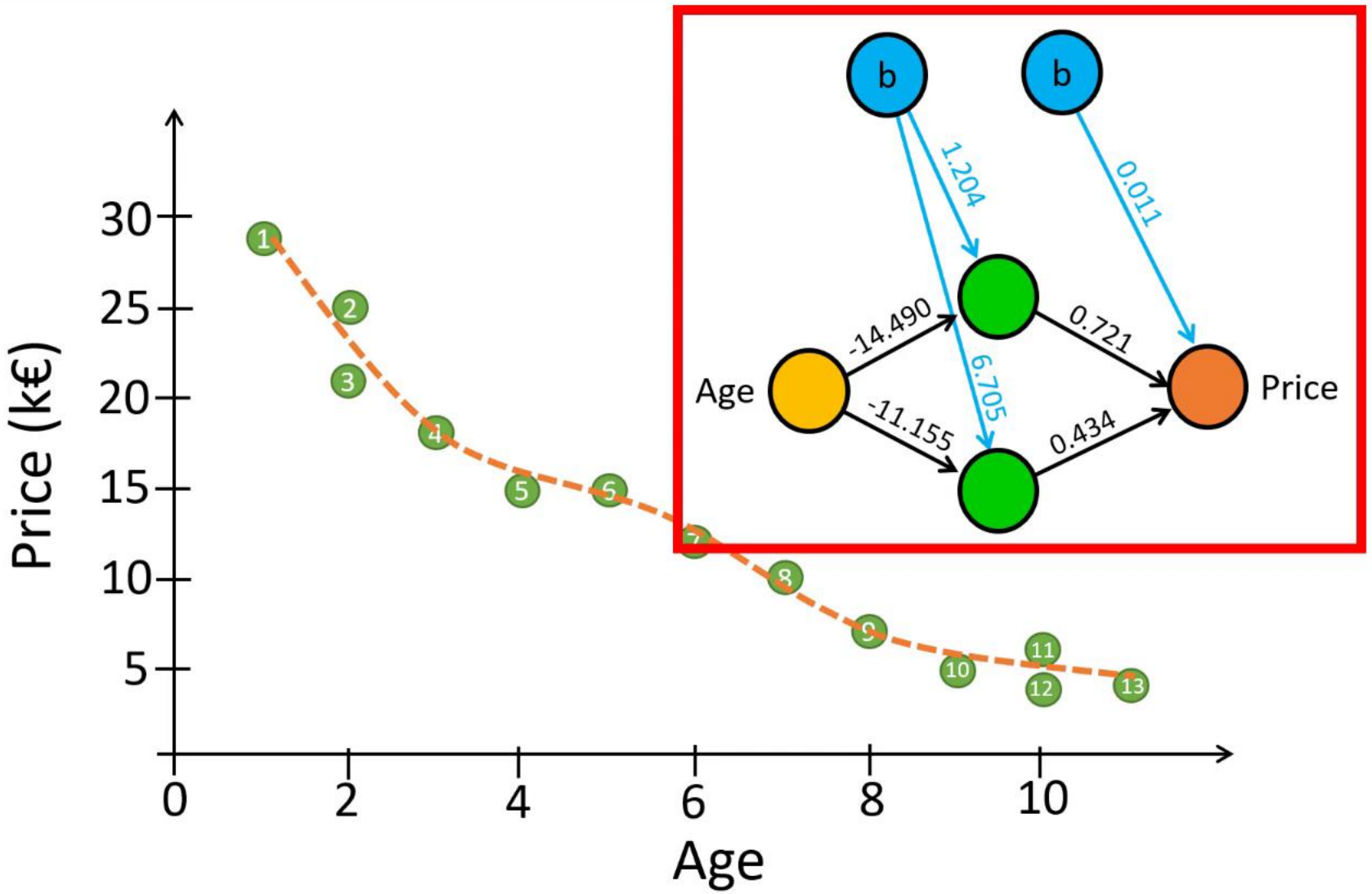
ANN



Car	Price (k€)	Age (years)	Norm Price	Norm Age
1	29	1	1.00	0.0
2	25	2	0.84	0.1
3	21	2	0.68	0.1
4	18	3	0.56	0.2
5	15	4	0.44	0.3
6	15	5	0.44	0.4
7	12	6	0.32	0.5
8	10	7	0.24	0.6
9	7	8	0.12	0.7
10	5	9	0.04	0.8
11	6	10	0.08	0.9
12	4	10	0.0	0.9
13	4	11	0.0	1.0

The following network was trained on the normalized data where the weights were optimized to these values.

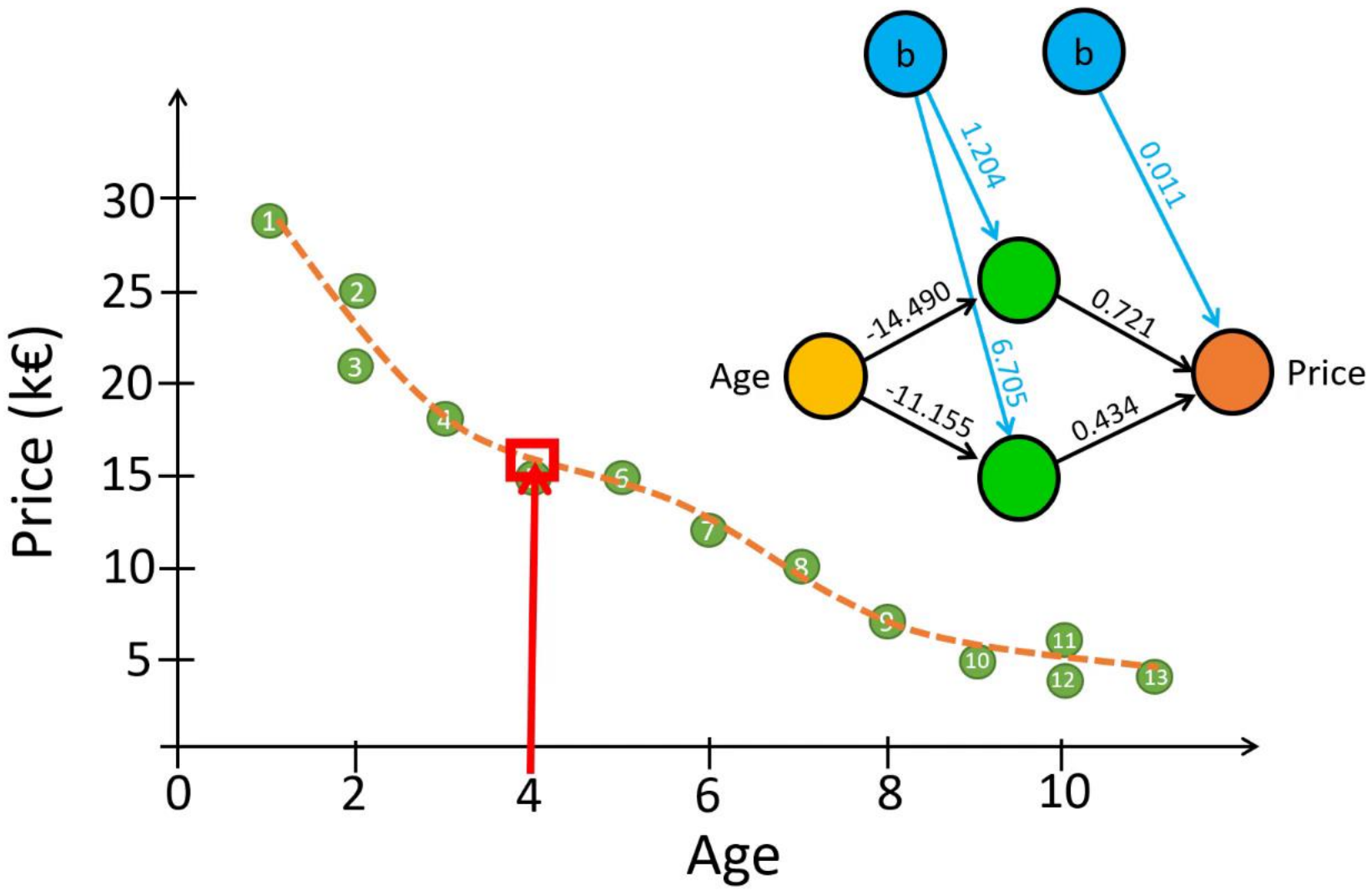
ANN



Car	Price (k€)	Age (years)	Norm Price	Norm Age
1	29	1	1.00	0.0
2	25	2	0.84	0.1
3	21	2	0.68	0.1
4	18	3	0.56	0.2
5	15	4	0.44	0.3
6	15	5	0.44	0.4
7	12	6	0.32	0.5
8	10	7	0.24	0.6
9	7	8	0.12	0.7
10	5	9	0.04	0.8
11	6	10	0.08	0.9
12	4	10	0.0	0.9
13	4	11	0.0	1.0

Note that you might obtain different values of the weight than the ones shown here because the values depend on the starting values you use as we discussed

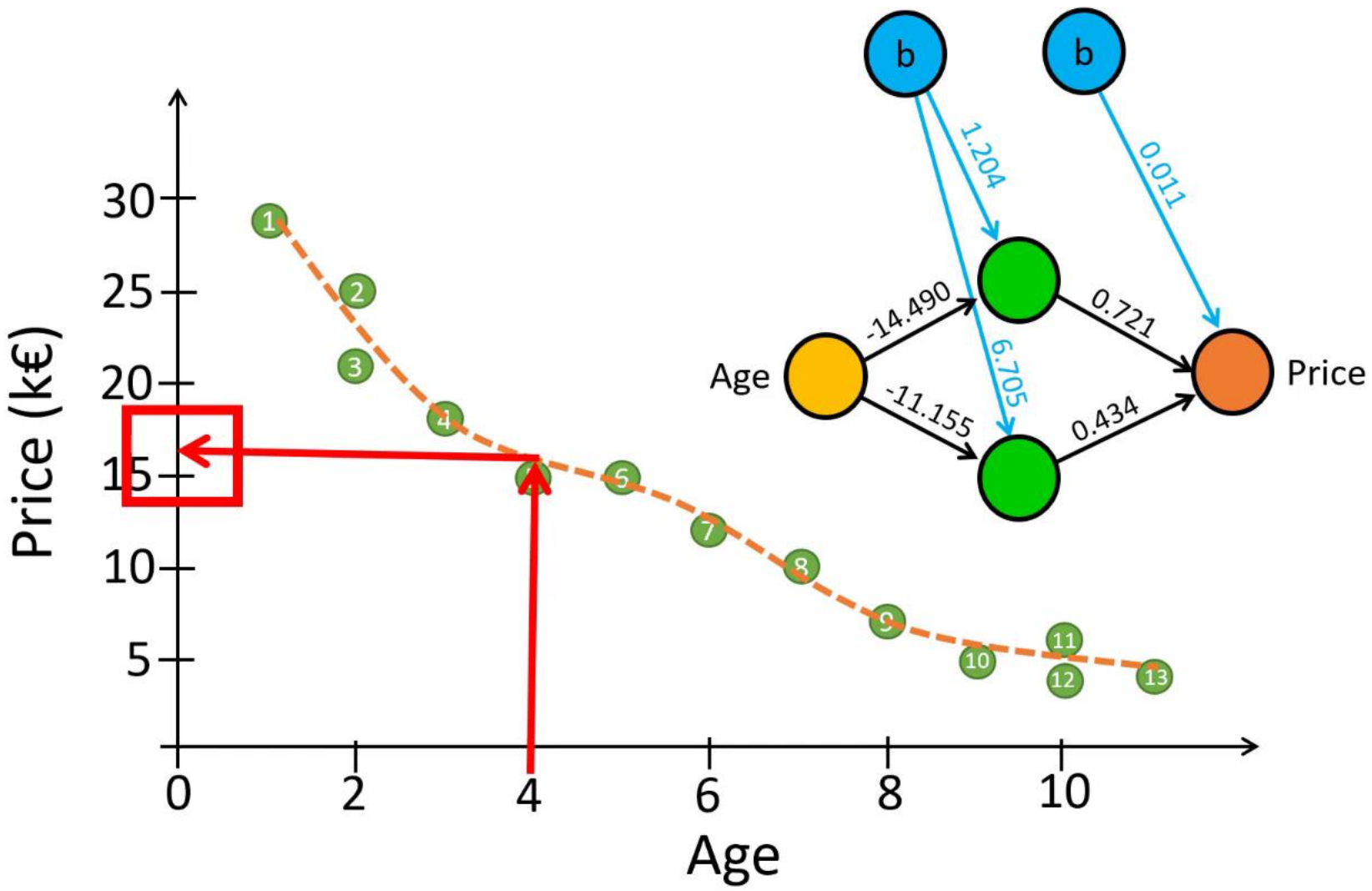
ANN



Car	Price (k€)	Age (years)	Norm Price	Norm Age
1	29	1	1.00	0.0
2	25	2	0.84	0.1
3	21	2	0.68	0.1
4	18	3	0.56	0.2
5	15	4	0.44	0.3
6	15	5	0.44	0.4
7	12	6	0.32	0.5
8	10	7	0.24	0.6
9	7	8	0.12	0.7
10	5	9	0.04	0.8
11	6	10	0.08	0.9
12	4	10	0.0	0.9
13	4	11	0.0	1.0

Suppose that we like to use the network to predict the price of a car that is 4 years old.

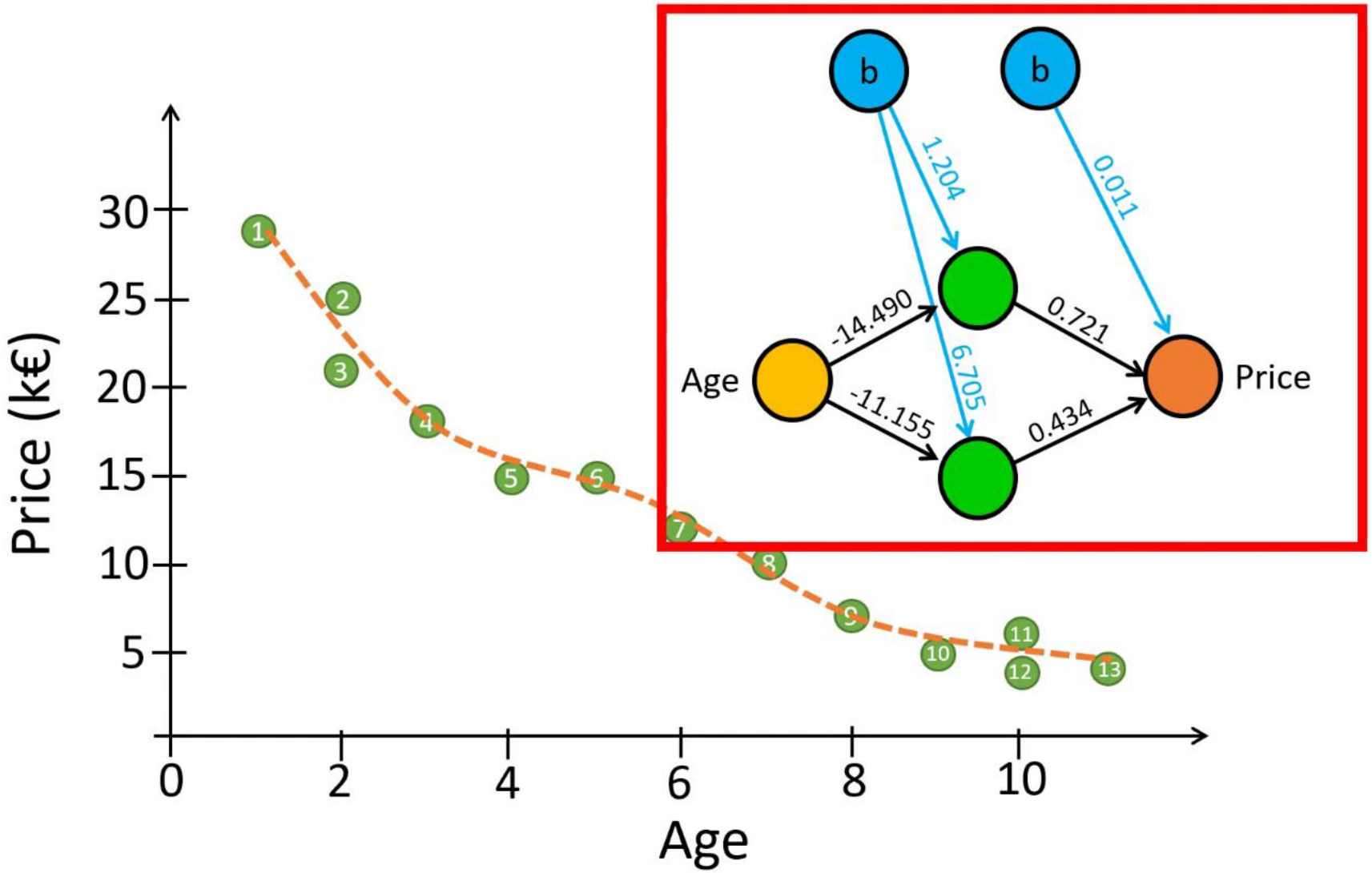
ANN



Car	Price (k€)	Age (years)	Norm Price	Norm Age
1	29	1	1.00	0.0
2	25	2	0.84	0.1
3	21	2	0.68	0.1
4	18	3	0.56	0.2
5	15	4	0.44	0.3
6	15	5	0.44	0.4
7	12	6	0.32	0.5
8	10	7	0.24	0.6
9	7	8	0.12	0.7
10	5	9	0.04	0.8
11	6	10	0.08	0.9
12	4	10	0.0	0.9
13	4	11	0.0	1.0

According to the curve, the predicted price seems to be around 16.000 euros.

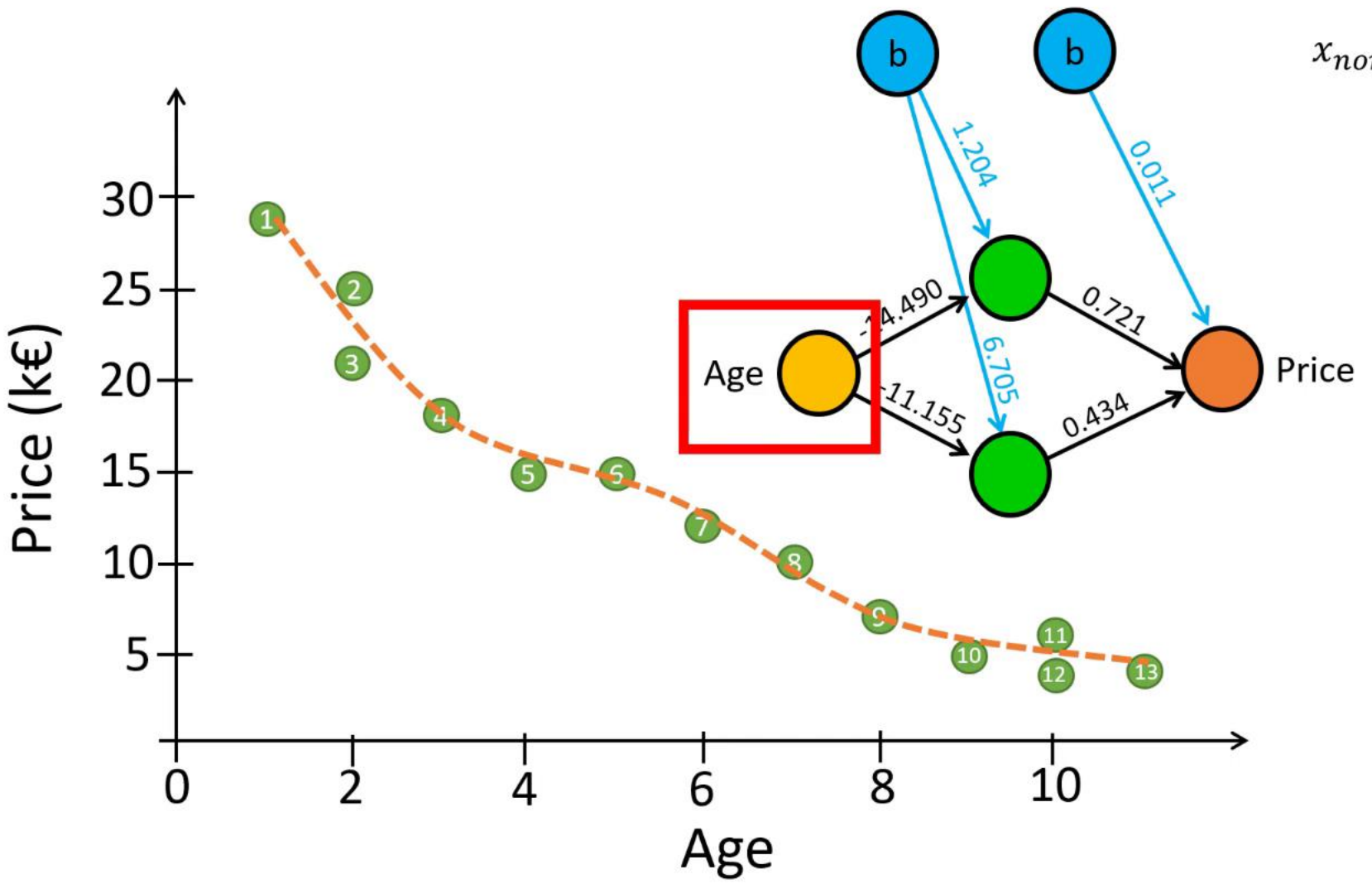
ANN



Car	Price (k€)	Age (years)	Norm Price	Norm Age
1	29	1	1.00	0.0
2	25	2	0.84	0.1
3	21	2	0.68	0.1
4	18	3	0.56	0.2
5	15	4	0.44	0.3
6	15	5	0.44	0.4
7	12	6	0.32	0.5
8	10	7	0.24	0.6
9	7	8	0.12	0.7
10	5	9	0.04	0.8
11	6	10	0.08	0.9
12	4	10	0.0	0.9
13	4	11	0.0	1.0

Let's use the network to predict more exactly.

ANN

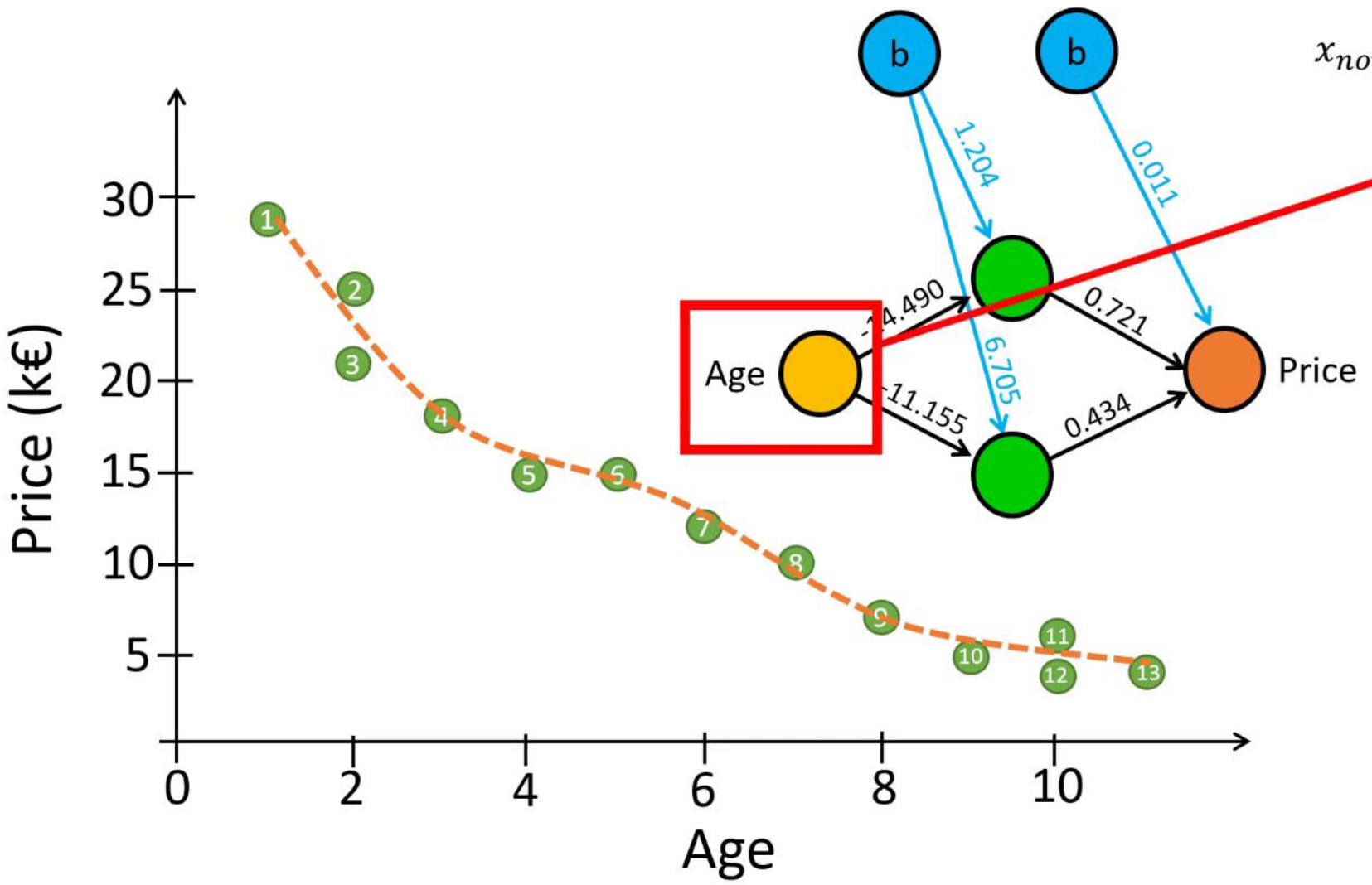


$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)} = \frac{4 - 1}{11 - 1} = 0.3$$

Car	Price (k€)	Age (years)	Norm Price	Norm Age
1	29	1	1.00	0.0
2	25	2	0.84	0.1
3	21	2	0.68	0.1
4	18	3	0.56	0.2
5	15	4	0.44	0.3
6	15	5	0.44	0.4
7	12	6	0.32	0.5
8	10	7	0.24	0.6
9	7	8	0.12	0.7
10	5	9	0.04	0.8
11	6	10	0.08	0.9
12	4	10	0.0	0.9
13	4	11	0.0	1.0

Since we have trained the network on normalized data, we need to plug in the corresponding normalized age.

ANN

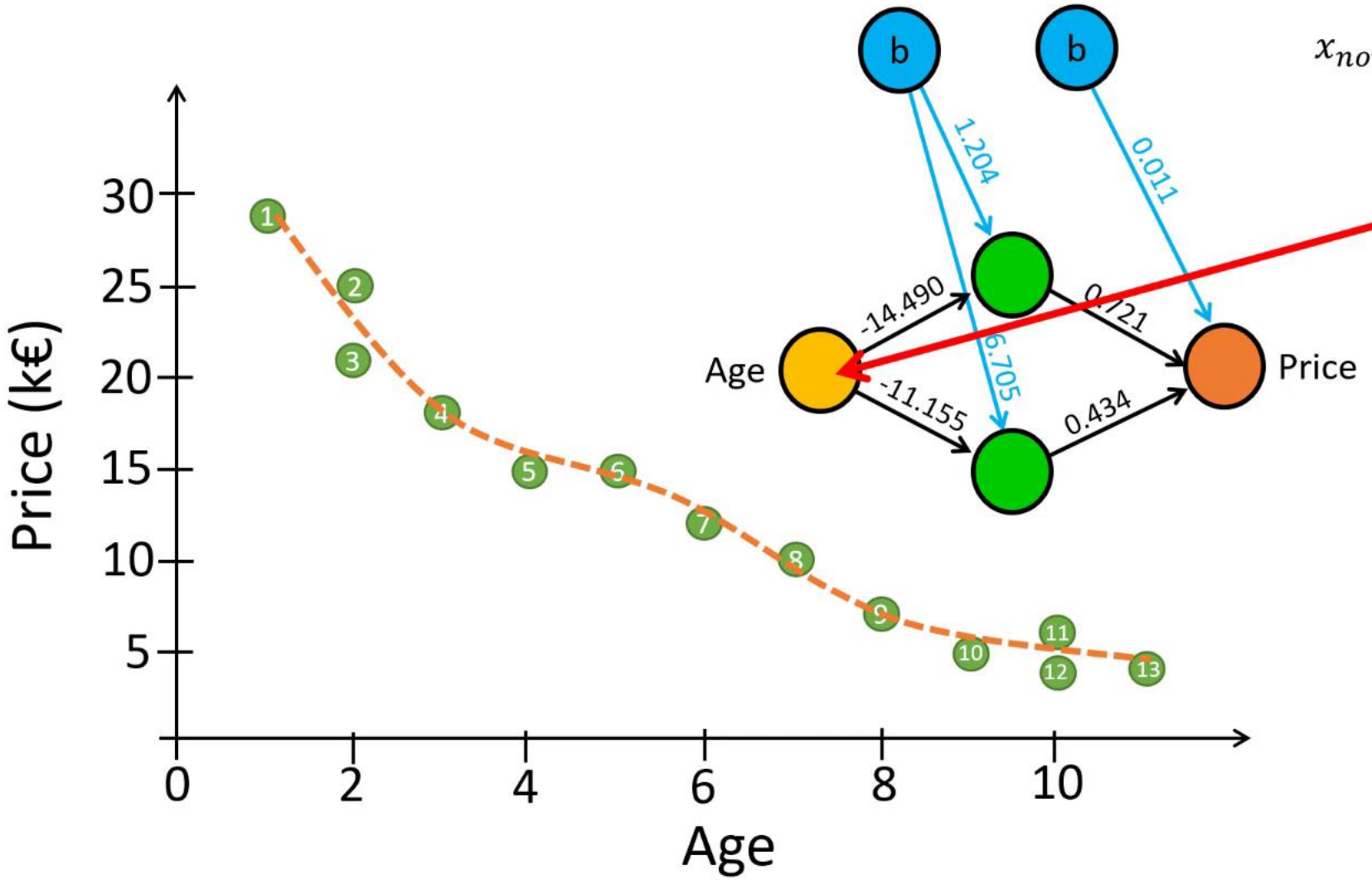


$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)} = \frac{4 - 1}{11 - 1} = 0.3$$

Car	Price (k€)	Age (years)	Norm Price	Norm Age
1	29	1	1.00	0.0
2	25	2	0.84	0.1
3	21	2	0.68	0.1
4	18	3	0.56	0.2
5	15	4	0.44	0.3
6	15	5	0.44	0.4
7	12	6	0.32	0.5
8	10	7	0.24	0.6
9	7	8	0.12	0.7
10	5	9	0.04	0.8
11	6	10	0.08	0.9
12	4	10	0.0	0.9
13	4	11	0.0	1.0

To predict the price of a car that is four years old,

ANN

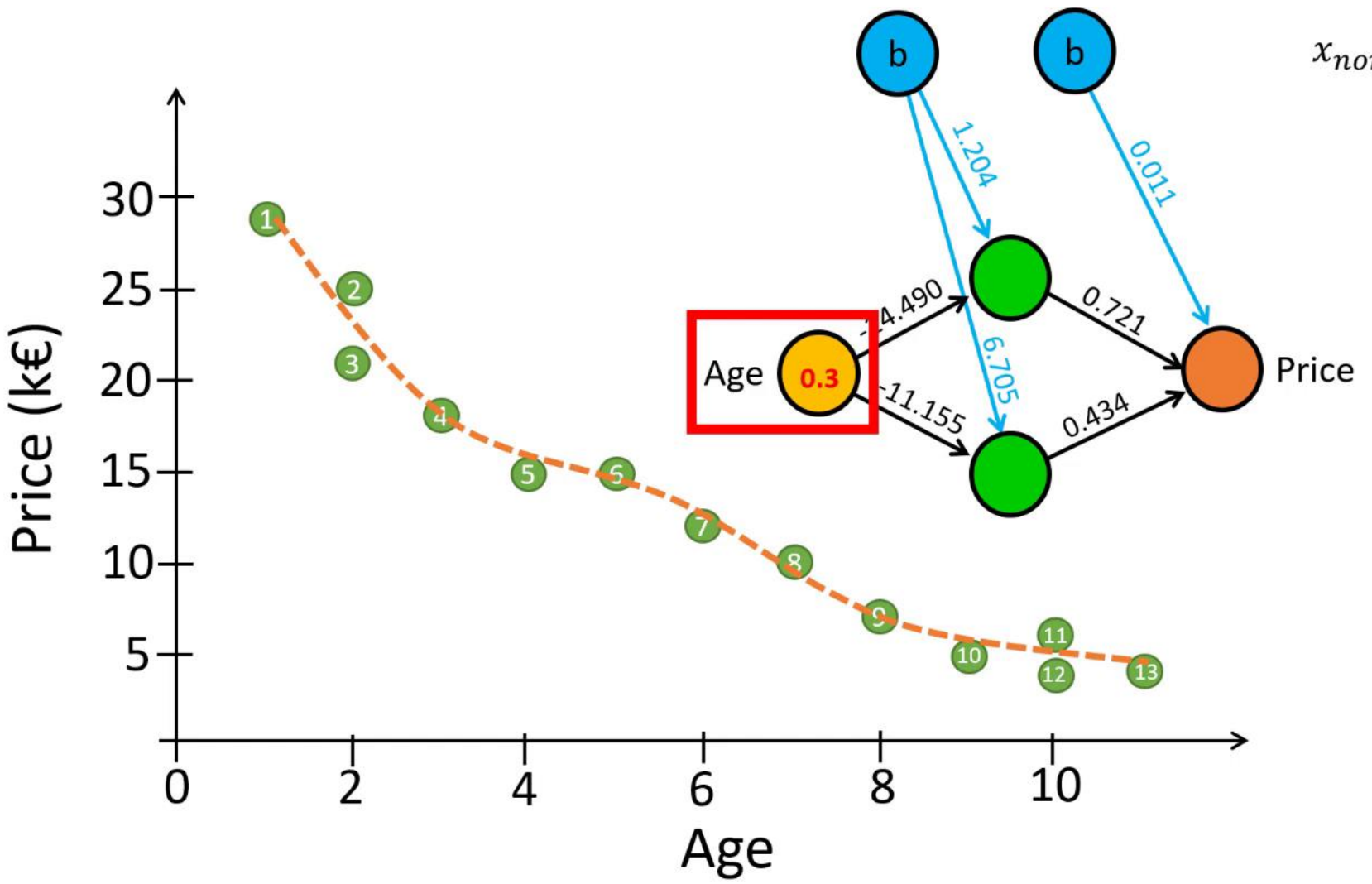


$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)} = \frac{4 - 1}{11 - 1} = 0.3$$

Car	Price (k€)	Age (years)	Norm Price	Norm Age
1	29	1	1.00	0.0
2	25	2	0.84	0.1
3	21	2	0.68	0.1
4	18	3	0.56	0.2
5	15	4	0.44	0.3
6	15	5	0.44	0.4
7	12	6	0.32	0.5
8	10	7	0.24	0.6
9	7	8	0.12	0.7
10	5	9	0.04	0.8
11	6	10	0.08	0.9
12	4	10	0.0	0.9
13	4	11	0.0	1.0

we plug in the corresponding normalized value of 0.3 in the network,

ANN

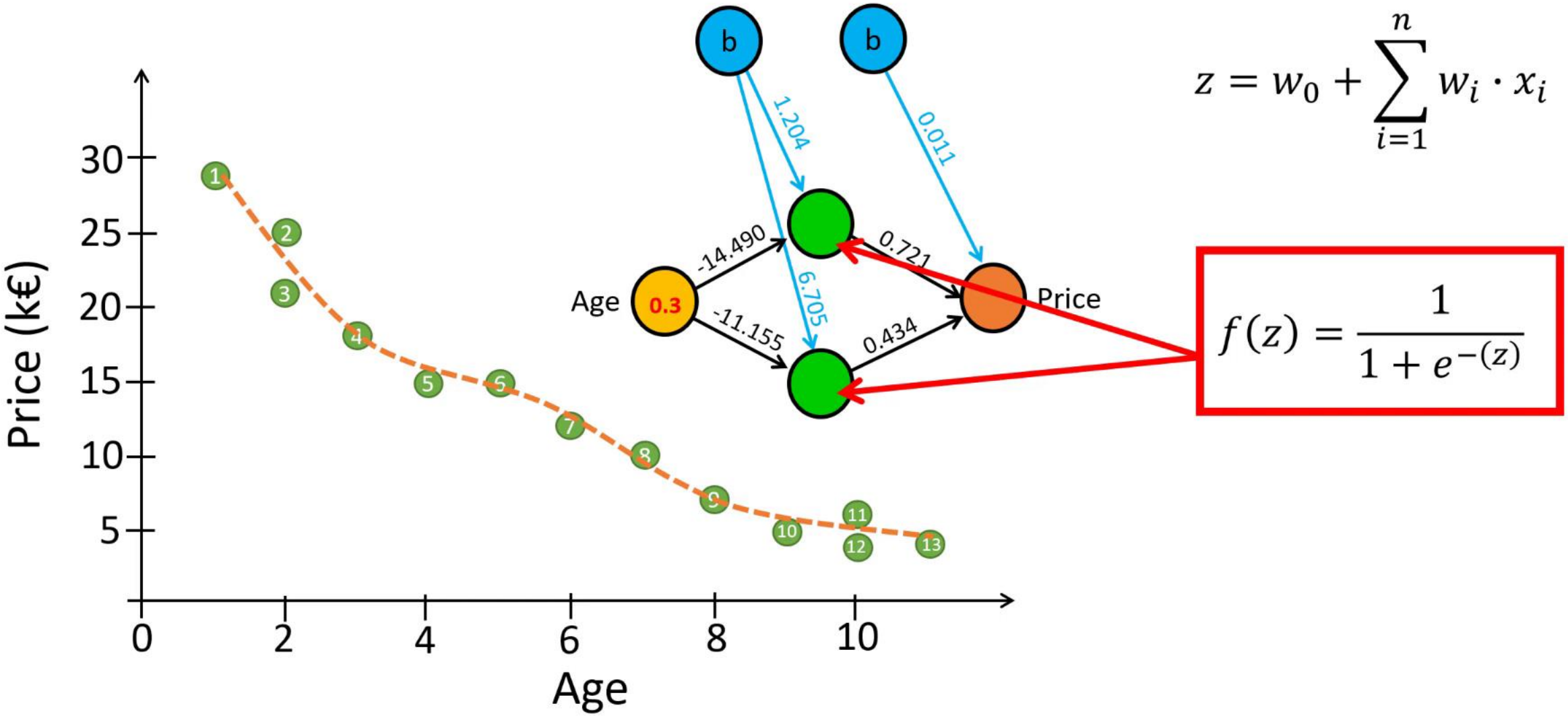


$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)} = \frac{4 - 1}{11 - 1} = 0.3$$

Car	Price (k€)	Age (years)	Norm Price	Norm Age
1	29	1	1.00	0.0
2	25	2	0.84	0.1
3	21	2	0.68	0.1
4	18	3	0.56	0.2
5	15	4	0.44	0.3
6	15	5	0.44	0.4
7	12	6	0.32	0.5
8	10	7	0.24	0.6
9	7	8	0.12	0.7
10	5	9	0.04	0.8
11	6	10	0.08	0.9
12	4	10	0.0	0.9
13	4	11	0.0	1.0

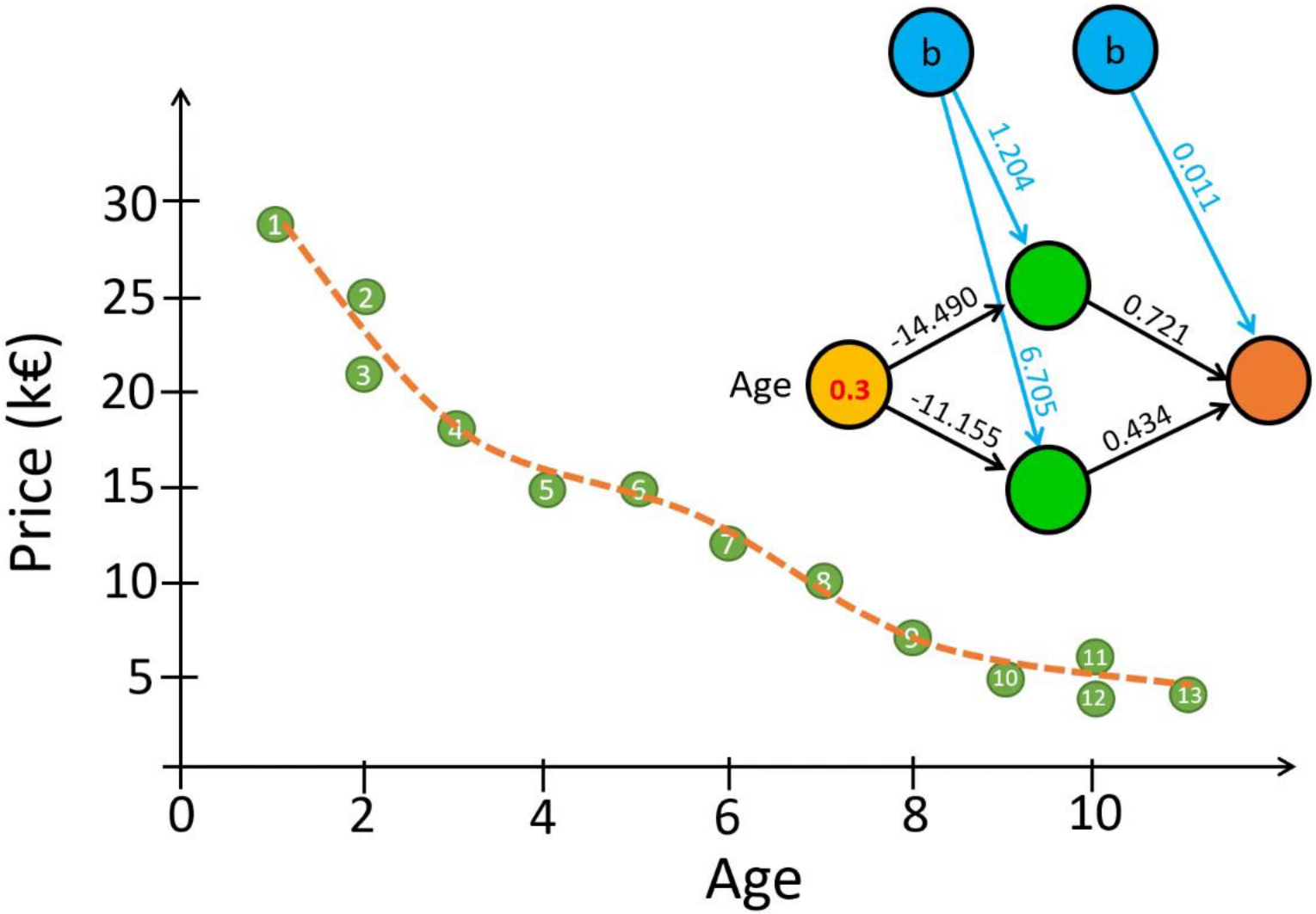
like this.

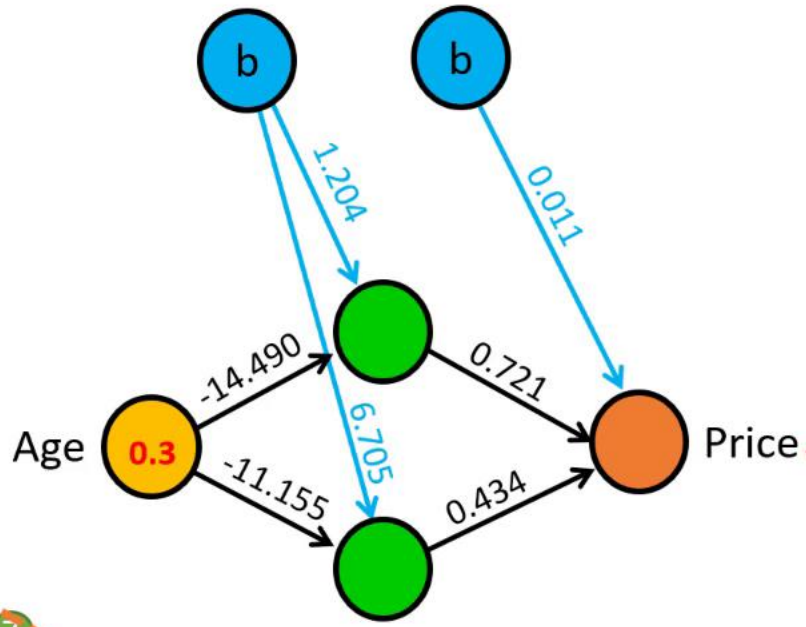
ANN



To generate a nonlinear function, I have here selected the sigmoid, or logistic, activation function for the hidden layer. Note that you can try other activation functions.

ANN



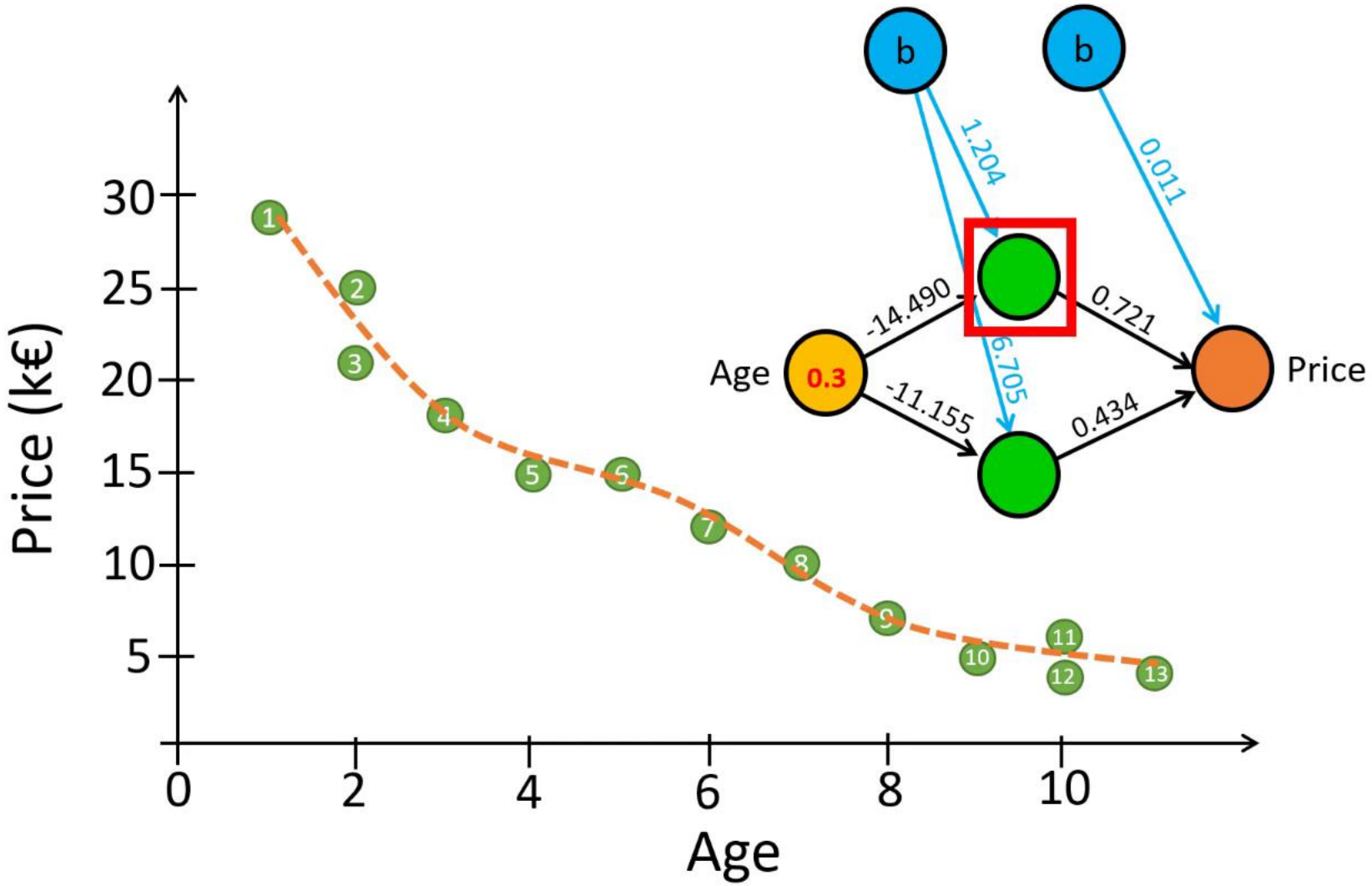


$$z = w_0 + \sum_{i=1}^{n} w_i \cdot x_i$$

$$f(z) = z$$

When the output is on a continuous scale, one usually selects the identity activation function for the output node.

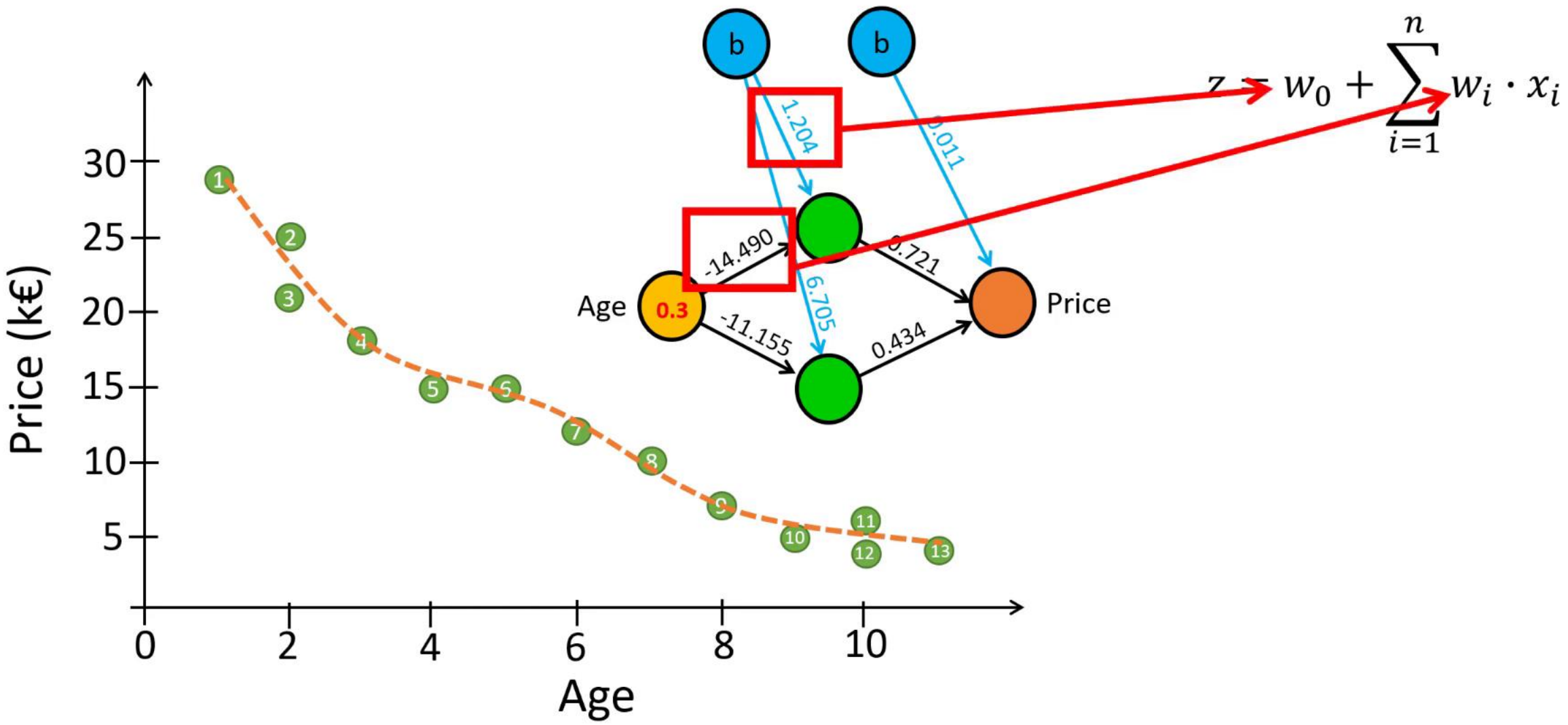
ANN



$$z = w_0 + \sum_{i=1}^{n} w_i \cdot x_i$$

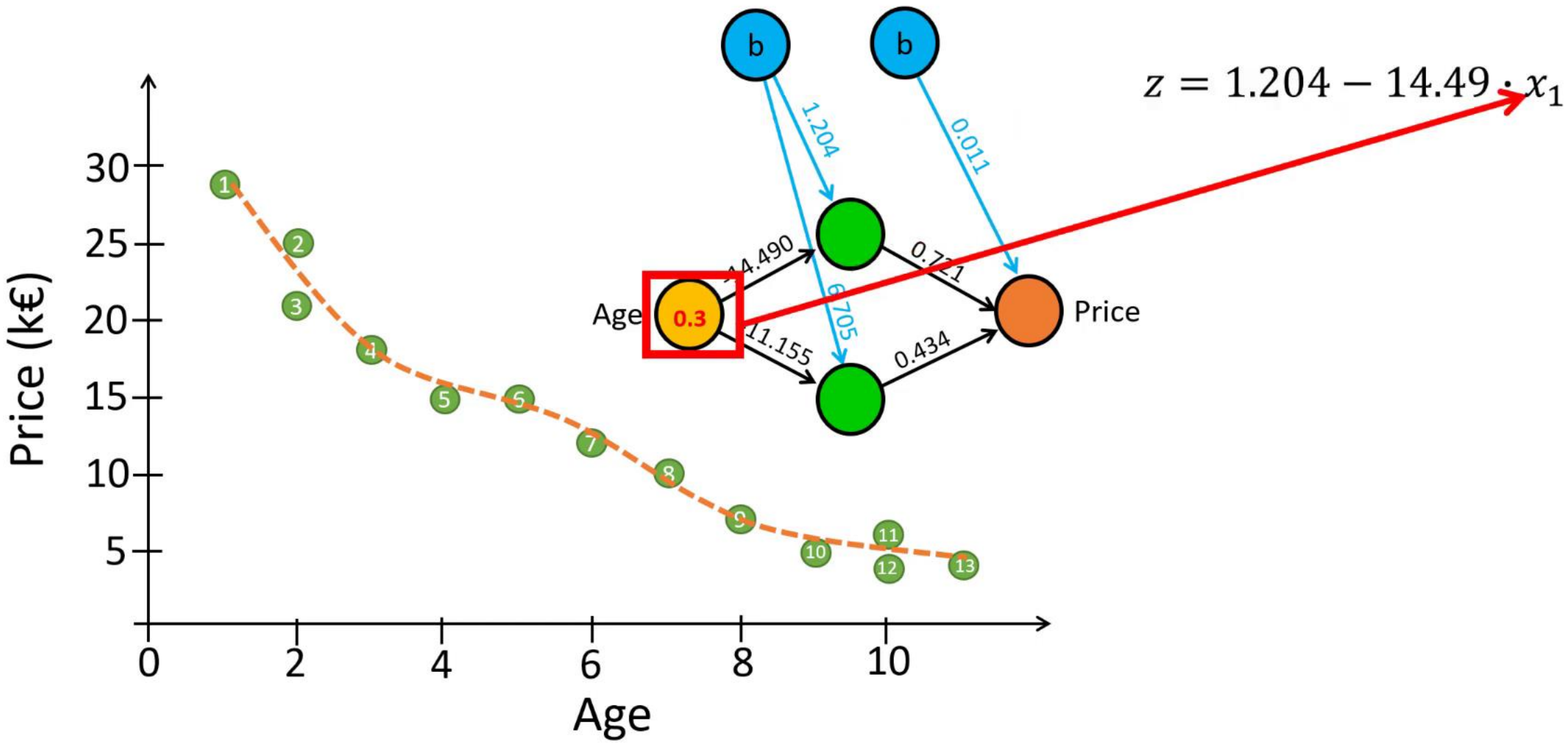
Let's calculate the value of this hidden node,

ANN



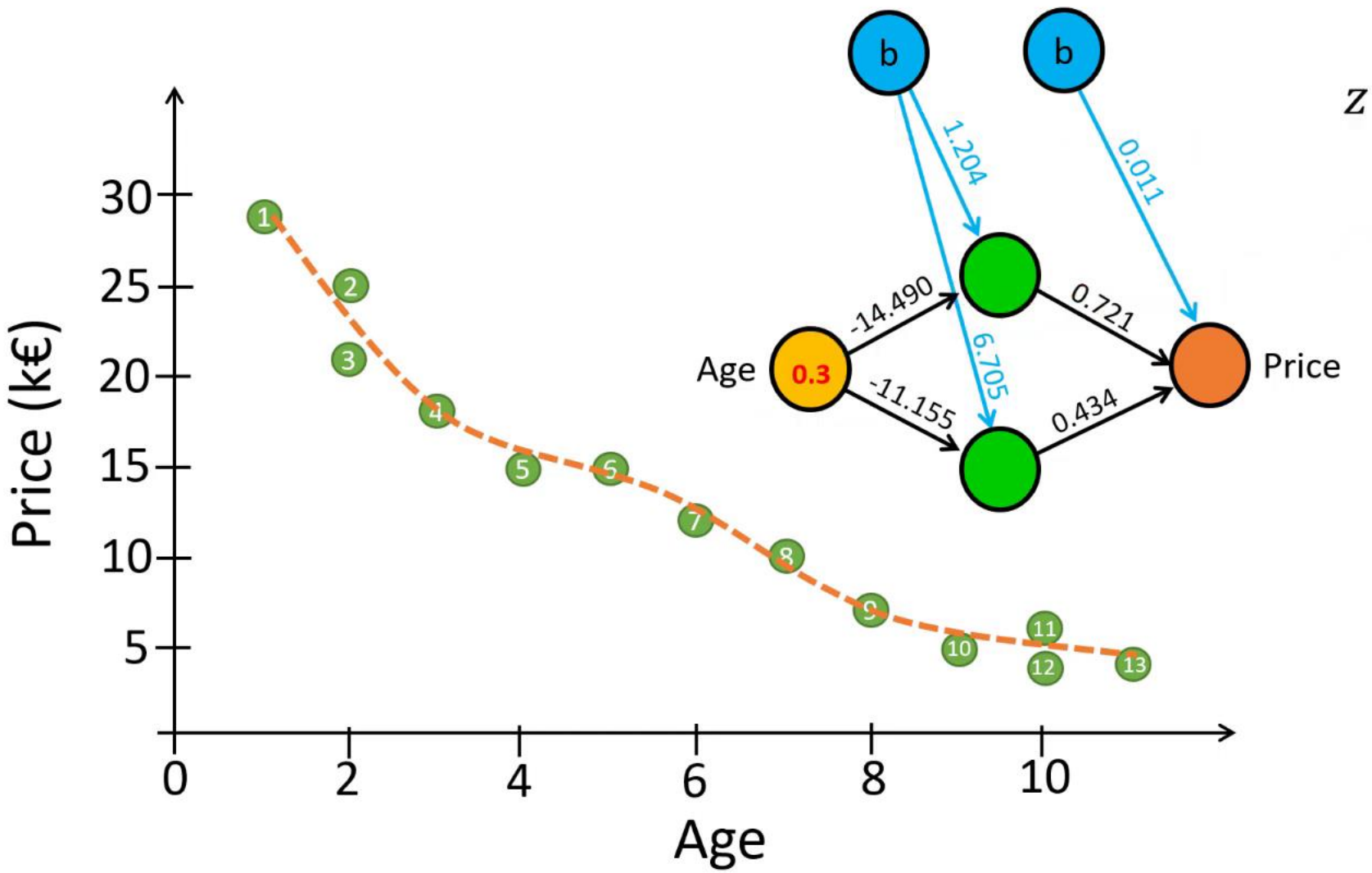
where we plug in the corresponding weights,

ANN



and the value of the input node,

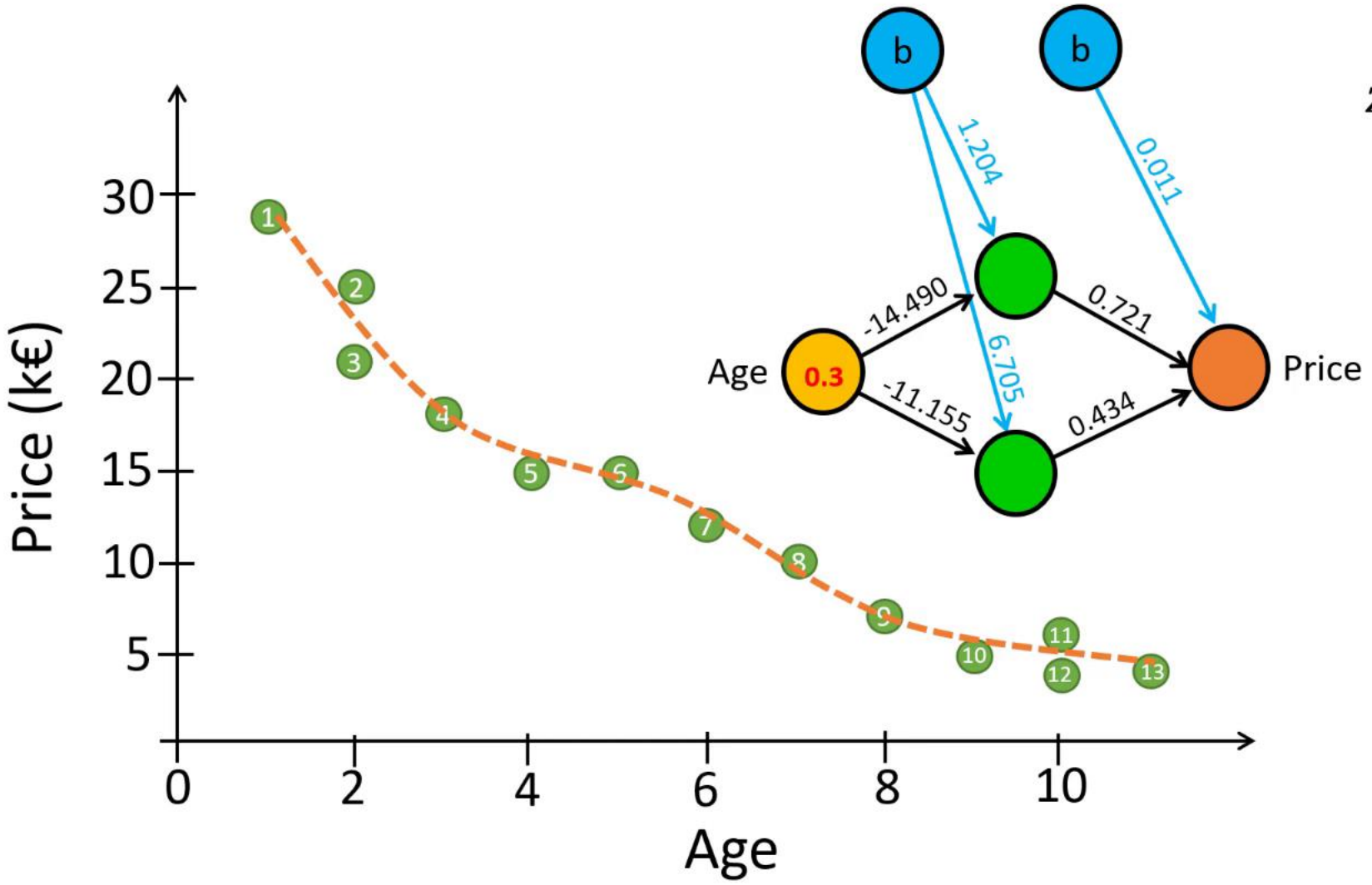
ANN



$$z = 1.204 - 14.49 \cdot 0.3 = \boxed{-3.14}$$

and do the math.

ANN

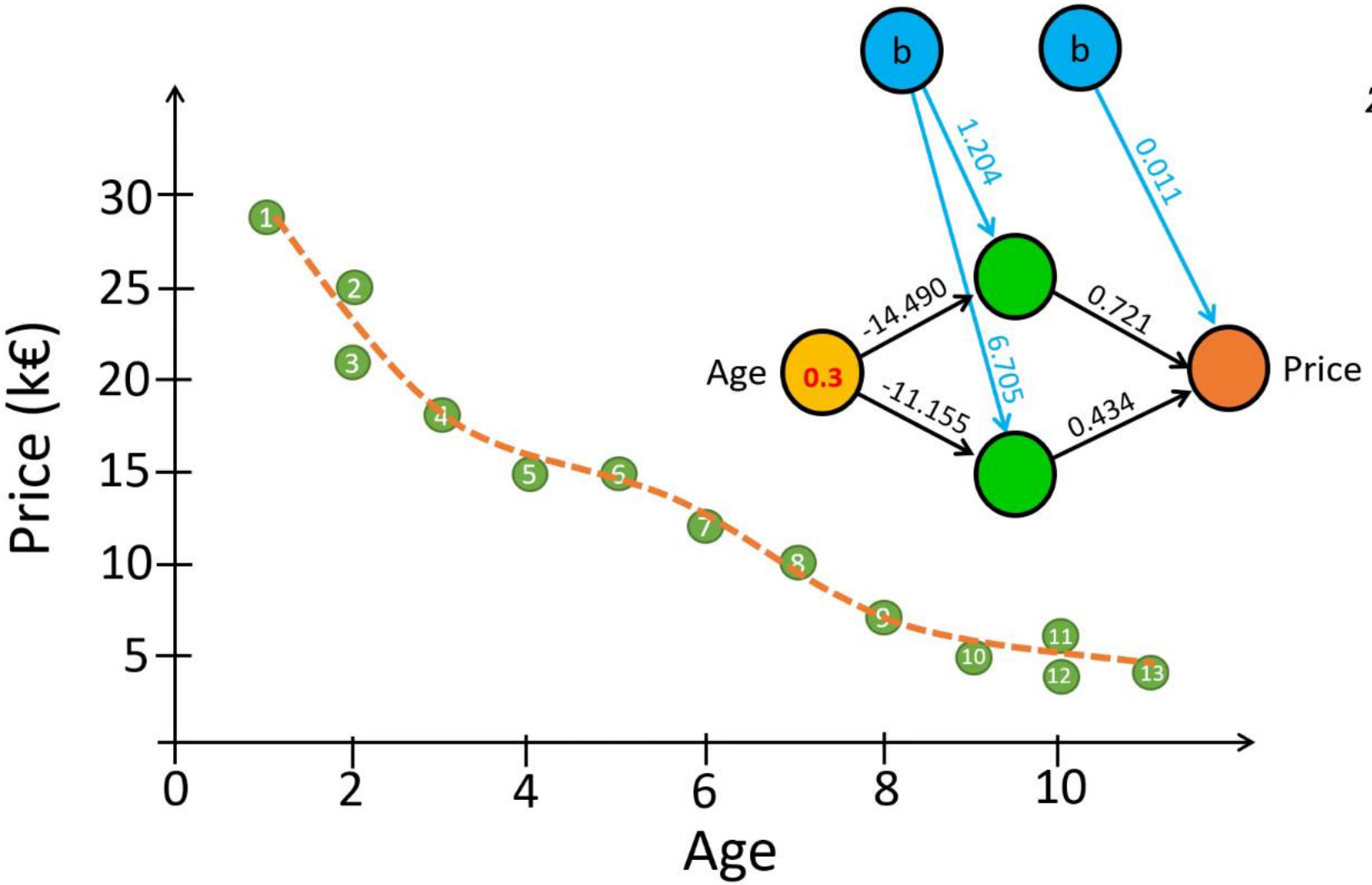


$$z = 1.204 - 14.49 \cdot 0.3 = -3.14$$

$$f(z) = \frac{1}{1 + e^{-(z)}}$$

We then plug in this value in the logistic activation function,

ANN

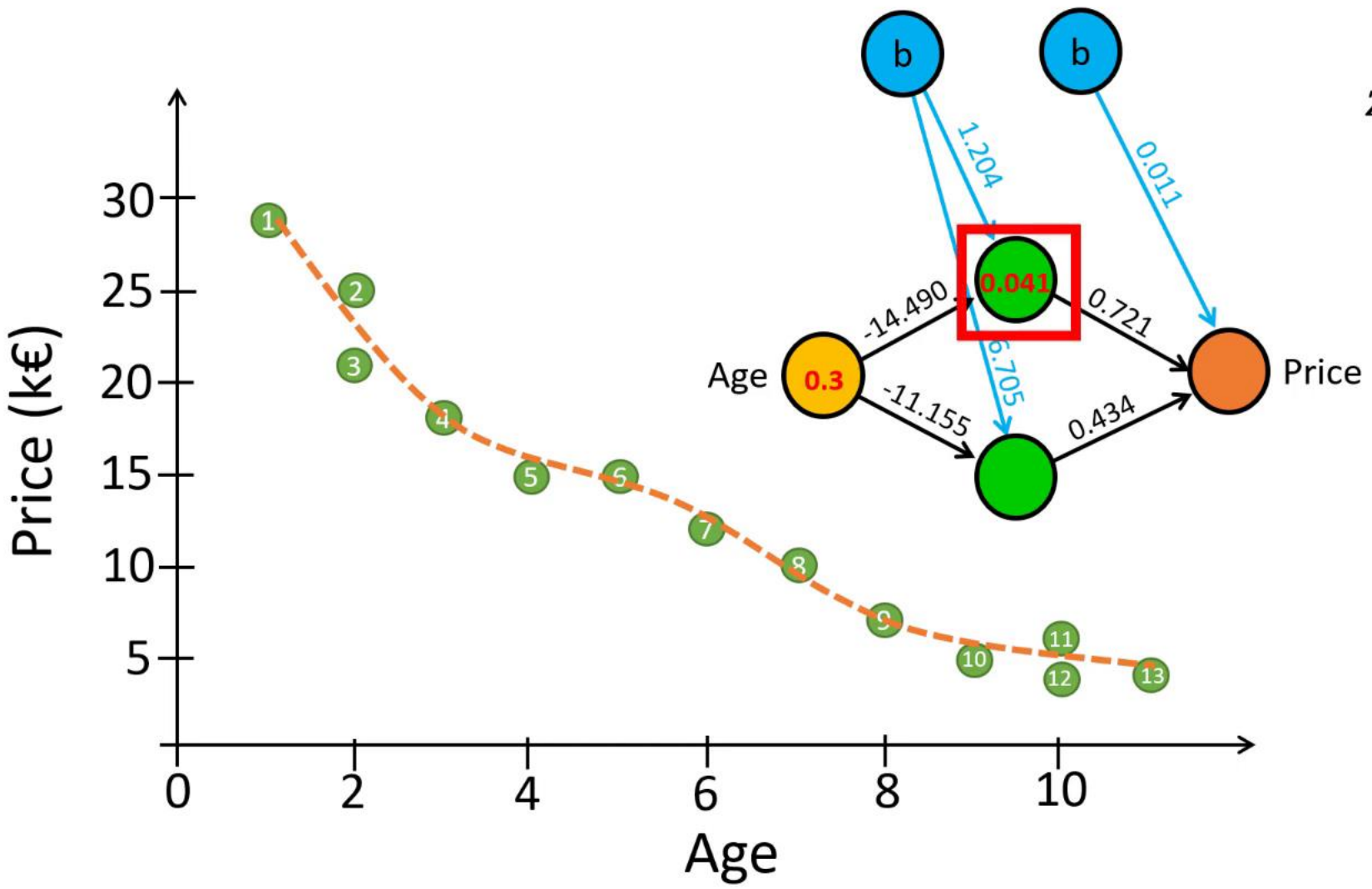


$$z = 1.204 - 14.49 \cdot 0.3 = -3.14$$

$$f(z) = \frac{1}{1 + e^{-(z)}} = 0.041$$

and do the math again.

ANN

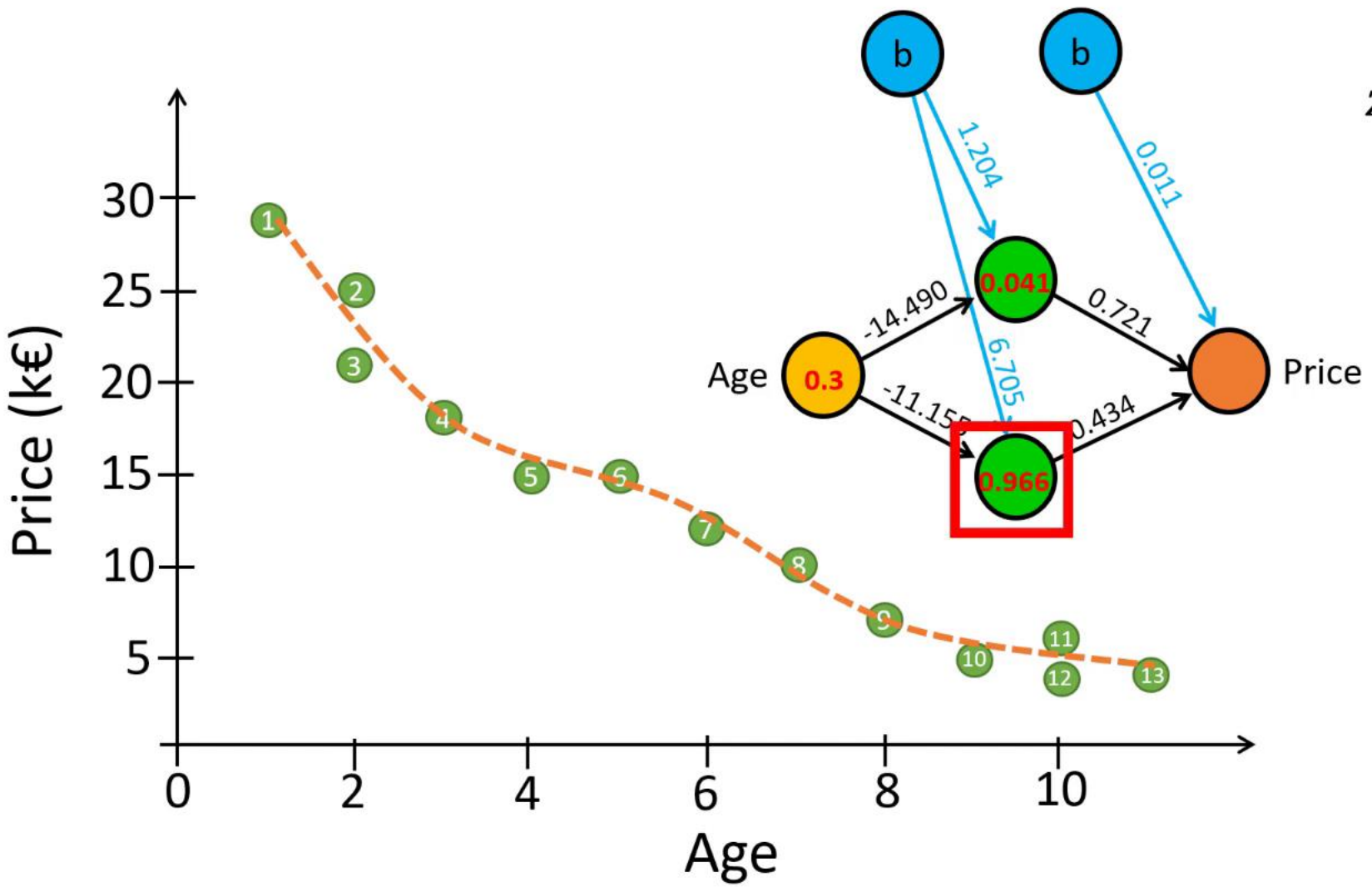


$$z = 1.204 - 14.49 \cdot 0.3 = -3.14$$

$$f(z) = \frac{1}{1 + e^{-(z)}} = 0.041$$

The value of this hidden node is therefore 0.041,

ANN

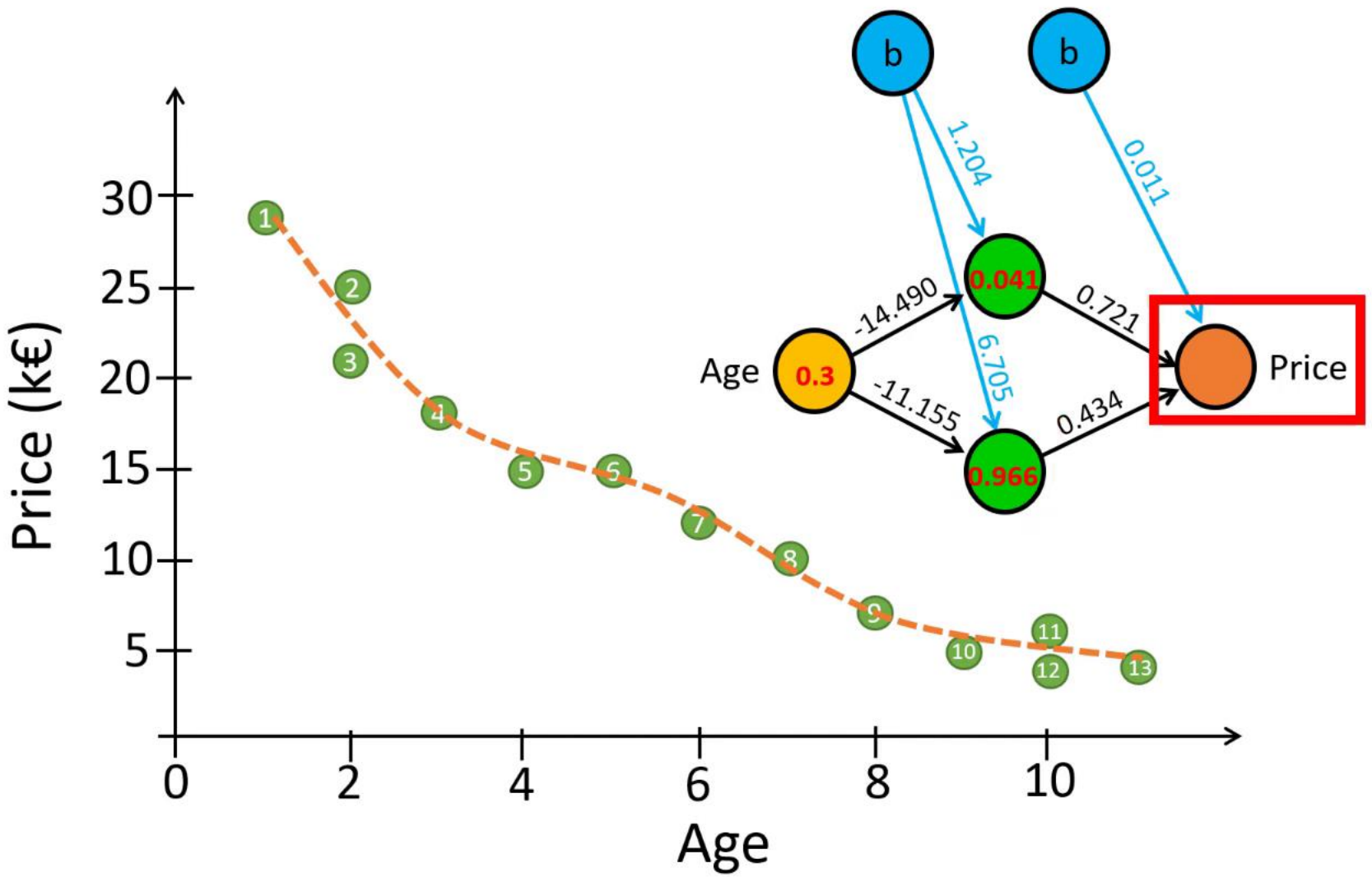


$$z = 6.705 - 11.15 \cdot 0.3 = 3.36$$

$$f(z) = \frac{1}{1 + e^{-(3.36)}} = 0.966$$

whereas the value of this hidden node is 0.966.

ANN

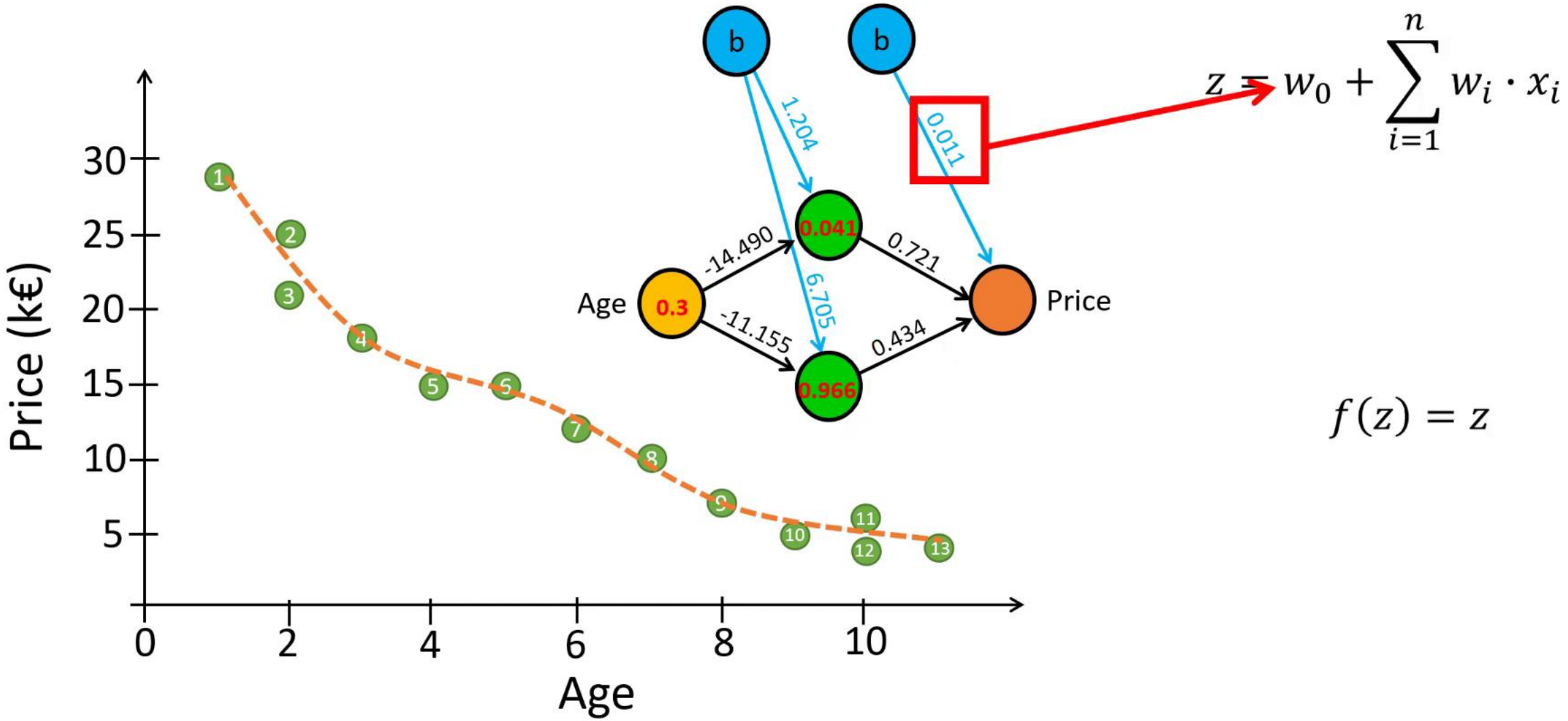


$$z = w_0 + \sum_{i=1}^{n} w_i \cdot x_i$$

$$f(z) = z$$

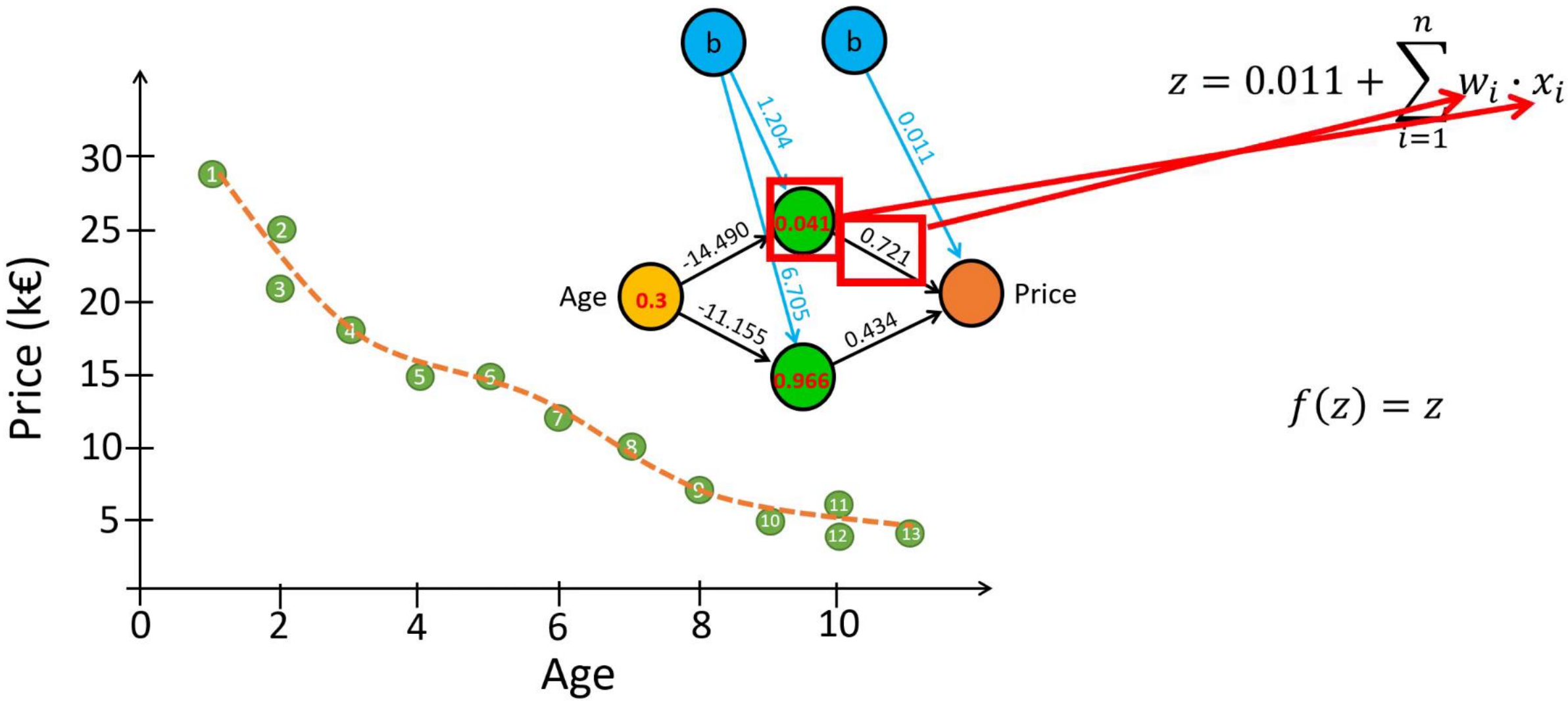
To calculate the value of the output node,

ANN



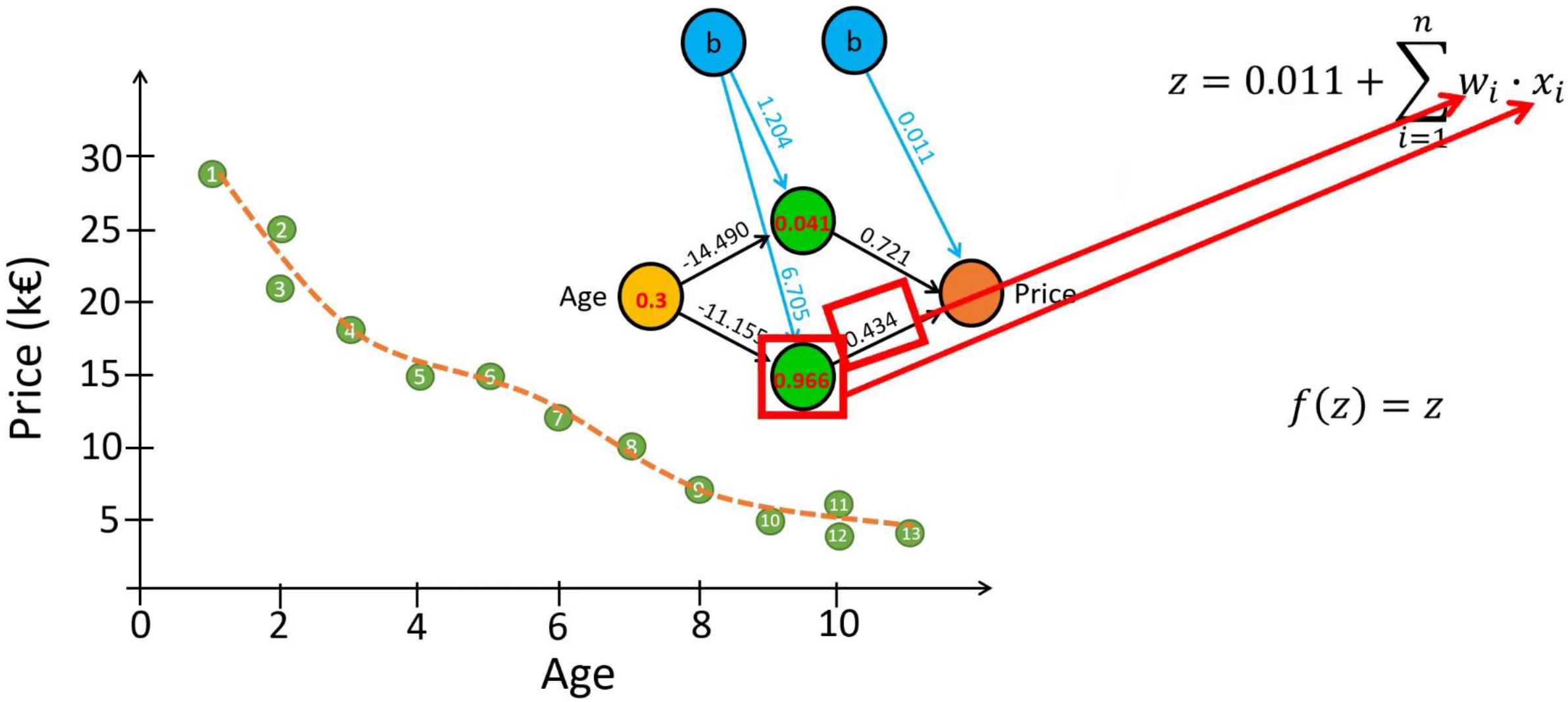
we plug in the value of the bias weight,

ANN



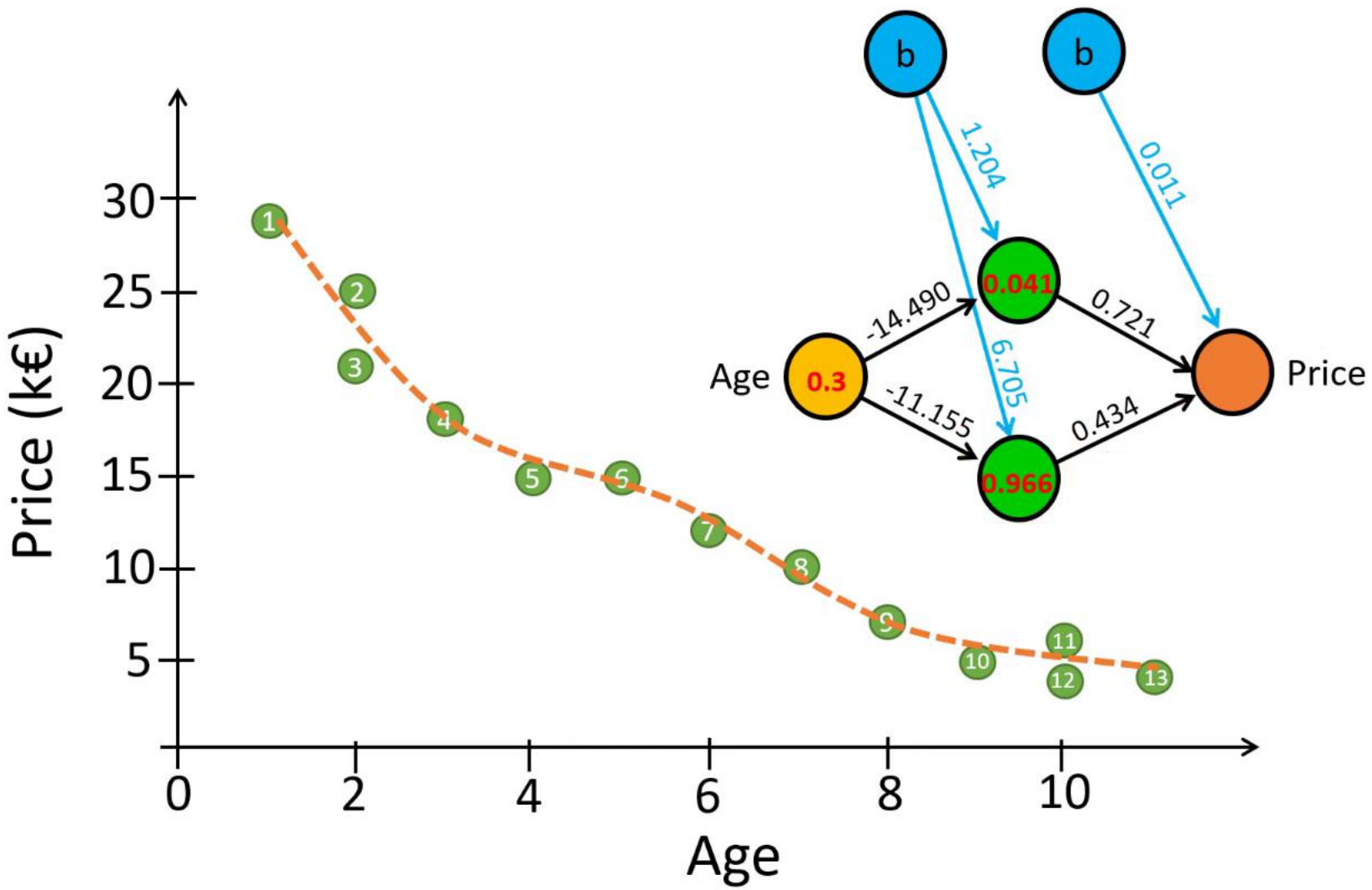
and the value of the first,

ANN



and second hidden node,

ANN

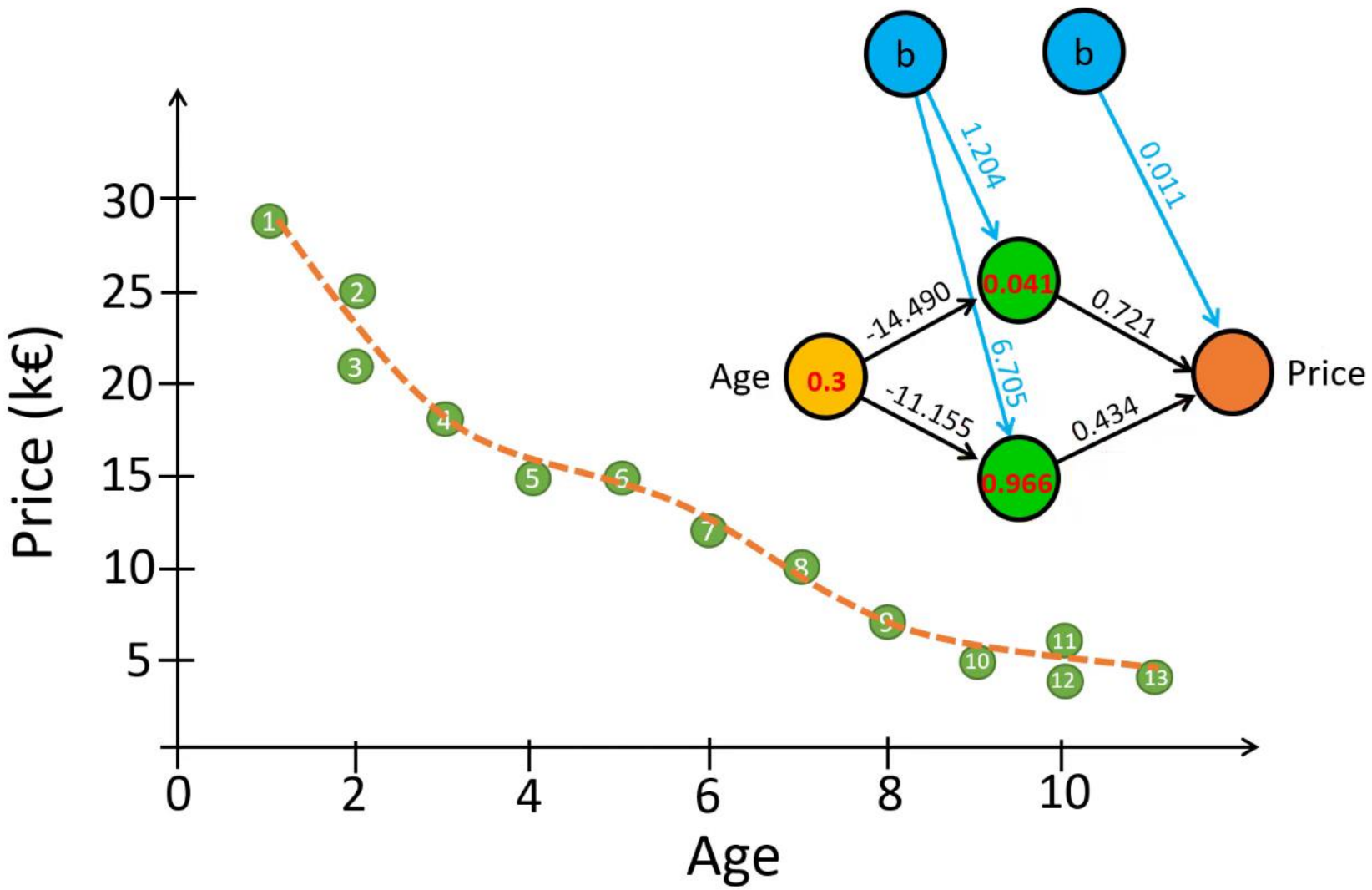


$$z = 0.011 + 0.721 \cdot 0.041 + 0.434 \cdot 0.966 = 0.46$$

$$f(z) = z$$

like this.

ANN

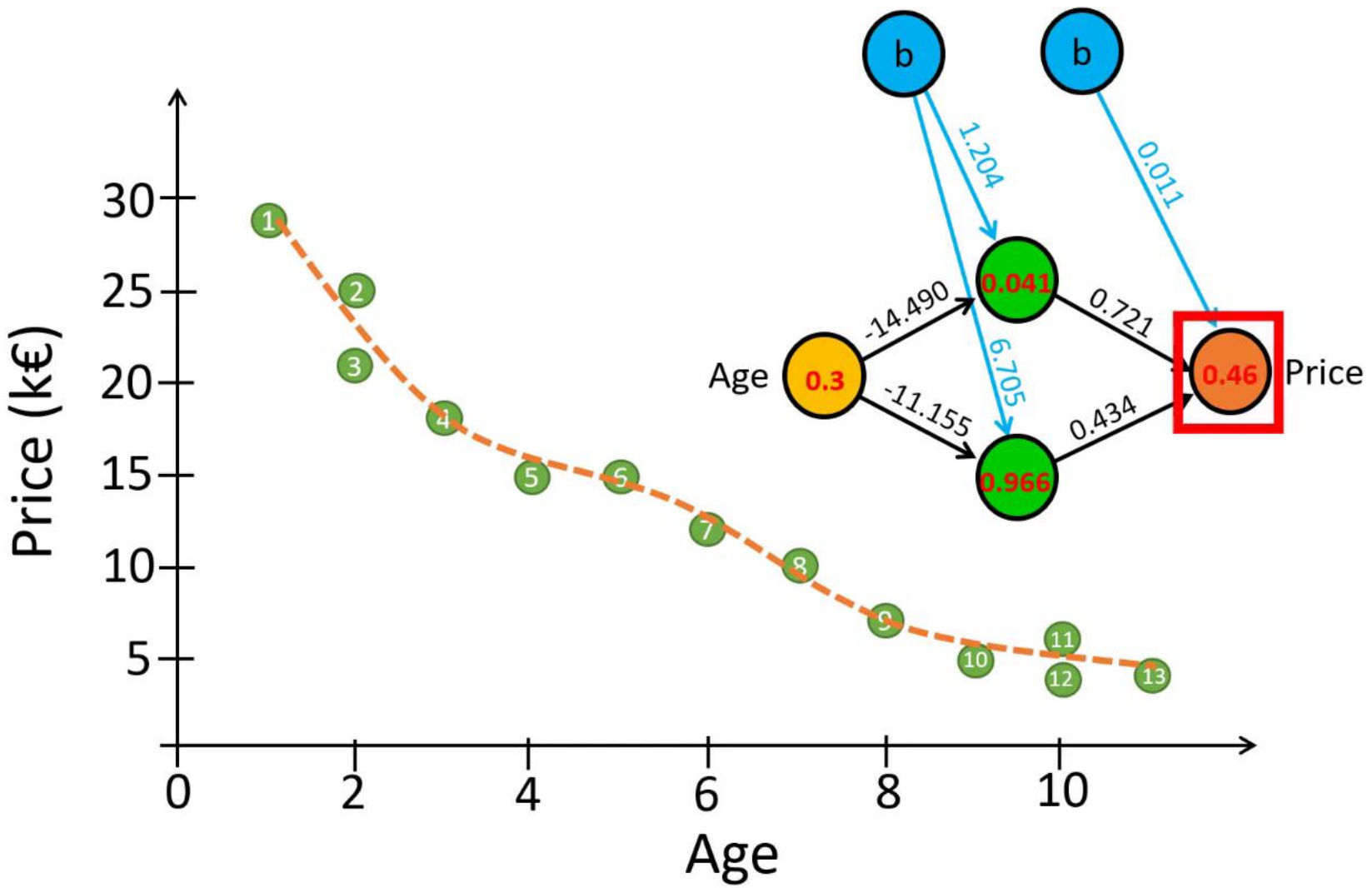


$$z = 0.011 + 0.721 \cdot 0.041 + 0.434 \cdot 0.966 = 0.46$$

$$f(z) = 0.46$$

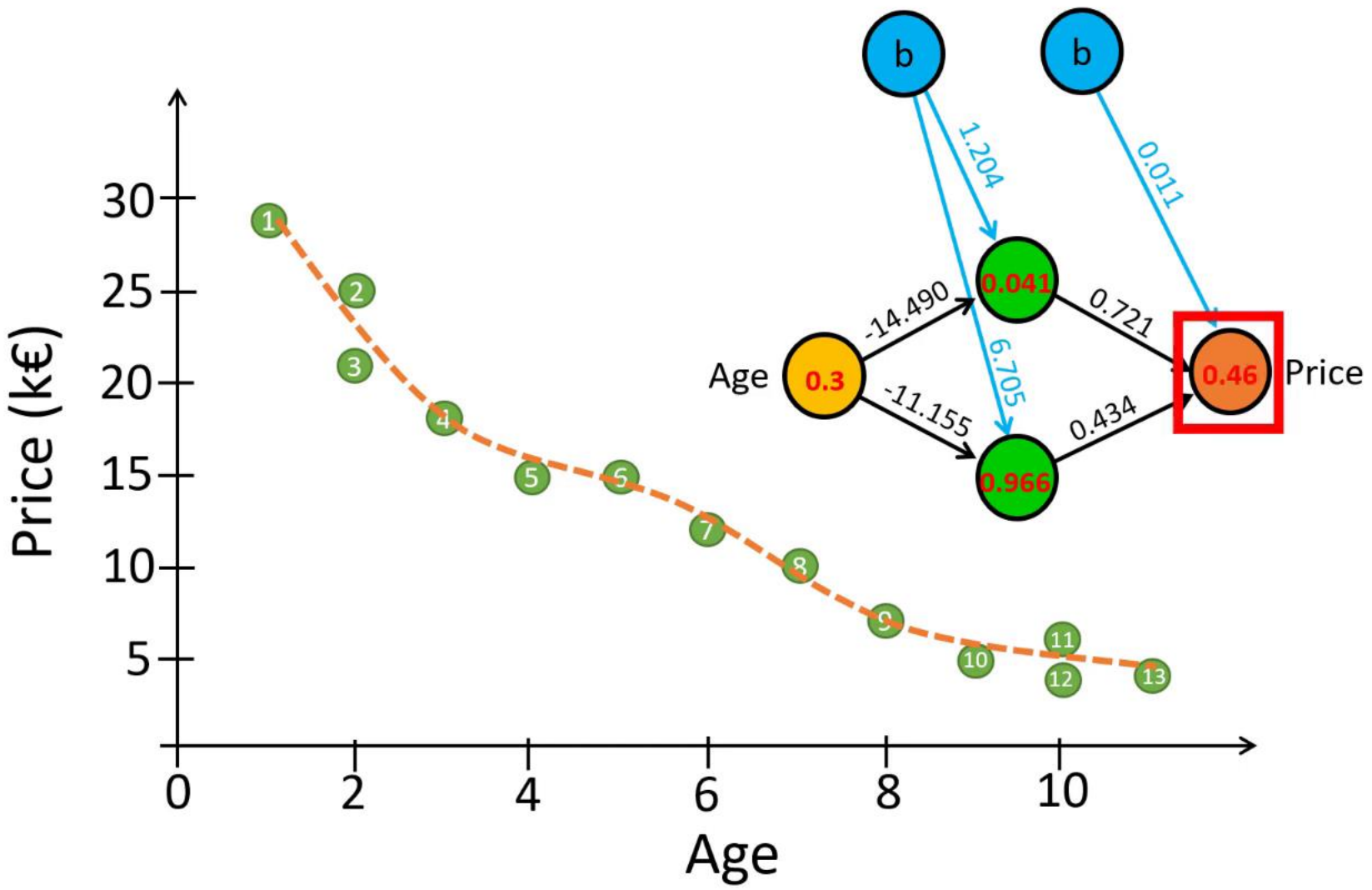
The value of the output node is therefore equal to 0.46.

ANN



Let's place this value in the output node.

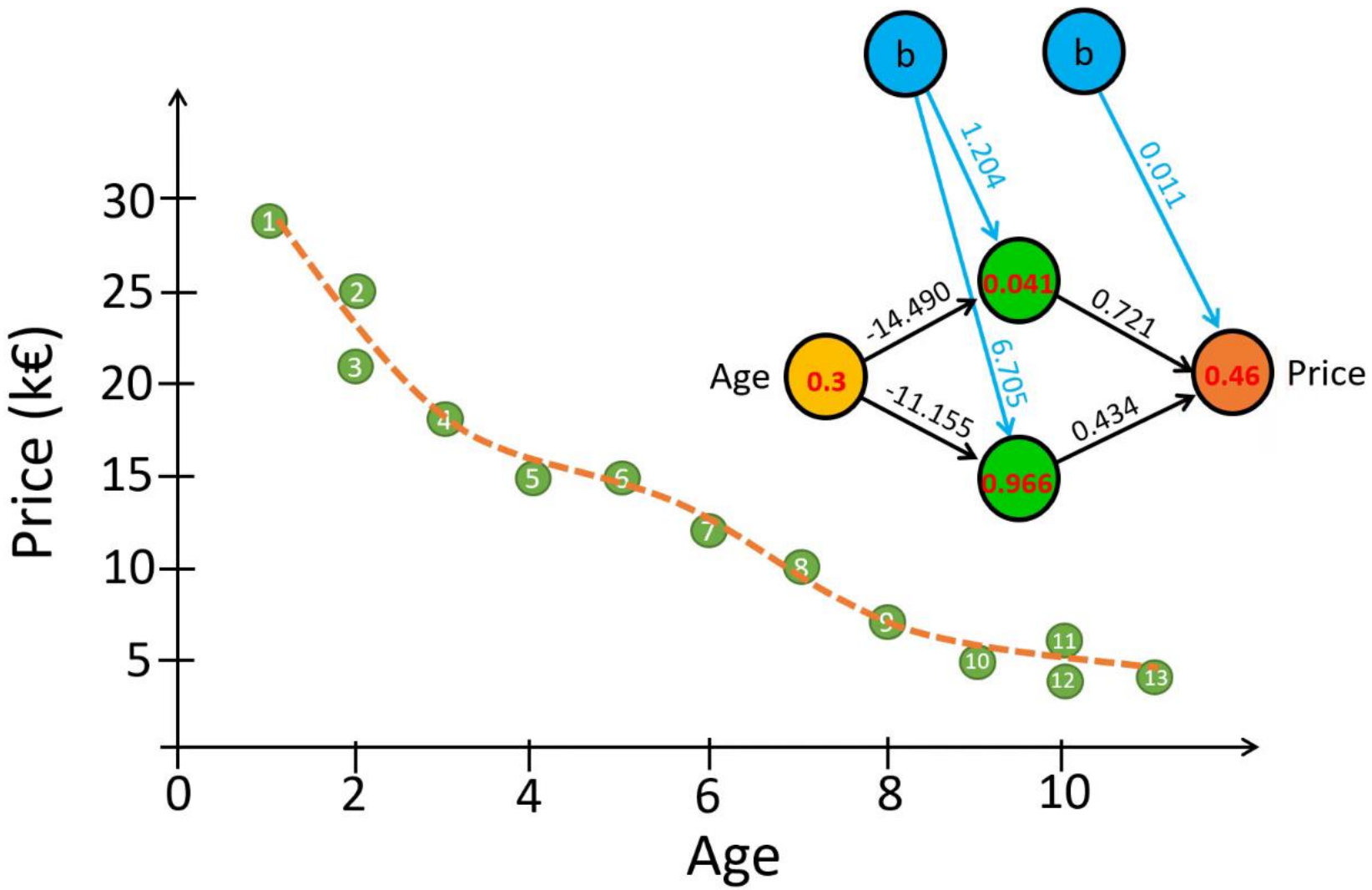
ANN



$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

To un-normalize this value,

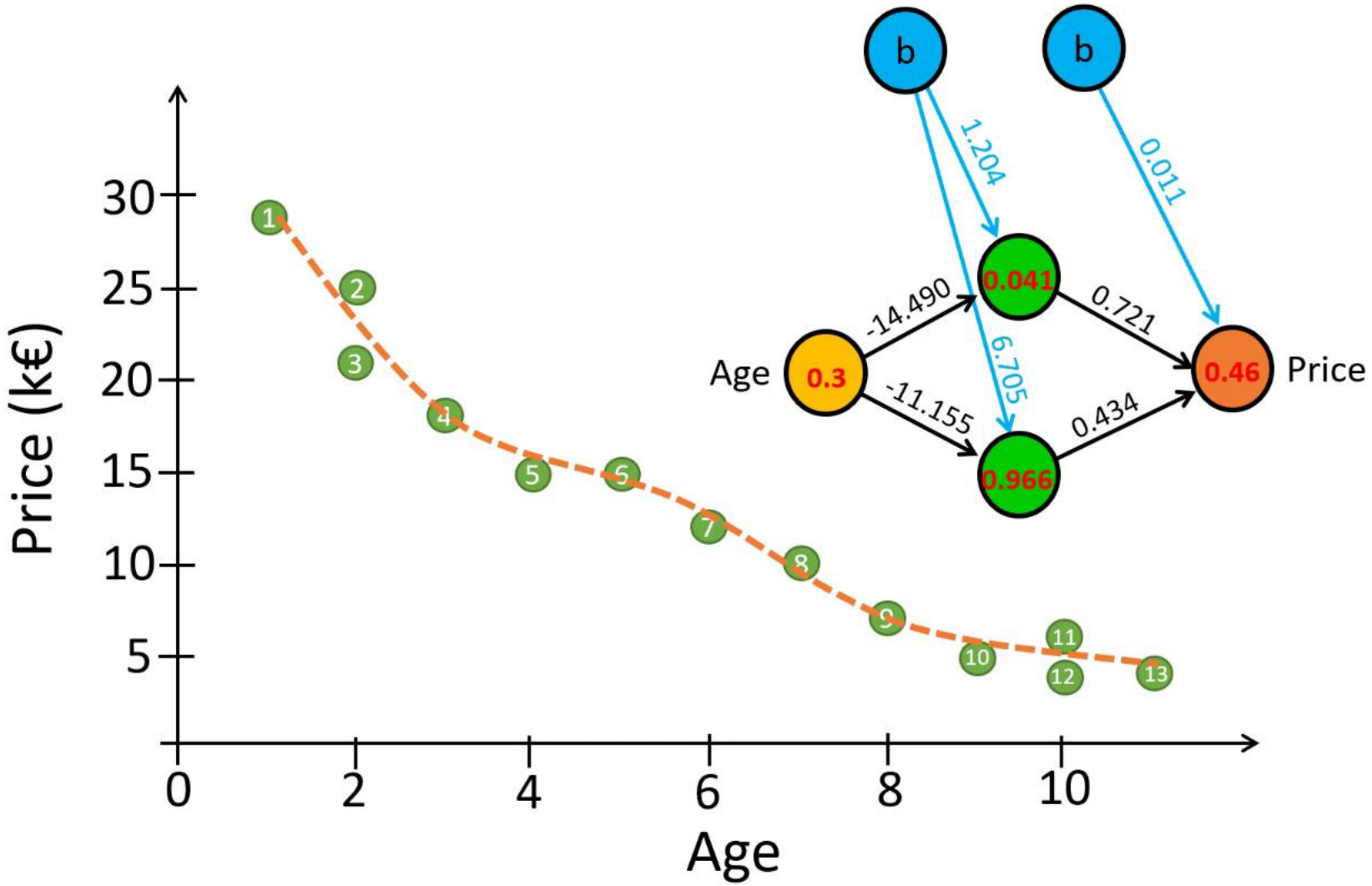
ANN



$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

we need to solve this equation for x,

ANN

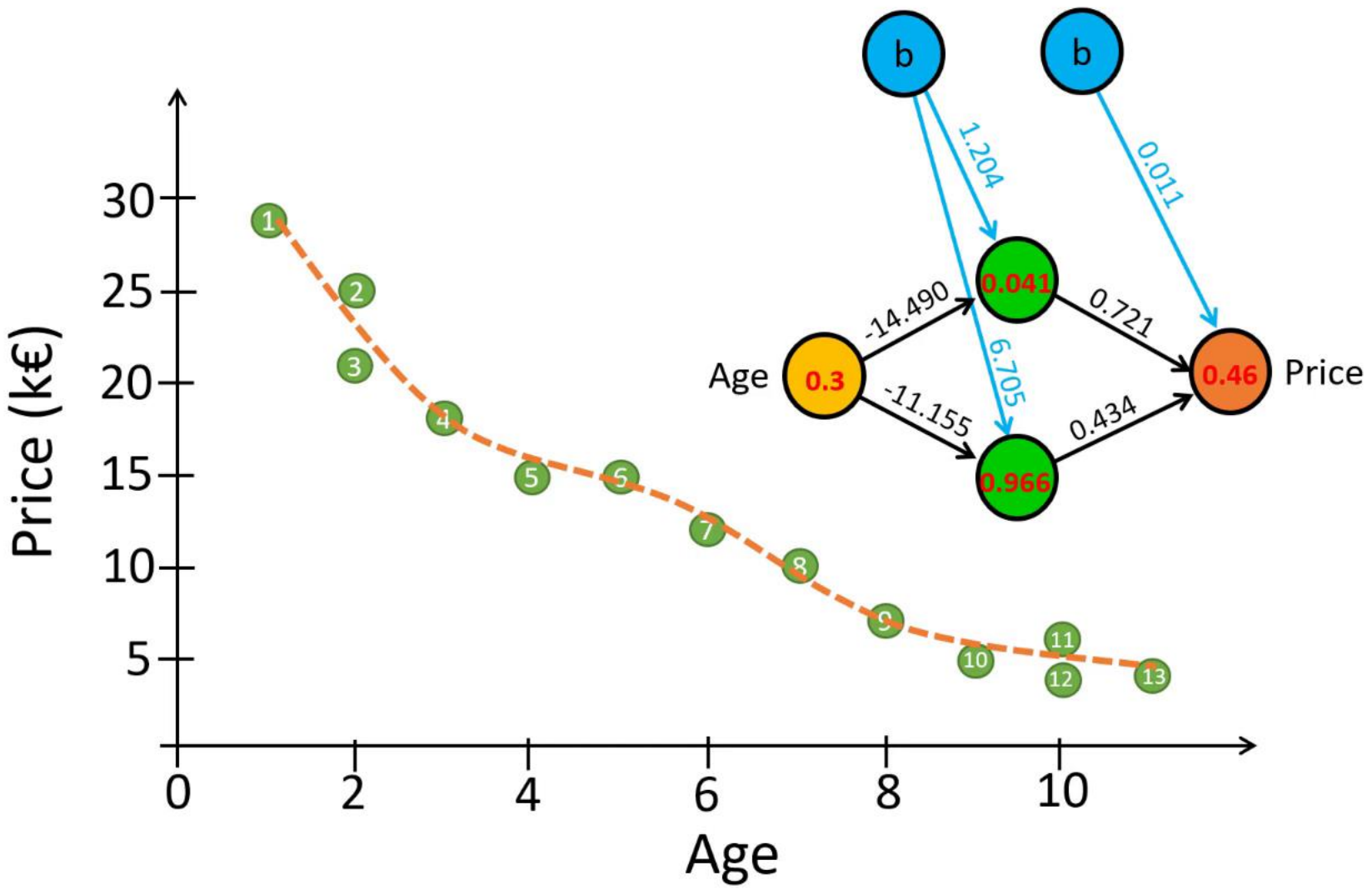


$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

$$x = x_{norm}(\max(x) - \min(x)) + \min(x)$$

like this.

ANN



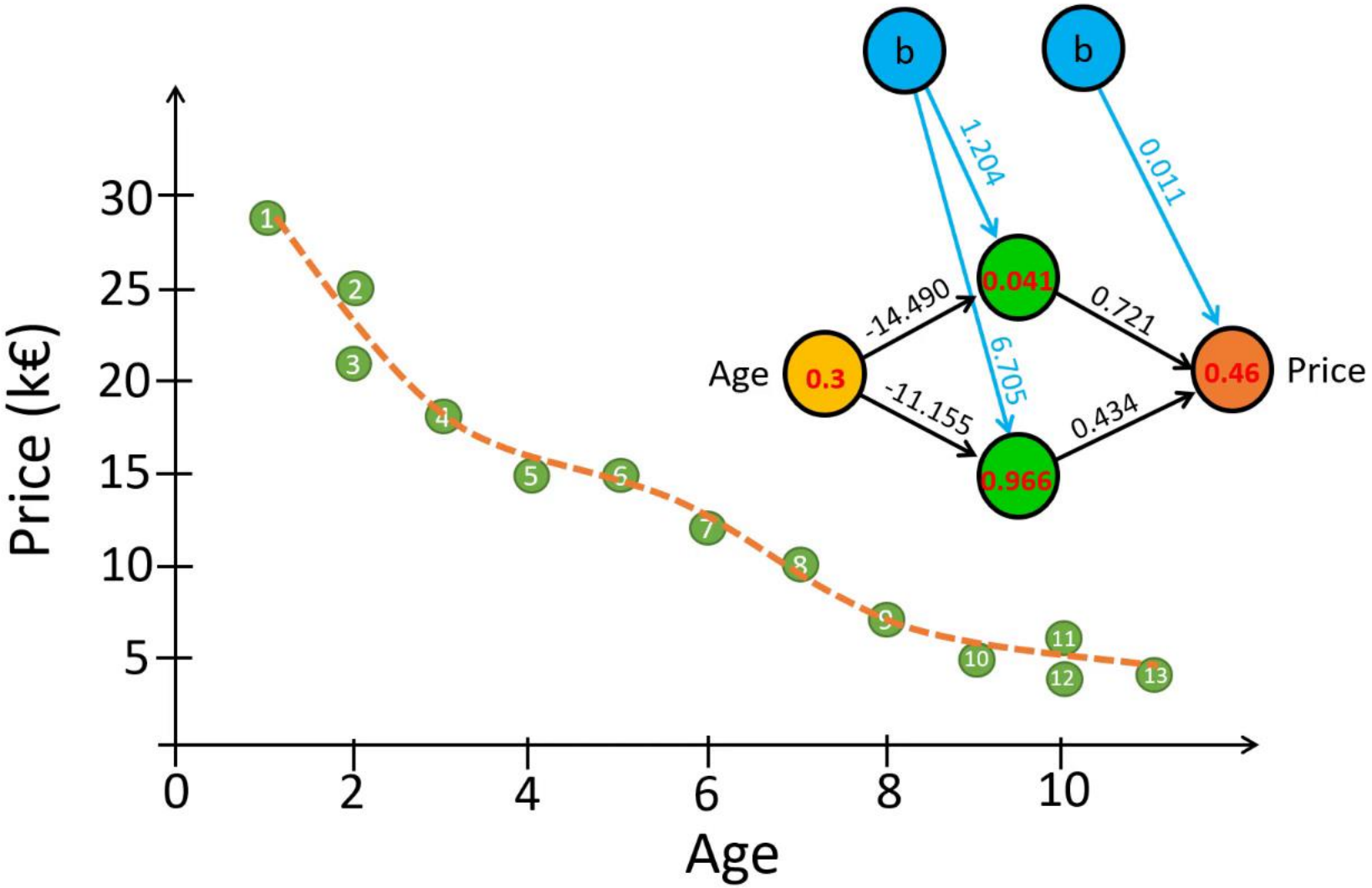
$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

$$x = x_{norm}(\max(x) - \min(x)) + \min(x)$$

$$x = x_{norm}(29 - 4) + 4$$

We then plug in the maximum,

ANN



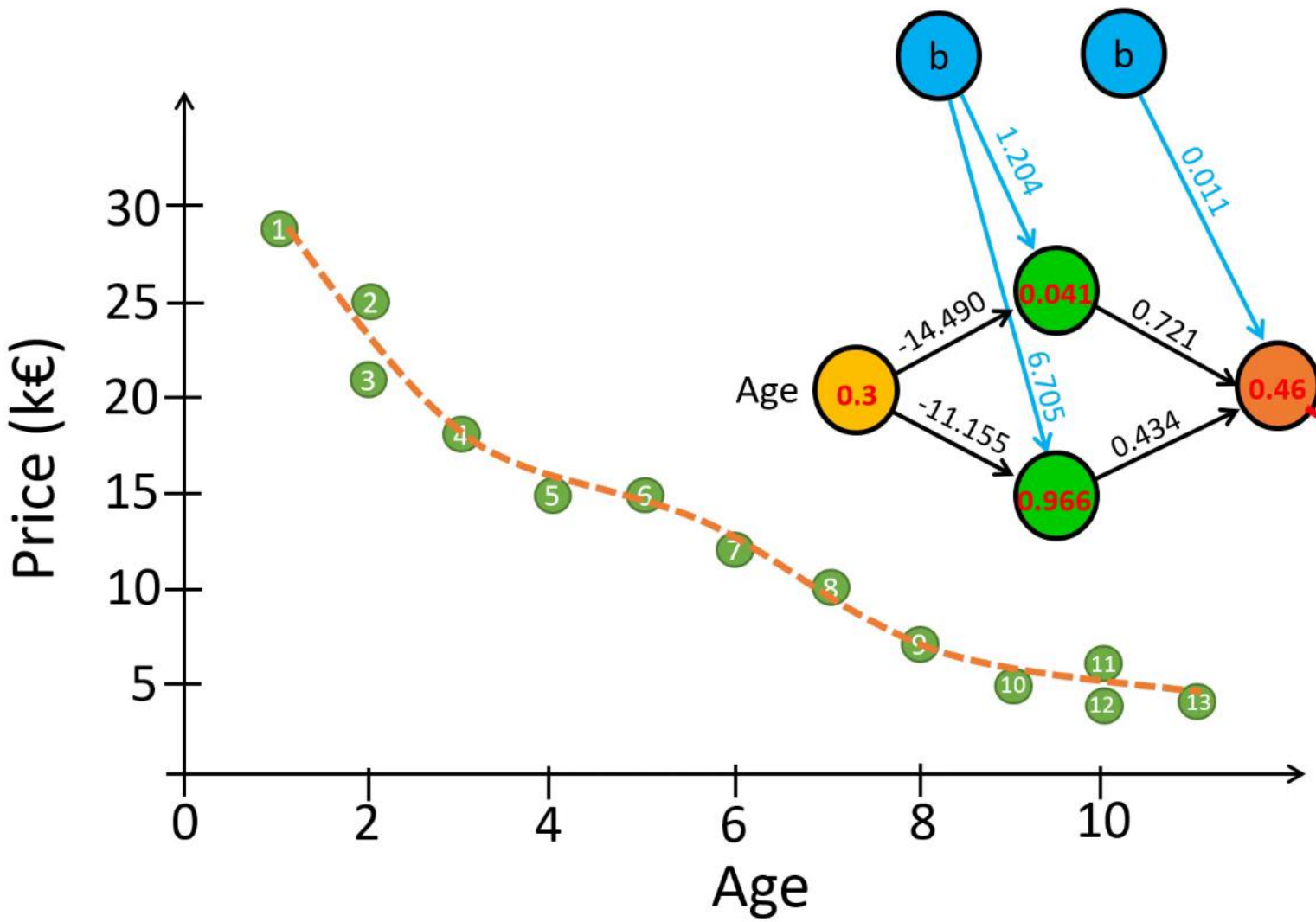
$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

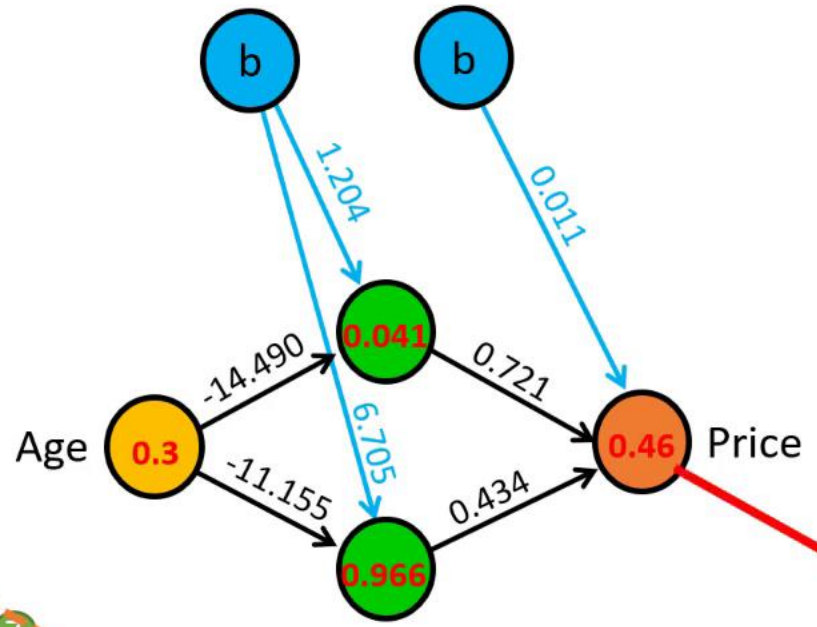
$$x = x_{norm}(\max(x) - \min(x)) + \min(x)$$

$$x = x_{norm}(29 - 4) + 4$$

and the minimum value from our training data,

ANN





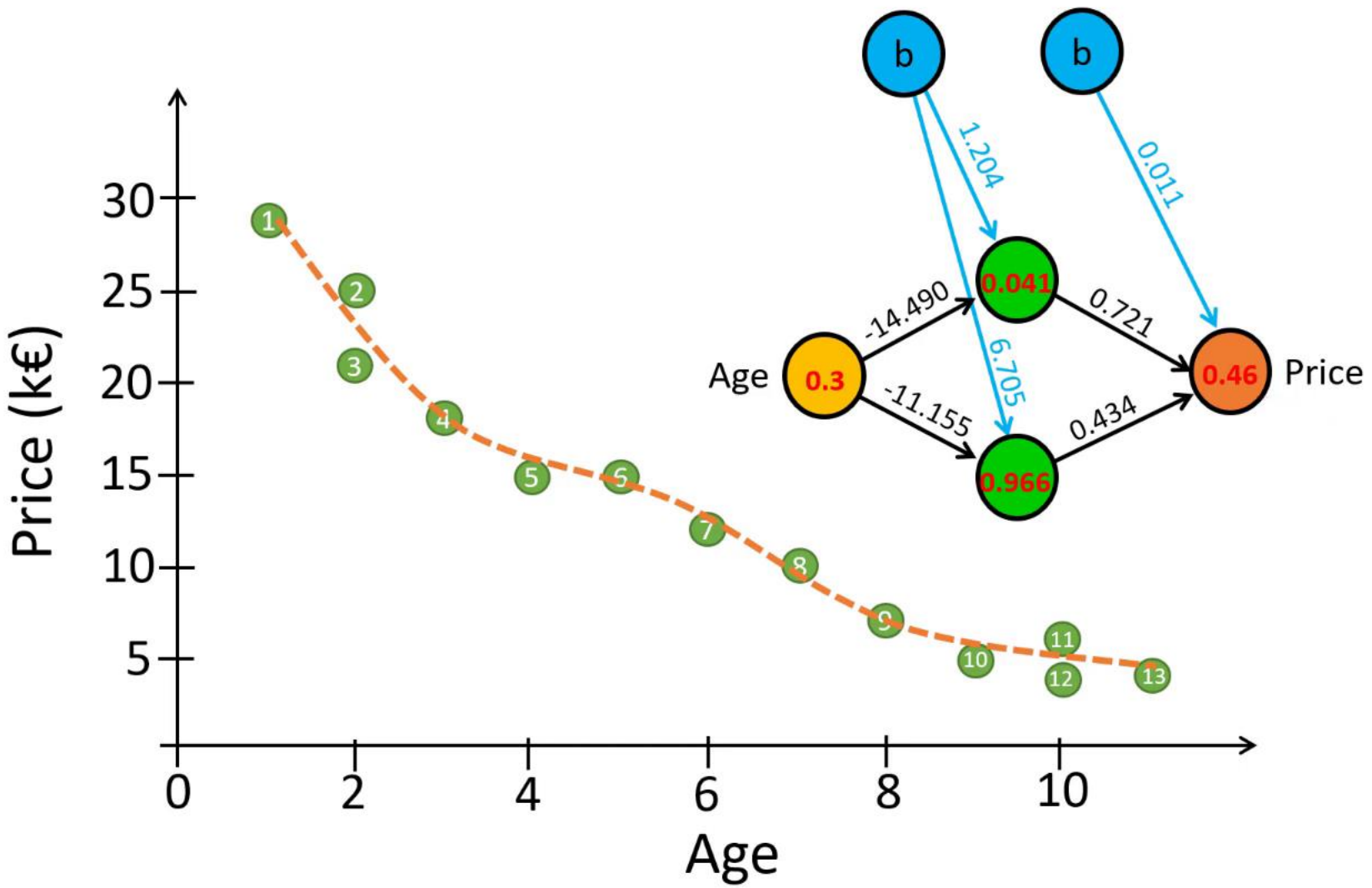
$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

$$x = x_{norm}(\max(x) - \min(x)) + \min(x)$$

$$x = 0.46 \cdot (29 - 4) + 4$$

and the output value from the network,

ANN



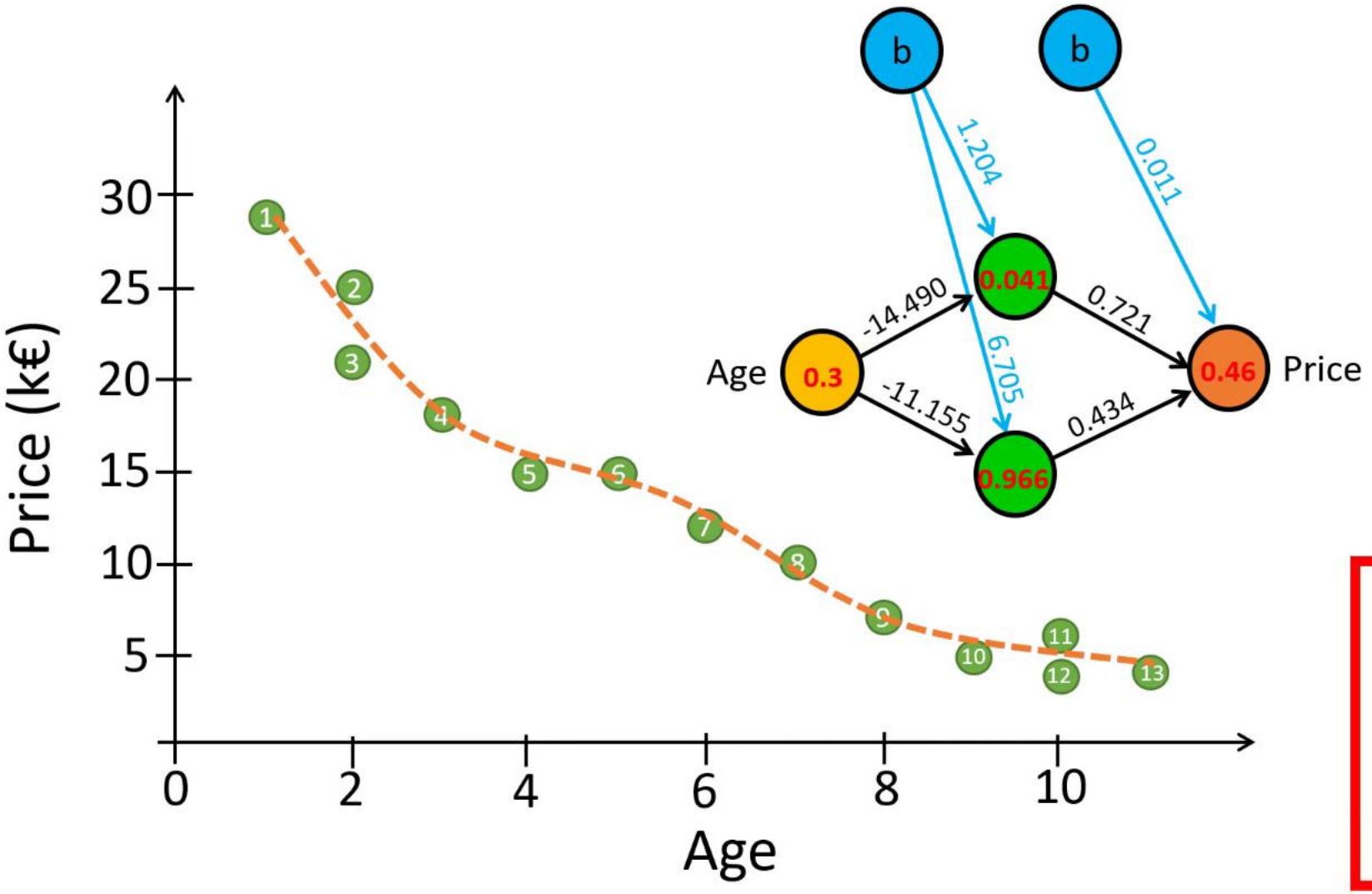
$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

$$x = x_{norm}(\max(x) - \min(x)) + \min(x)$$

$$x = 0.46 \cdot (29 - 4) + 4 = 15.5$$

and do the math.

ANN



$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

$$x = x_{norm}(\max(x) - \min(x)) + \min(x)$$

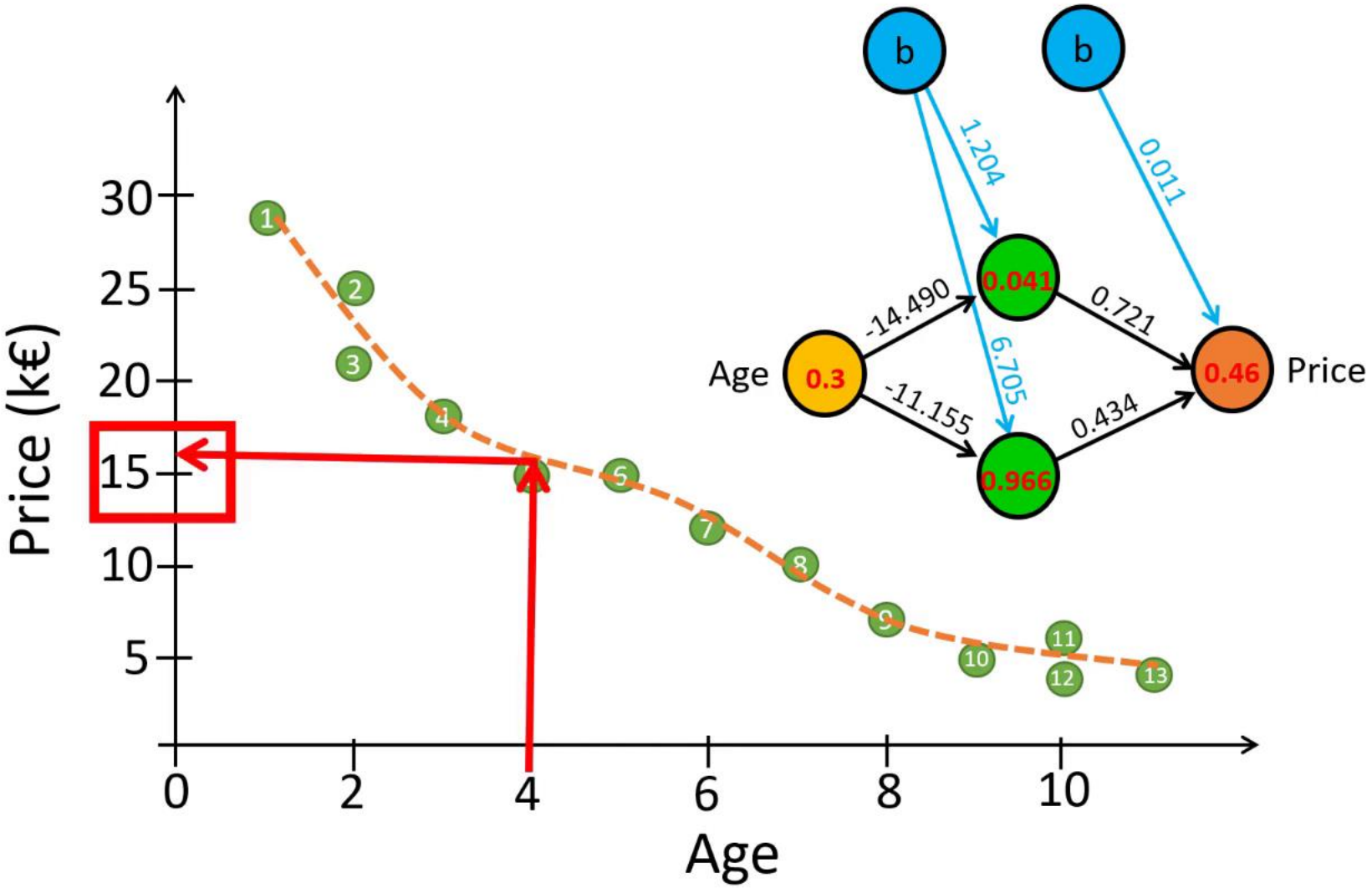
$$x = 0.46 \cdot (29 - 4) + 4 = 15.5$$



Age = 4

The estimated price of our four years old car is therefore about 15.500 euros,

ANN



$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

$$x = x_{norm}(\max(x) - \min(x)) + \min(x)$$

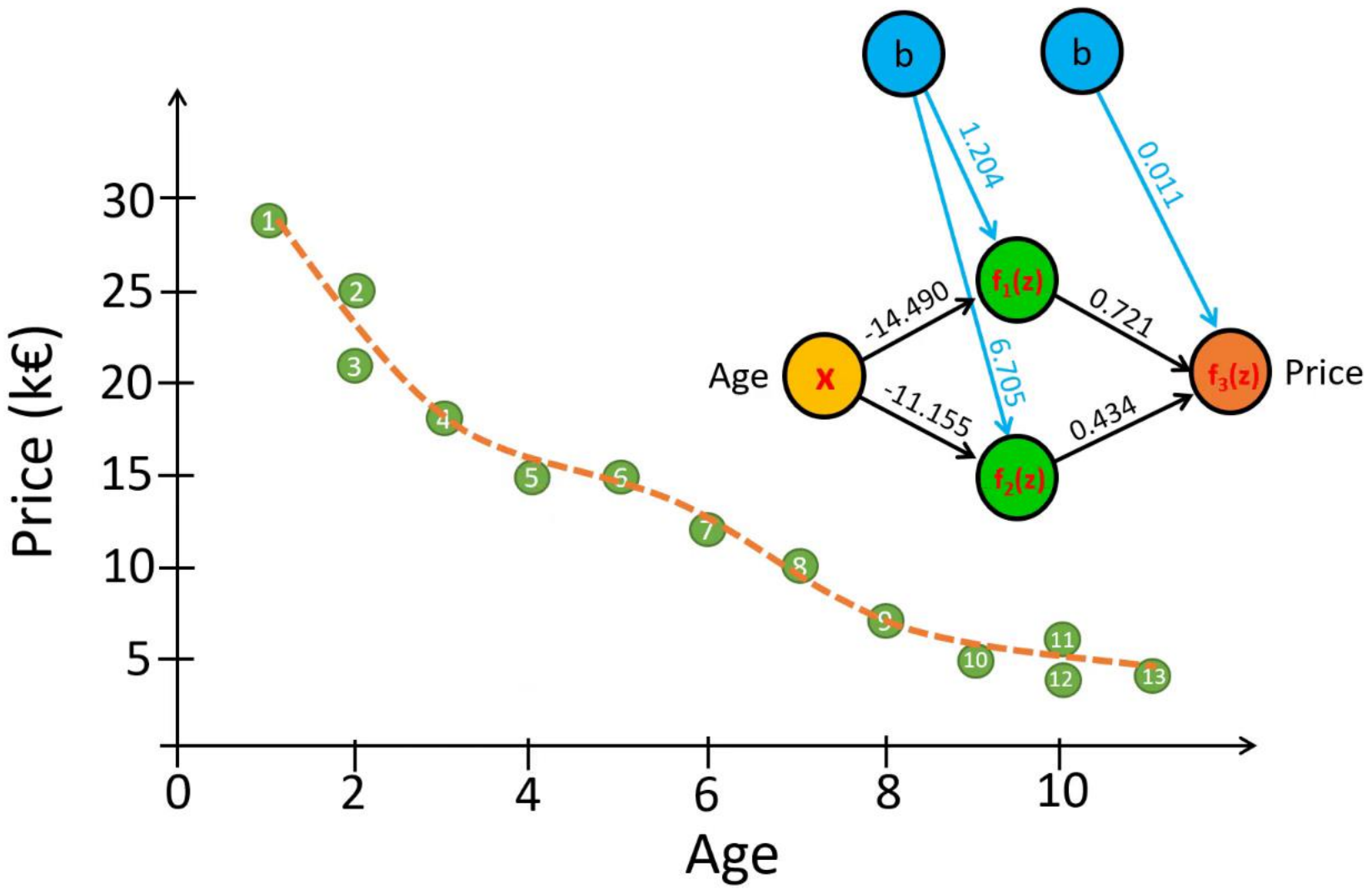
$$x = 0.46 \cdot (29 - 4) + 4 = 15.5$$



Age = 4

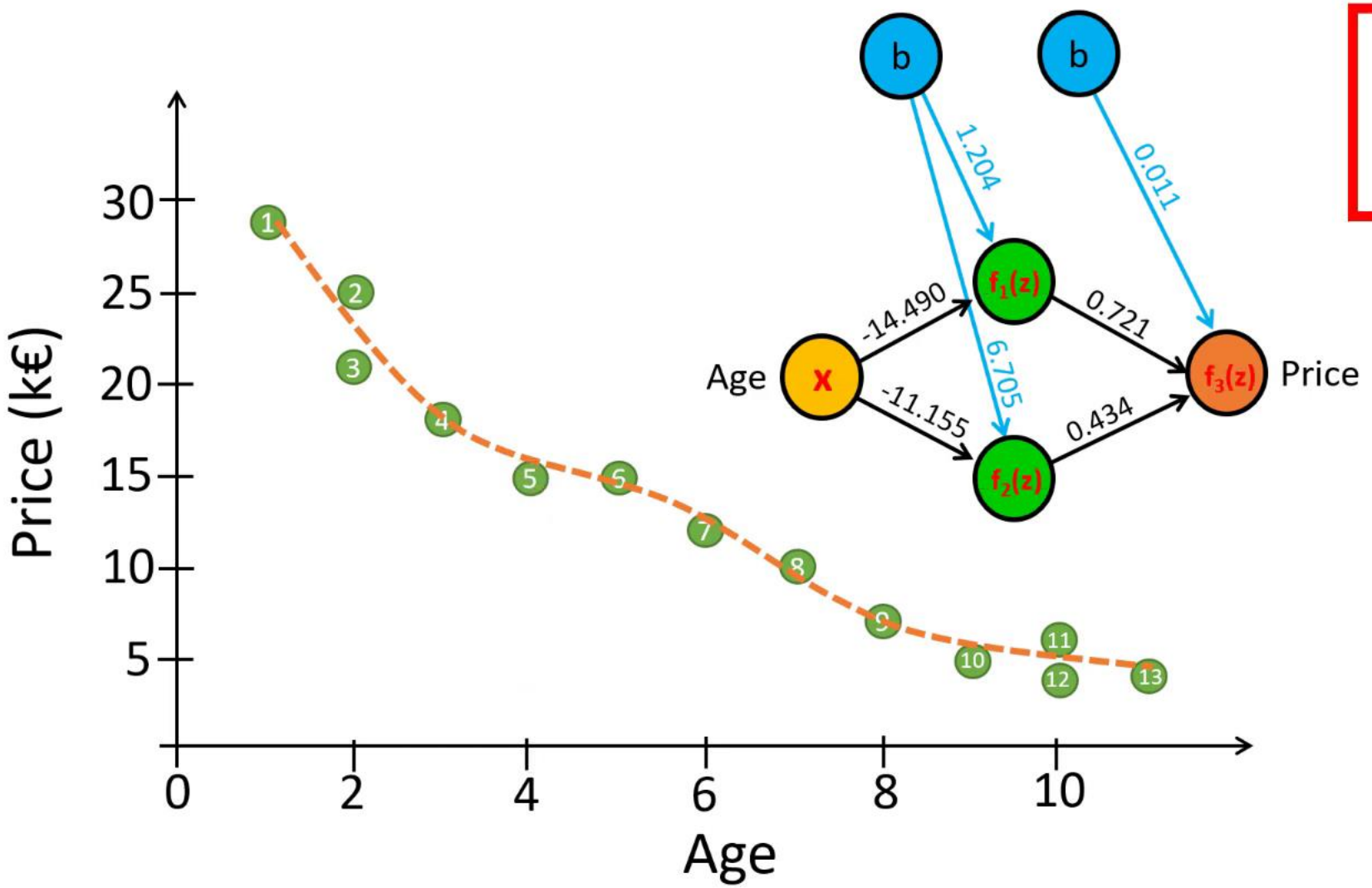
which corresponds to the height of the curve.

ANN



We will now see how we can generate the curve that is shown in this plot.

ANN

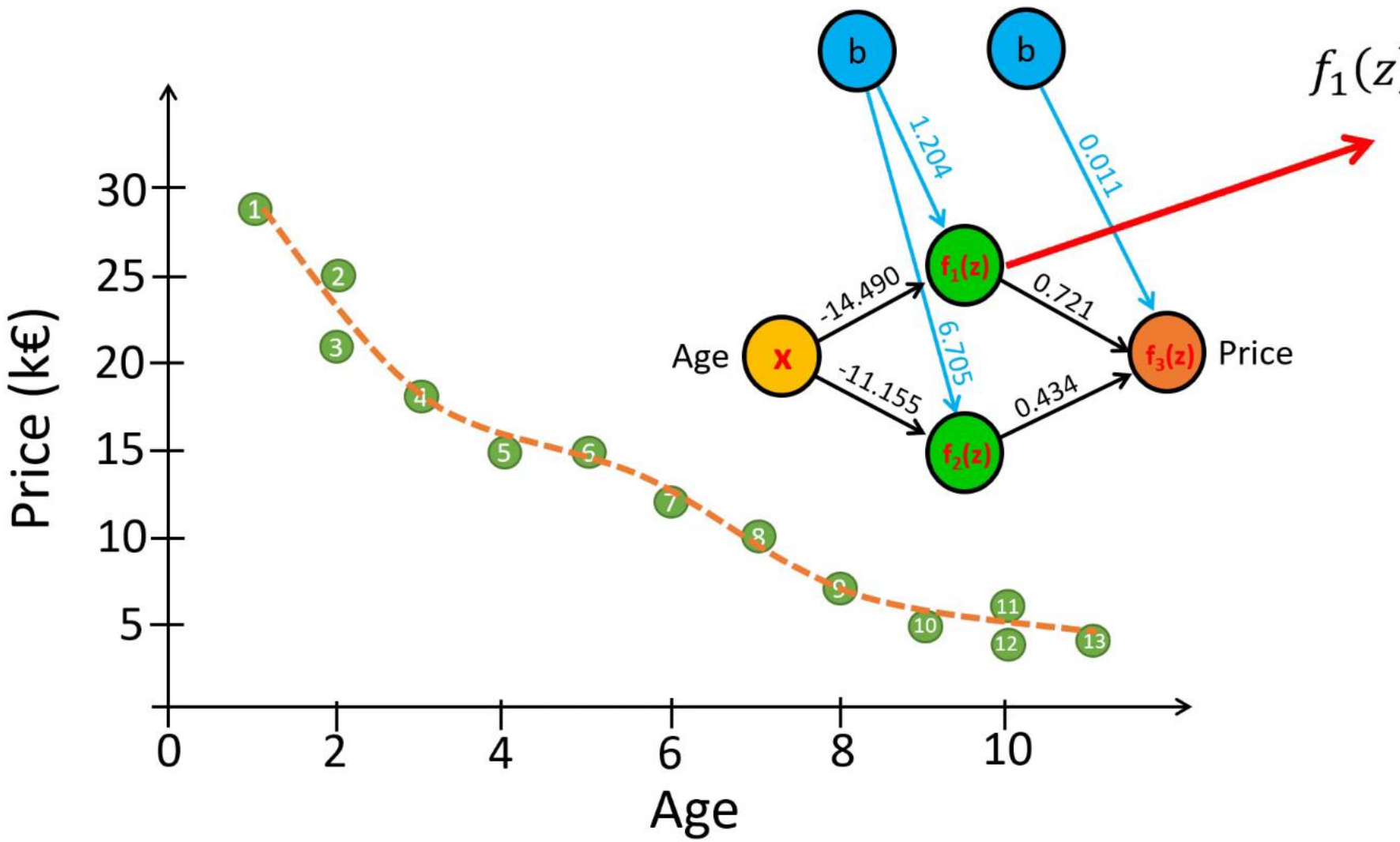


$$z_1 = w_0 + \sum_{i=1}^{n} w_i \cdot x_i$$

$$f(z) = \frac{1}{1 + e^{-(z)}}$$

If we plug in this equation into the logistic activation function,

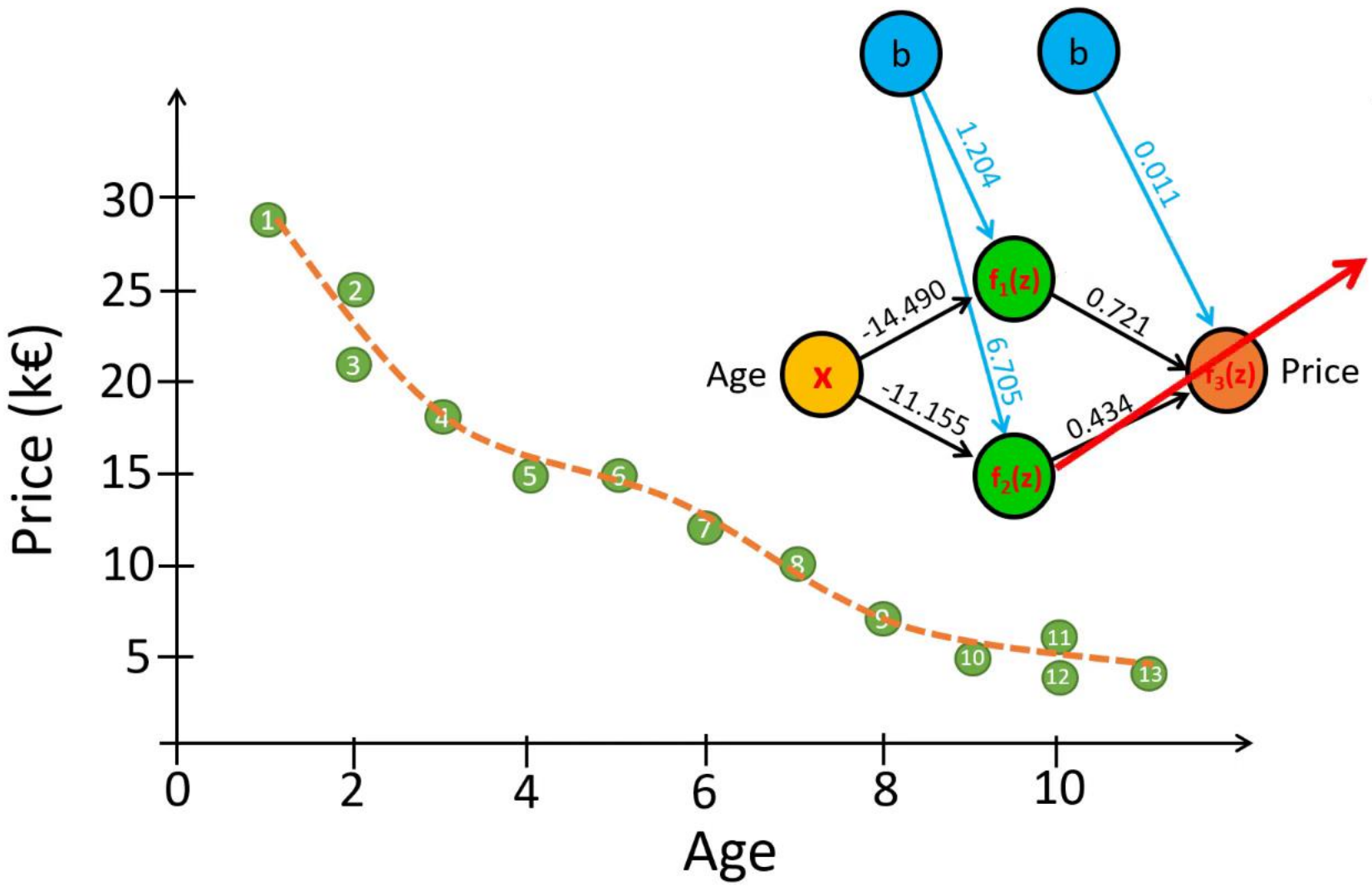
ANN



$$f_1(z) = \frac{1}{1 + e^{-(1.204 - 14.490x)}}$$

we can define the equation that computes the value of the first,

ANN

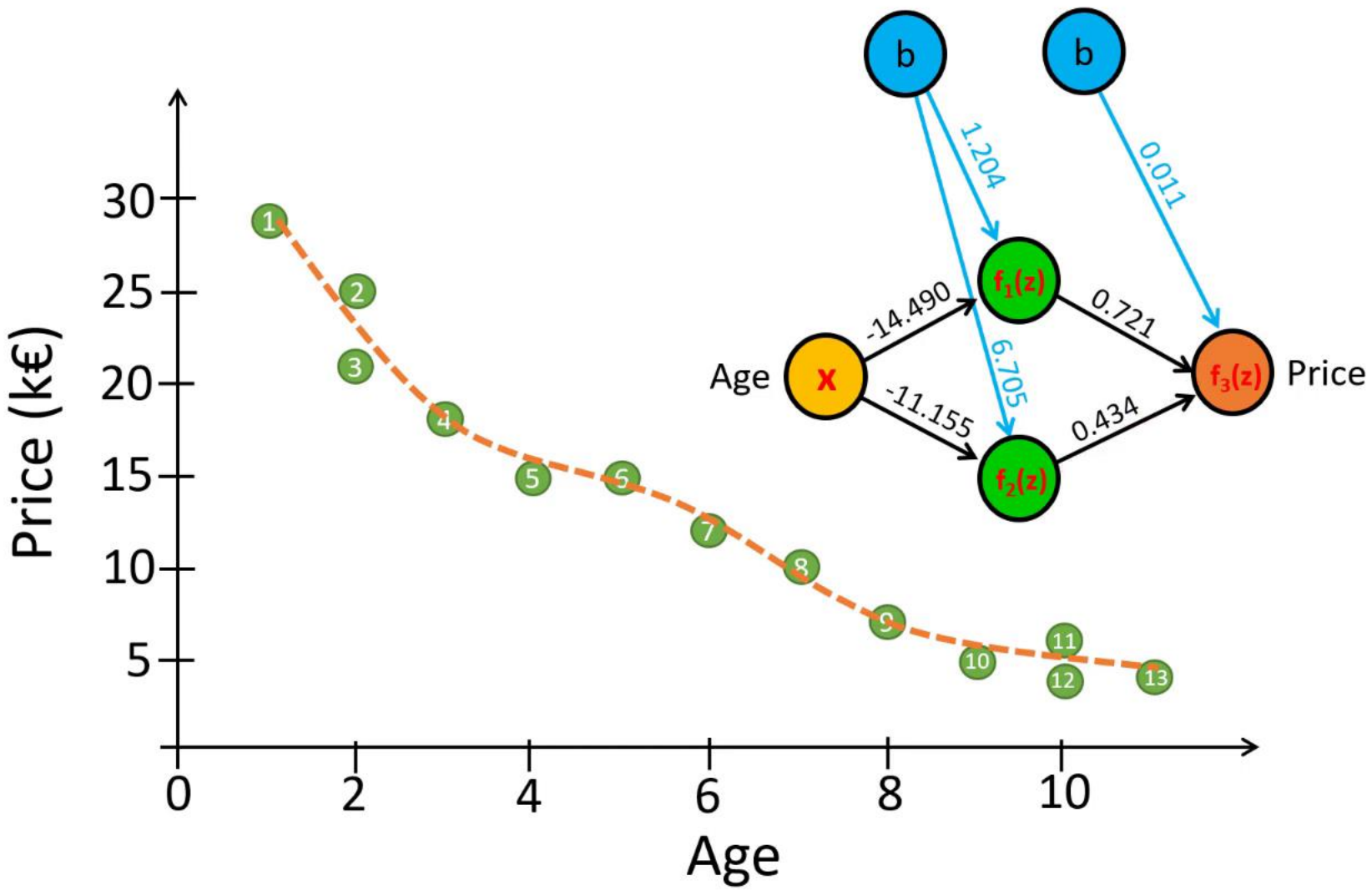


$$f_1(z) = \frac{1}{1 + e^{-(1.204 - 14.490x)}}$$

$$f_2(z) = \frac{1}{1 + e^{-(6.705 - 11.155x)}}$$

and the second hidden node.

ANN



$$f_1(z) = \frac{1}{1 + e^{-(1.204 - 14.490x)}}$$

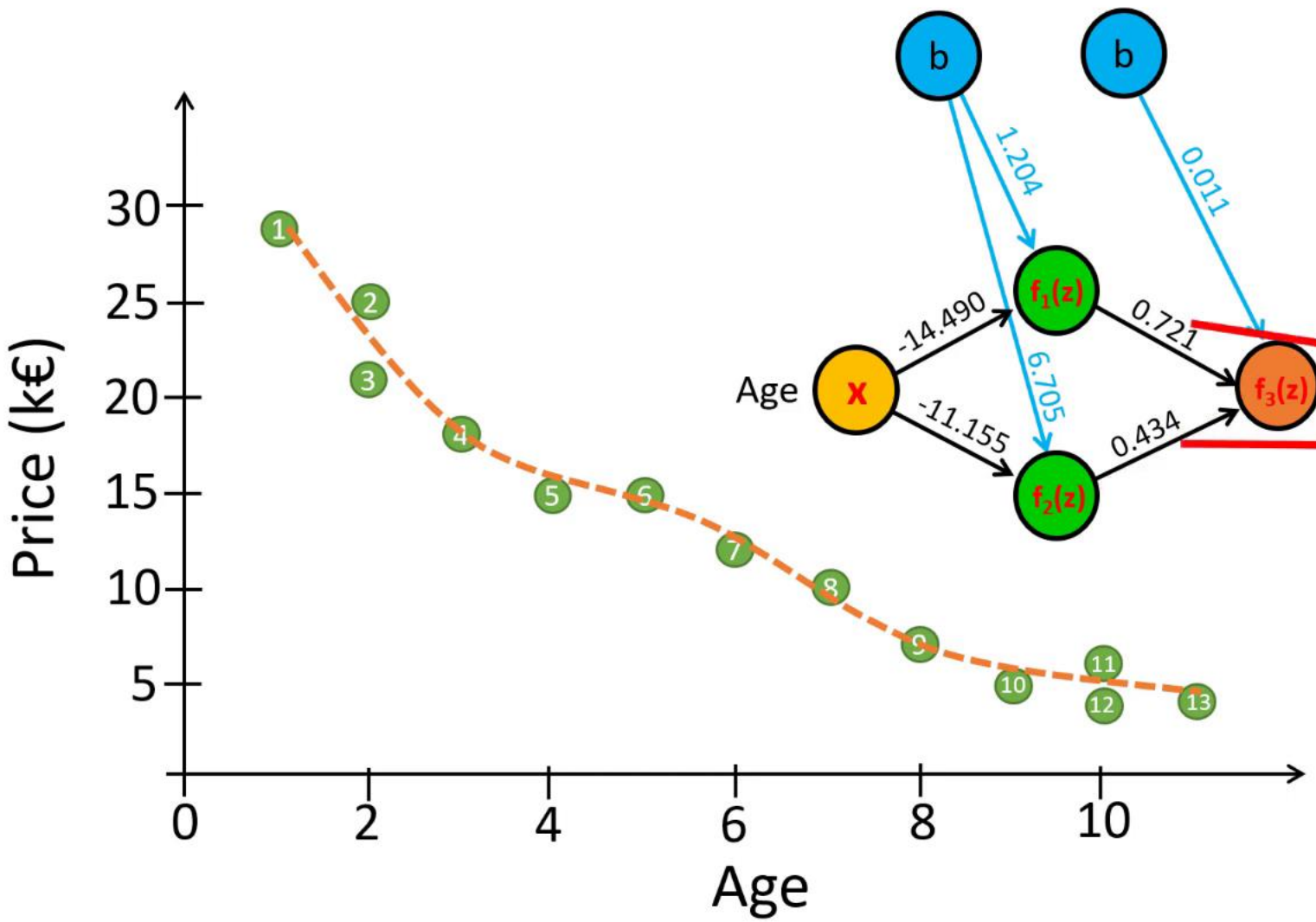
$$f_2(z) = \frac{1}{1 + e^{-(6.705 - 11.155x)}}$$

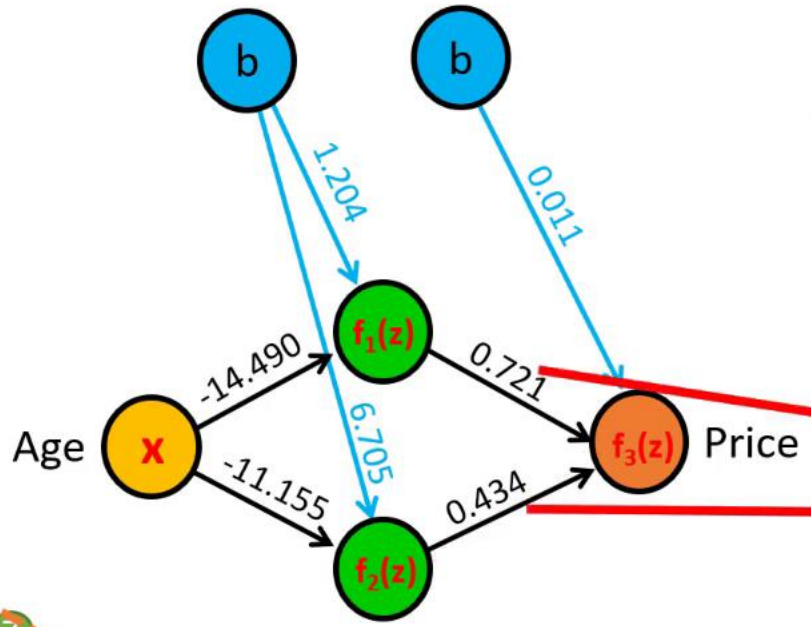
$$z = w_0 + \sum_{i=1}^{n} w_i \cdot x_i$$

$$f(z) = z$$

Then we plug in these functions here,

ANN





$$f_1(z) = \frac{1}{1 + e^{-(1.204 - 14.490x)}}$$

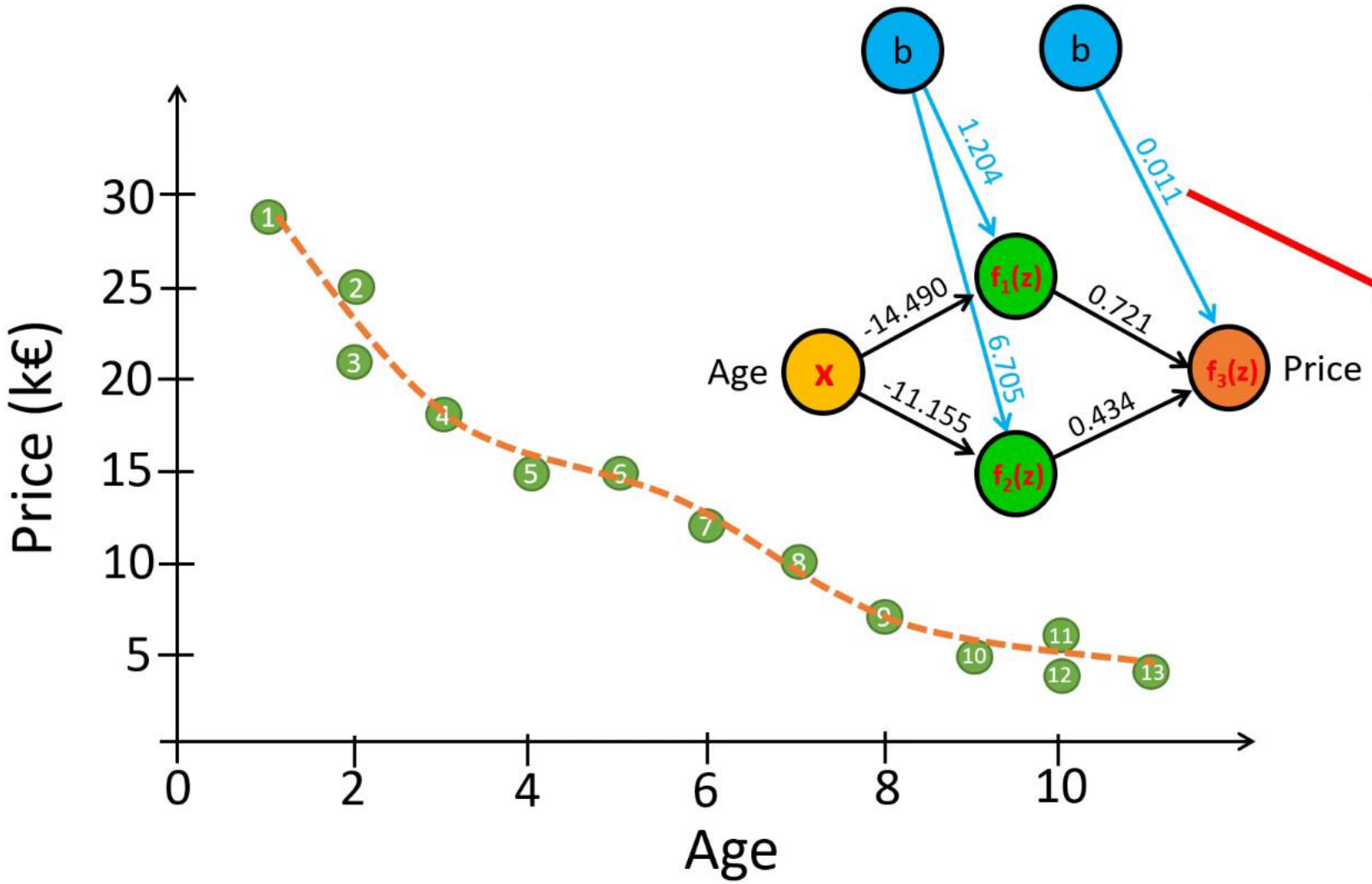
$$f_2(z) = \frac{1}{1 + e^{-(6.705 - 11.155x)}}$$

$$z = w_0 + \sum_{i=1}^{n} w_i \cdot x_i$$

$$f(z) = z$$

and the corresponding weights,

ANN



$$f_1(z) = \frac{1}{1 + e^{-(1.204 - 14.490x)}}$$

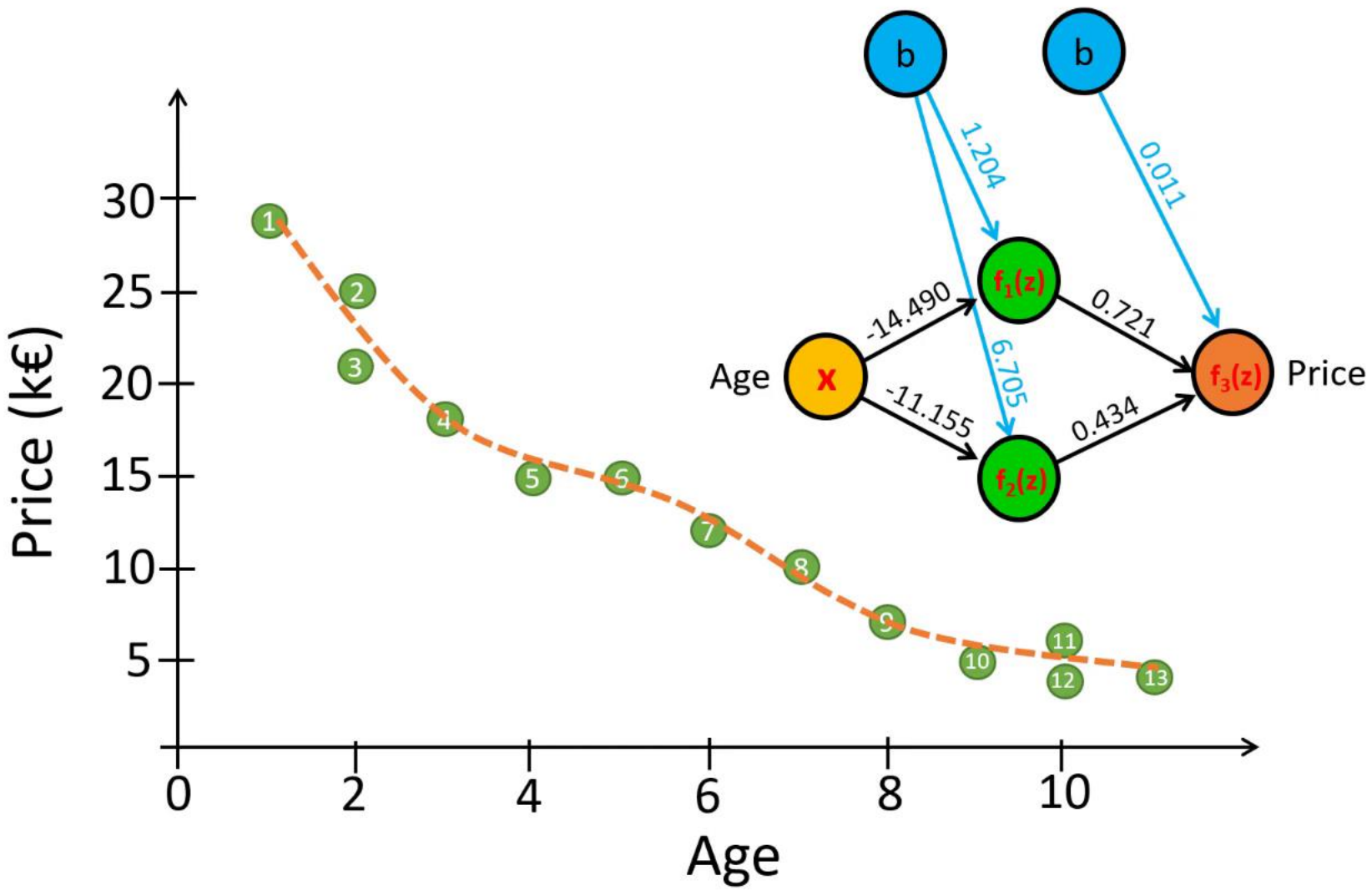
$$f_2(z) = \frac{1}{1 + e^{-(6.705 - 11.155x)}}$$

$$z = w_0 + \sum_{i=1}^{n} w_i \cdot x_i$$

$$f(z) = z$$

and the bias weight,

ANN



$$f_1(z) = \frac{1}{1 + e^{-(1.204 - 14.490x)}}$$

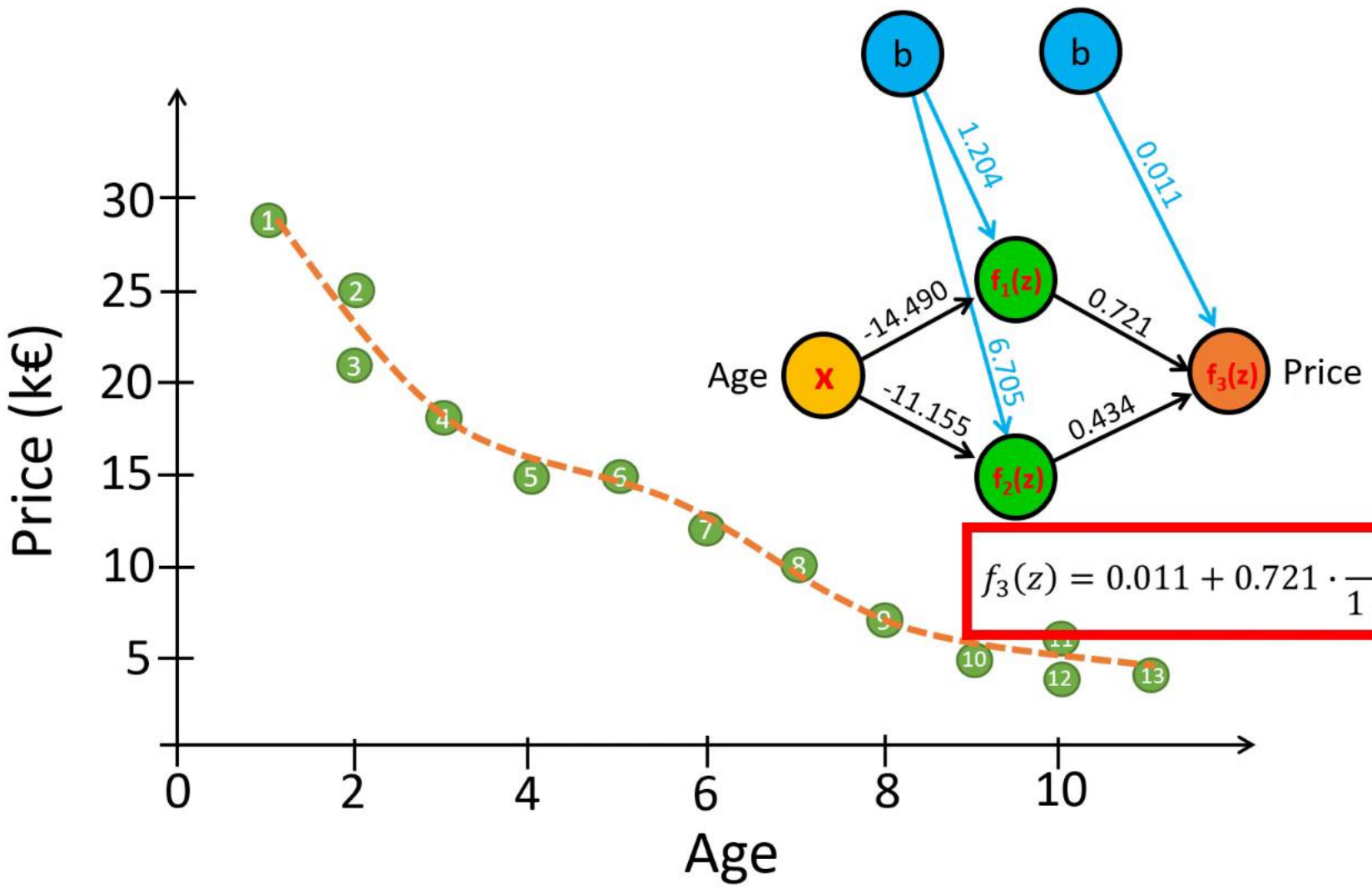
$$f_2(z) = \frac{1}{1 + e^{-(6.705 - 11.155x)}}$$

$$z = w_0 + \sum_{i=1}^{n} w_i \cdot x_i$$

$$f(z) = z$$

into the identity function,

ANN



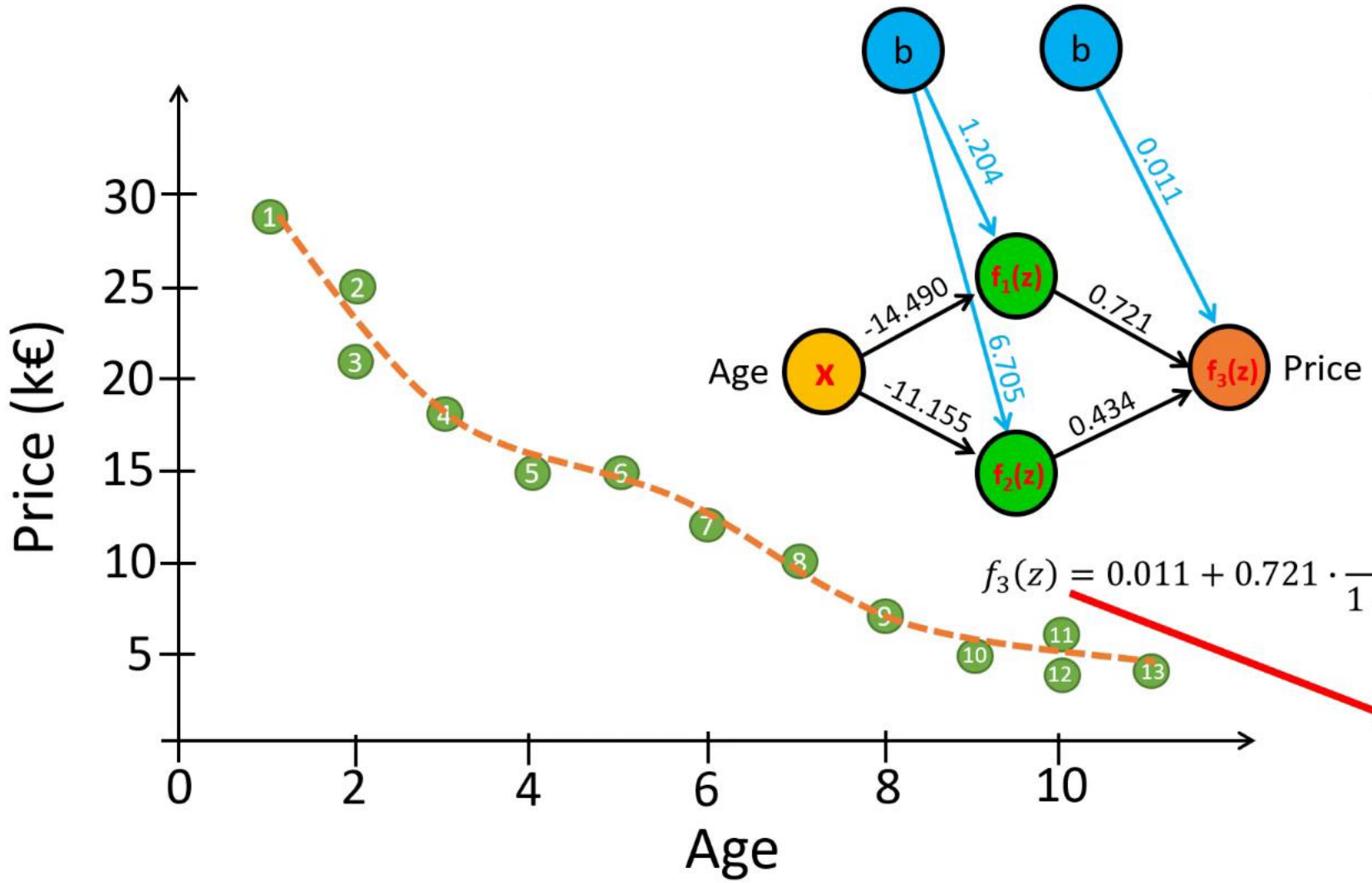
$$f_1(z) = \frac{1}{1 + e^{-(1.204 - 14.490x)}}$$

$$f_2(z) = \frac{1}{1 + e^{-(6.705 - 11.155x)}}$$

$$f_3(z) = 0.011 + 0.721 \cdot \frac{1}{1 + e^{-(1.204 - 14.490x)}} + 0.434 \frac{1}{1 + e^{-(6.705 - 11.155x)}}$$

like this. This will give us the following nonlinear function.

ANN



$$f_1(z) = \frac{1}{1 + e^{-(1.204 - 14.490x)}}$$

$$f_2(z) = \frac{1}{1 + e^{-(6.705 - 11.155x)}}$$

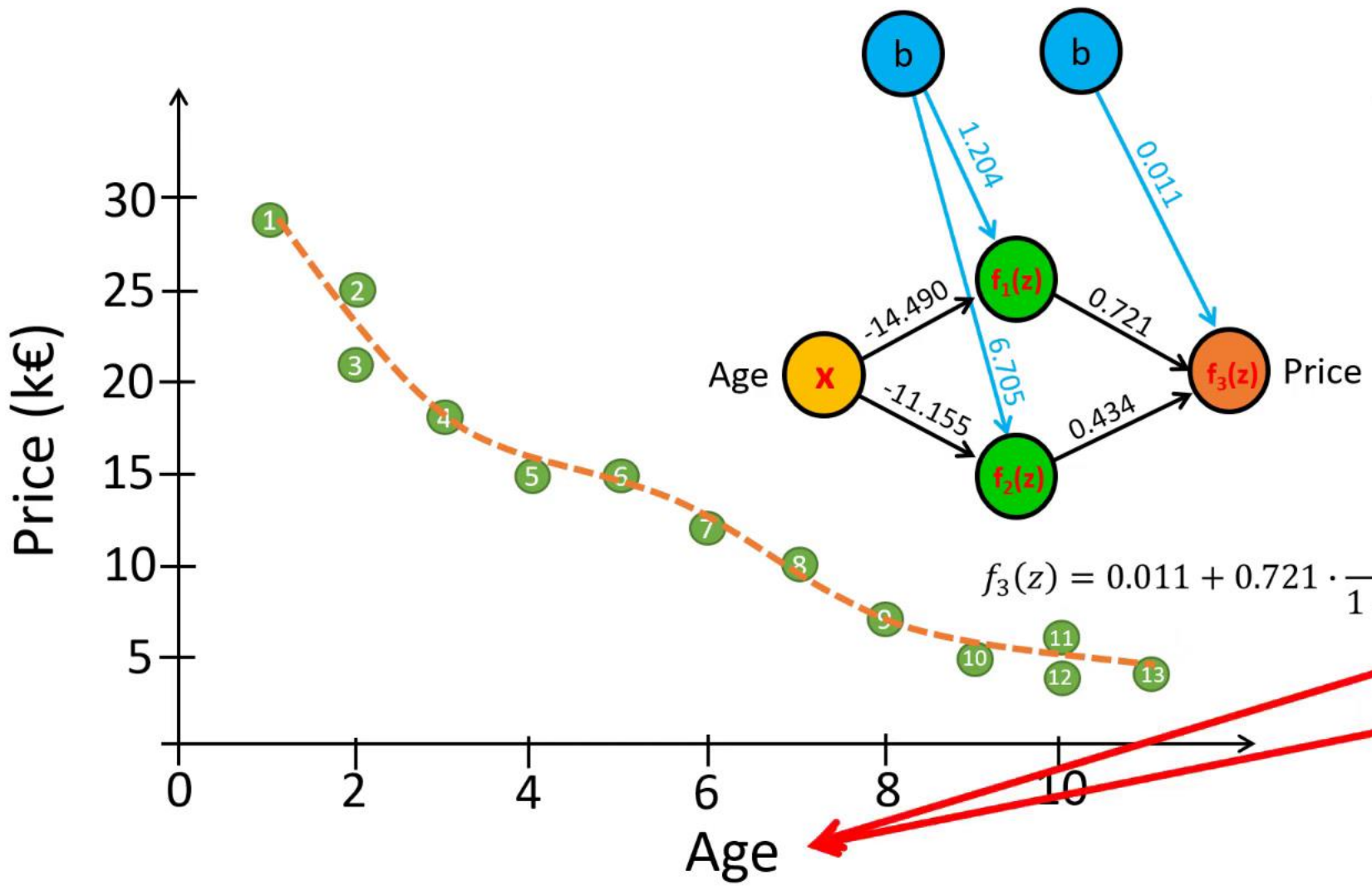
$$f_3(z) = 0.011 + 0.721 \cdot \frac{1}{1 + e^{-(1.204 - 14.490x)}} + 0.434 \frac{1}{1 + e^{-(6.705 - 11.155x)}}$$

$$x = x_{norm}(\max(x) - \min(x)) + \min(x)$$

$$x = x_{norm}(29 - 4) + 4$$

If we now plug in the value of the output node, in this equation, to un-normalize the data,

ANN



$$f_1(z) = \frac{1}{1 + e^{-(1.204 - 14.490x)}}$$

$$f_2(z) = \frac{1}{1 + e^{-(6.705 - 11.155x)}}$$

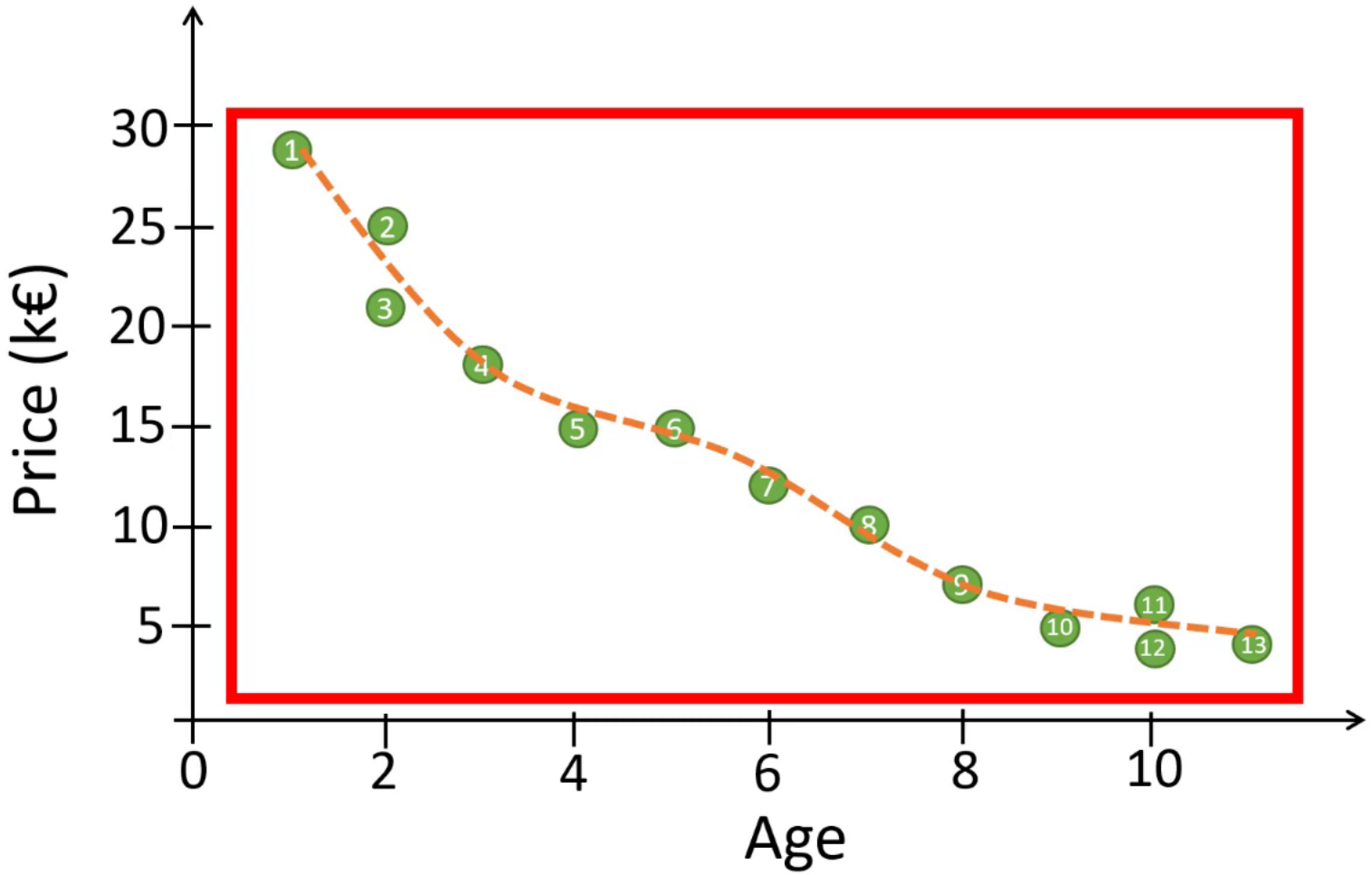
$$f_3(z) = 0.011 + 0.721 \cdot \frac{1}{1 + e^{-(1.204 - 14.490x)}} + 0.434 \frac{1}{1 + e^{-(6.705 - 11.155x)}}$$

$$x = x_{norm}(\max(x) - \min(x)) + \min(x)$$

$$x = x_{norm}(29 - 4) + 4$$

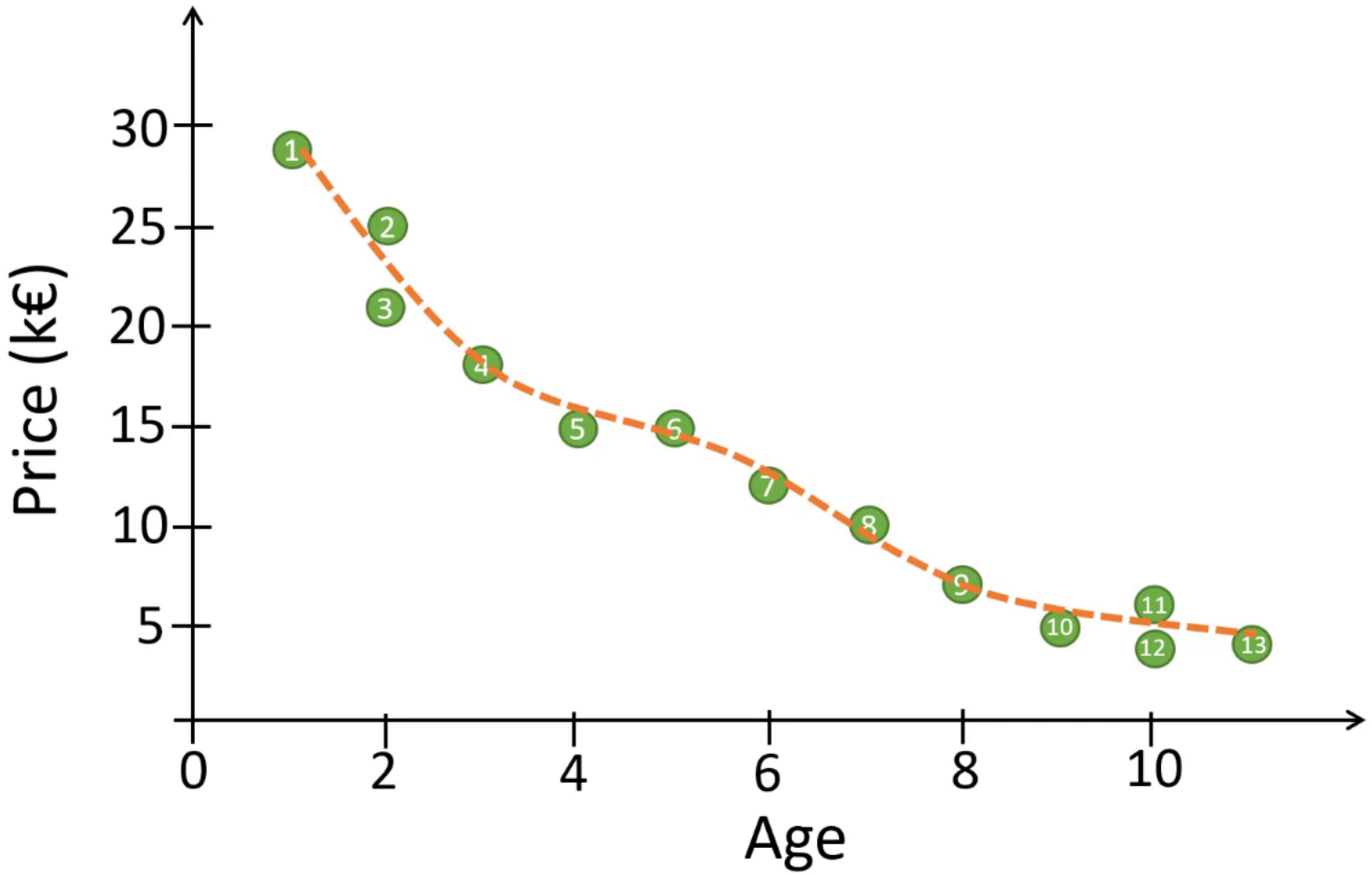
for a range of different values of the Age between zero and one,

ANN



then we will be able to reproduce the curve. Nice!

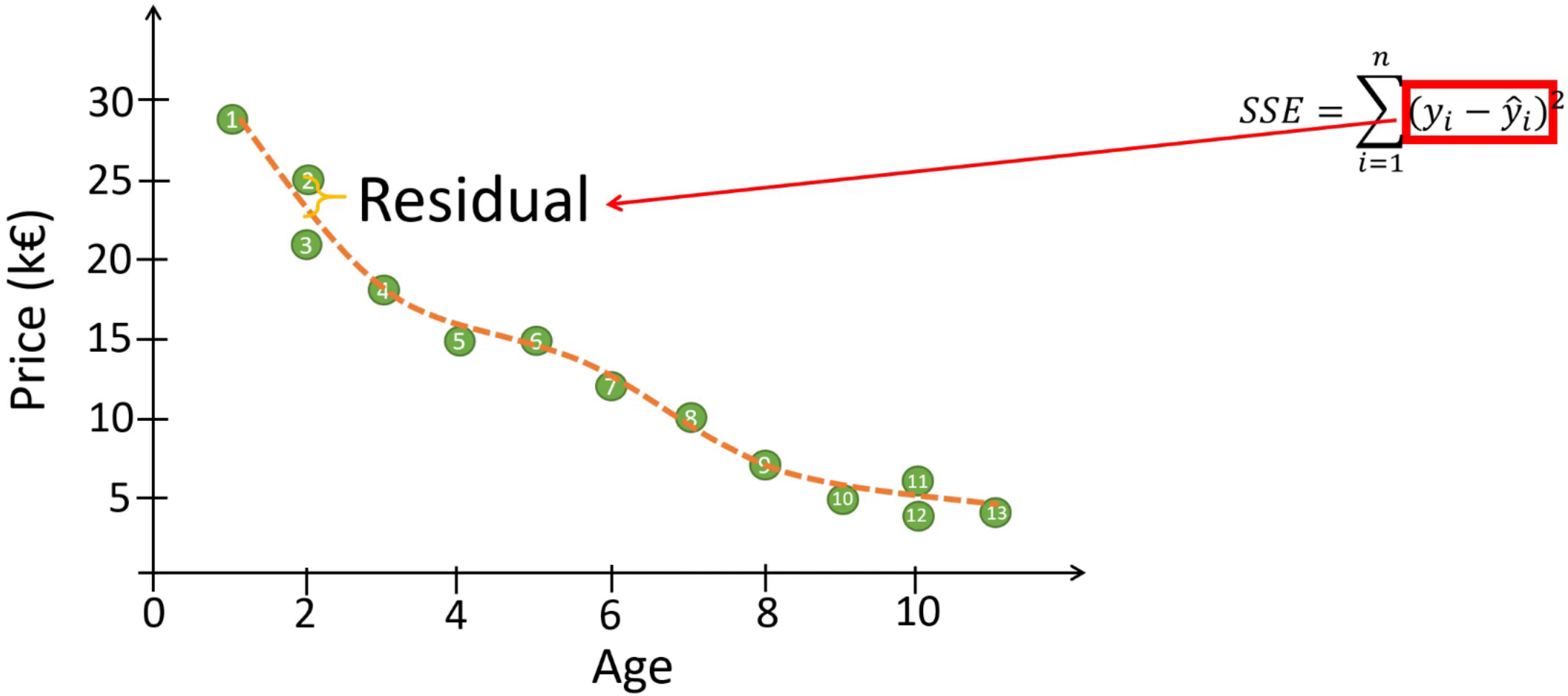
SSE



$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

To evaluate how well the network fits the training data, we can calculate the sum of squared residuals, or errors.

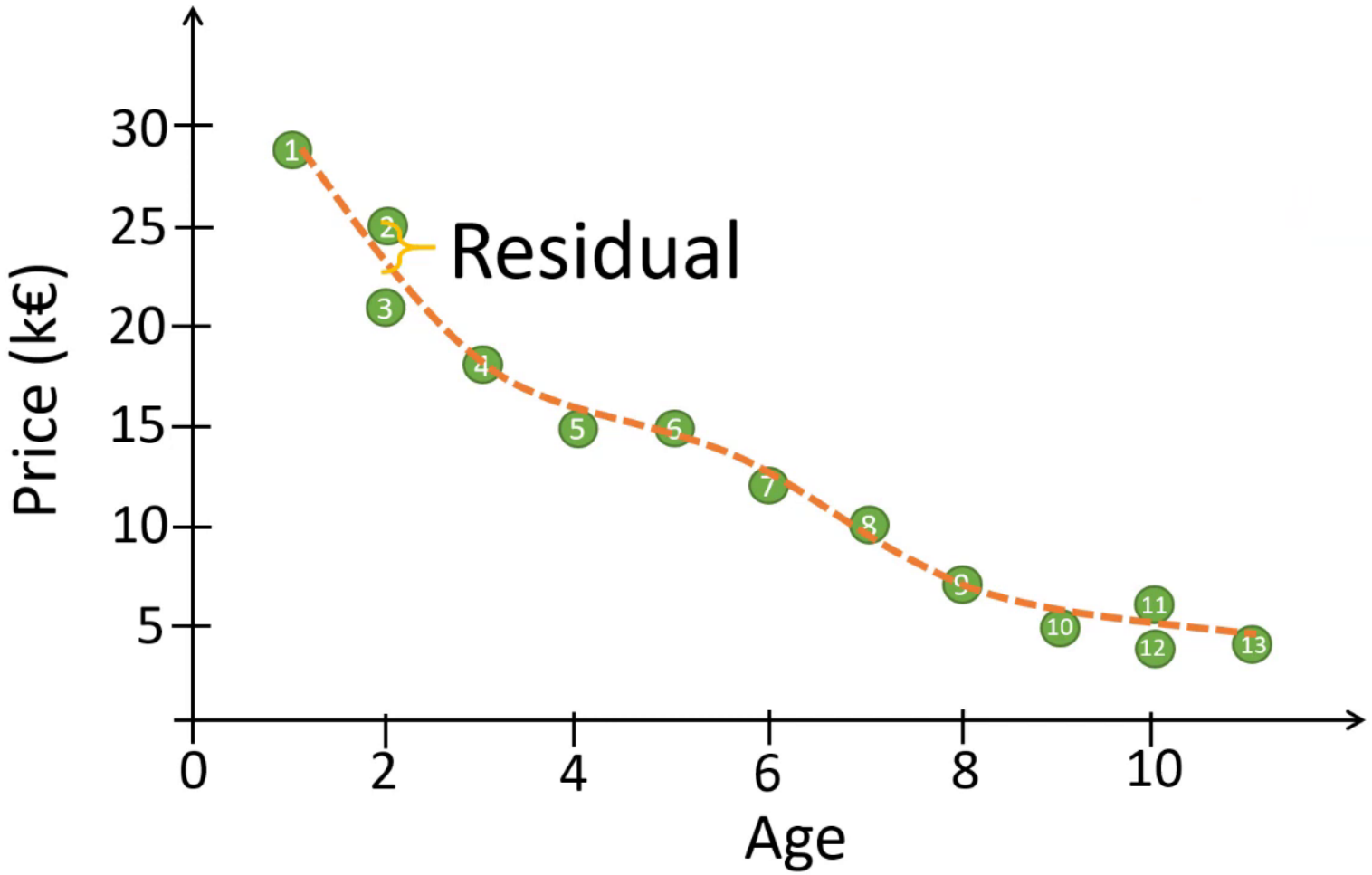
SSE



$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

A residual is the difference between the observed price and the predicted price according to the curve.

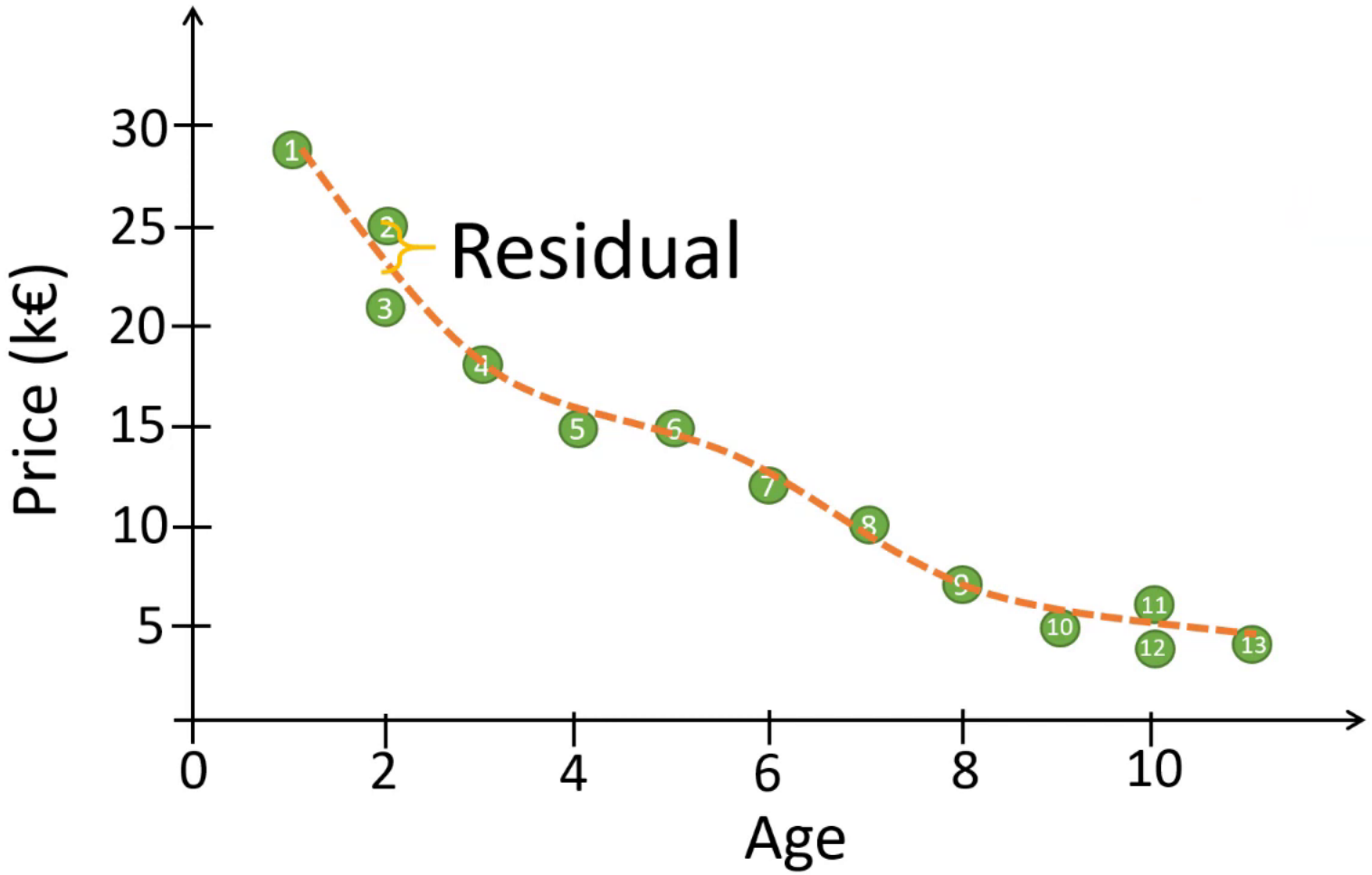
SSE



$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

If we would sum all the,

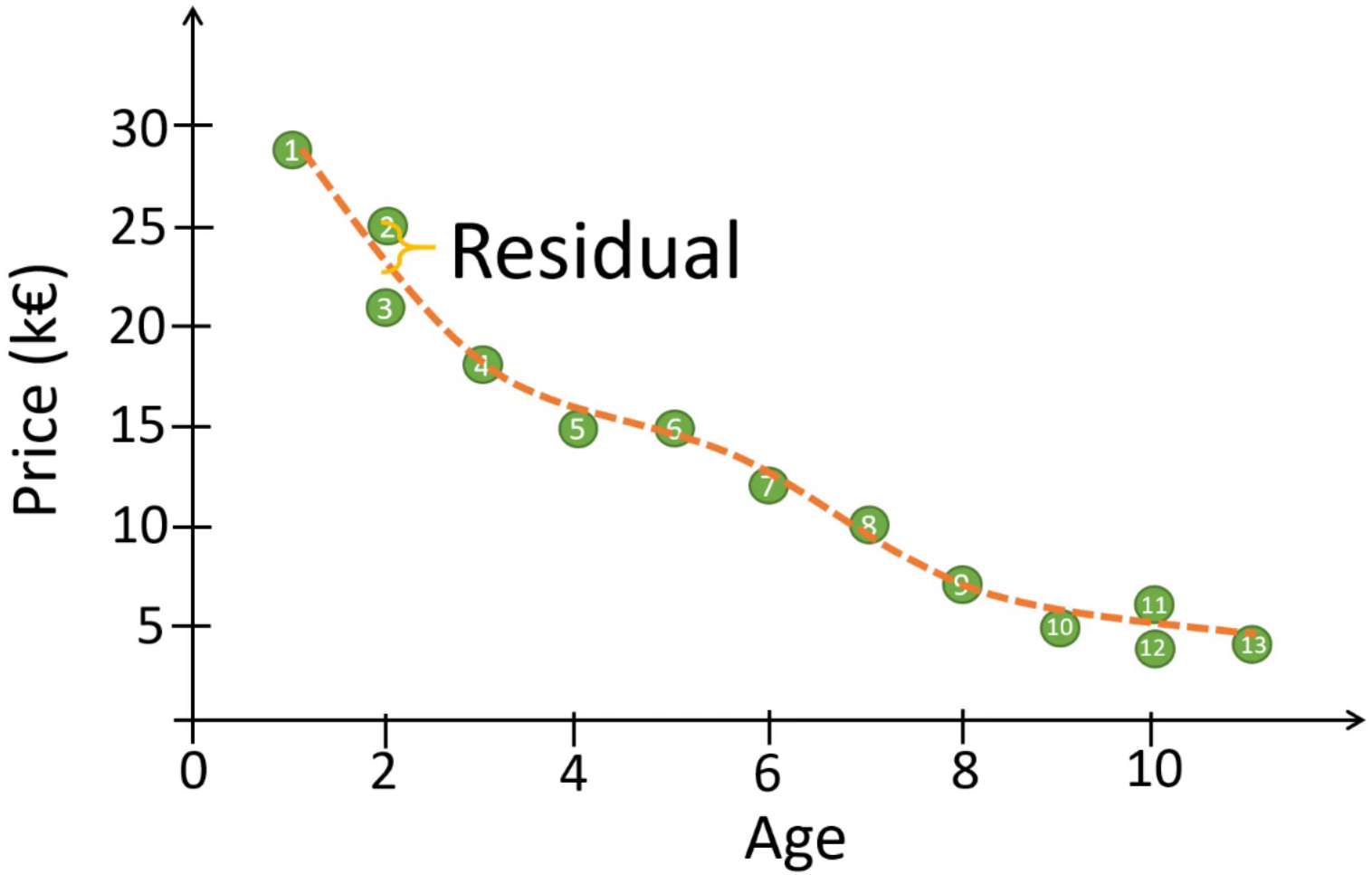
SSE



$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

residuals,

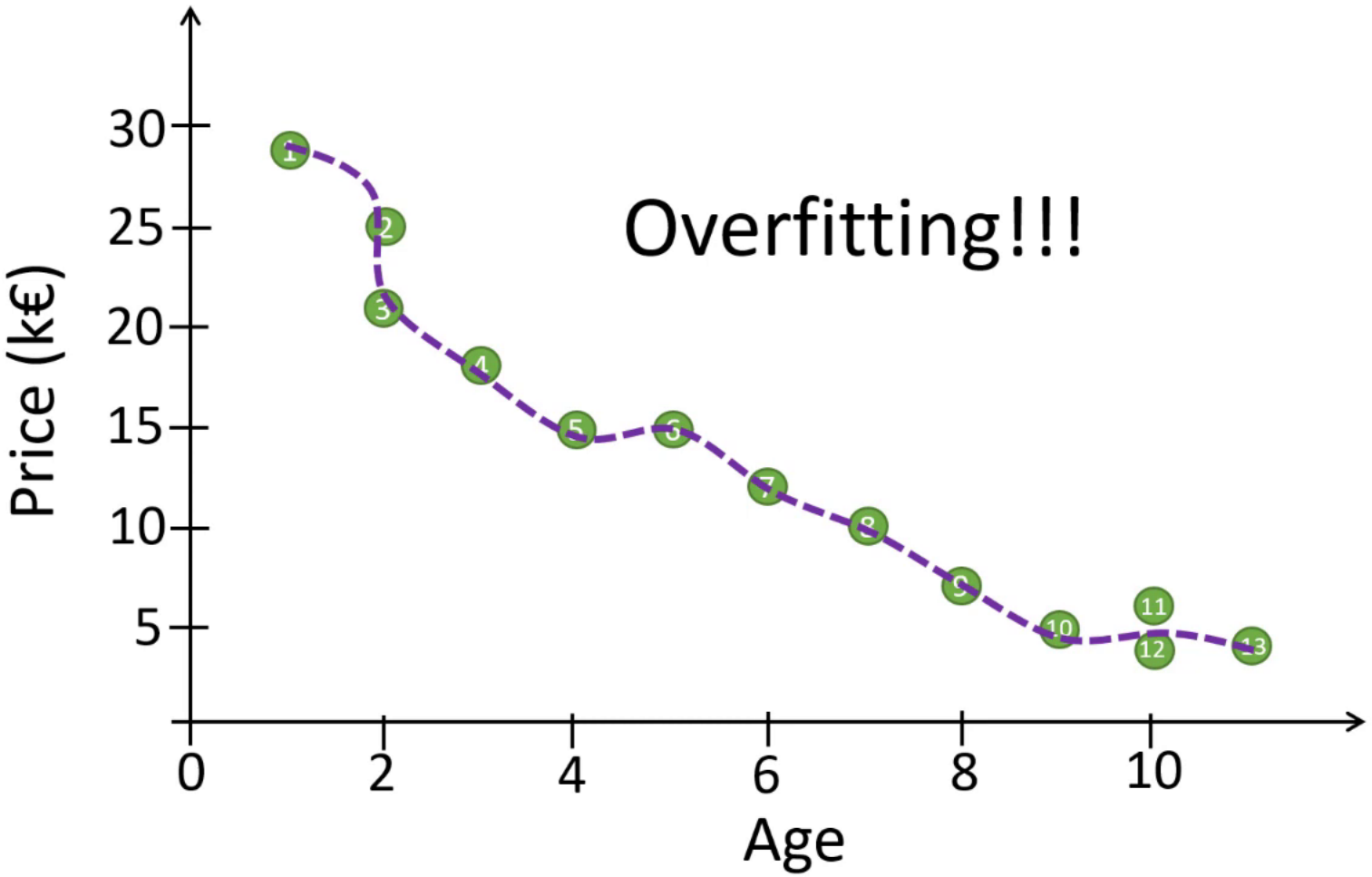
SSE



$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = 11.69$$

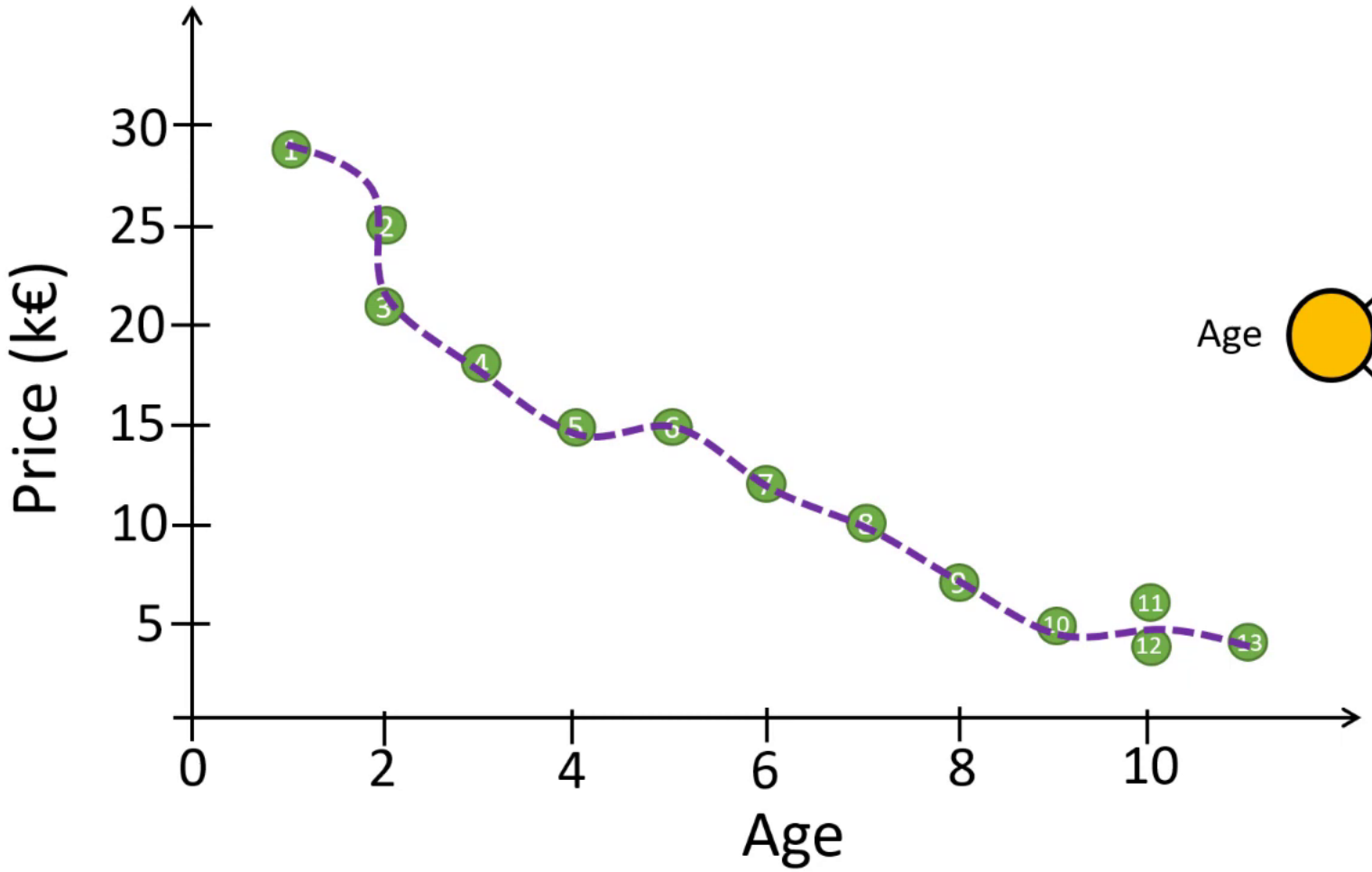
then we would get a value of about 11.69. The smaller this value is, the better the model fits the data.

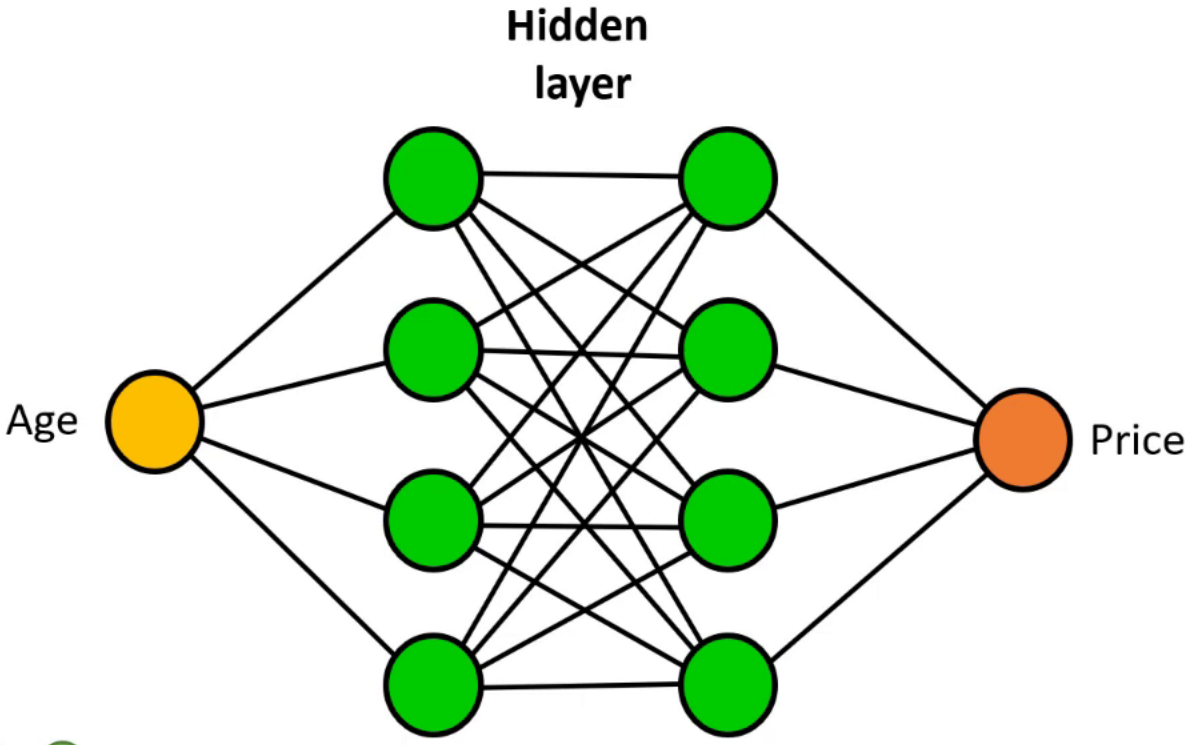
Overfitting



However, one problem is that we may encounter overfitting.

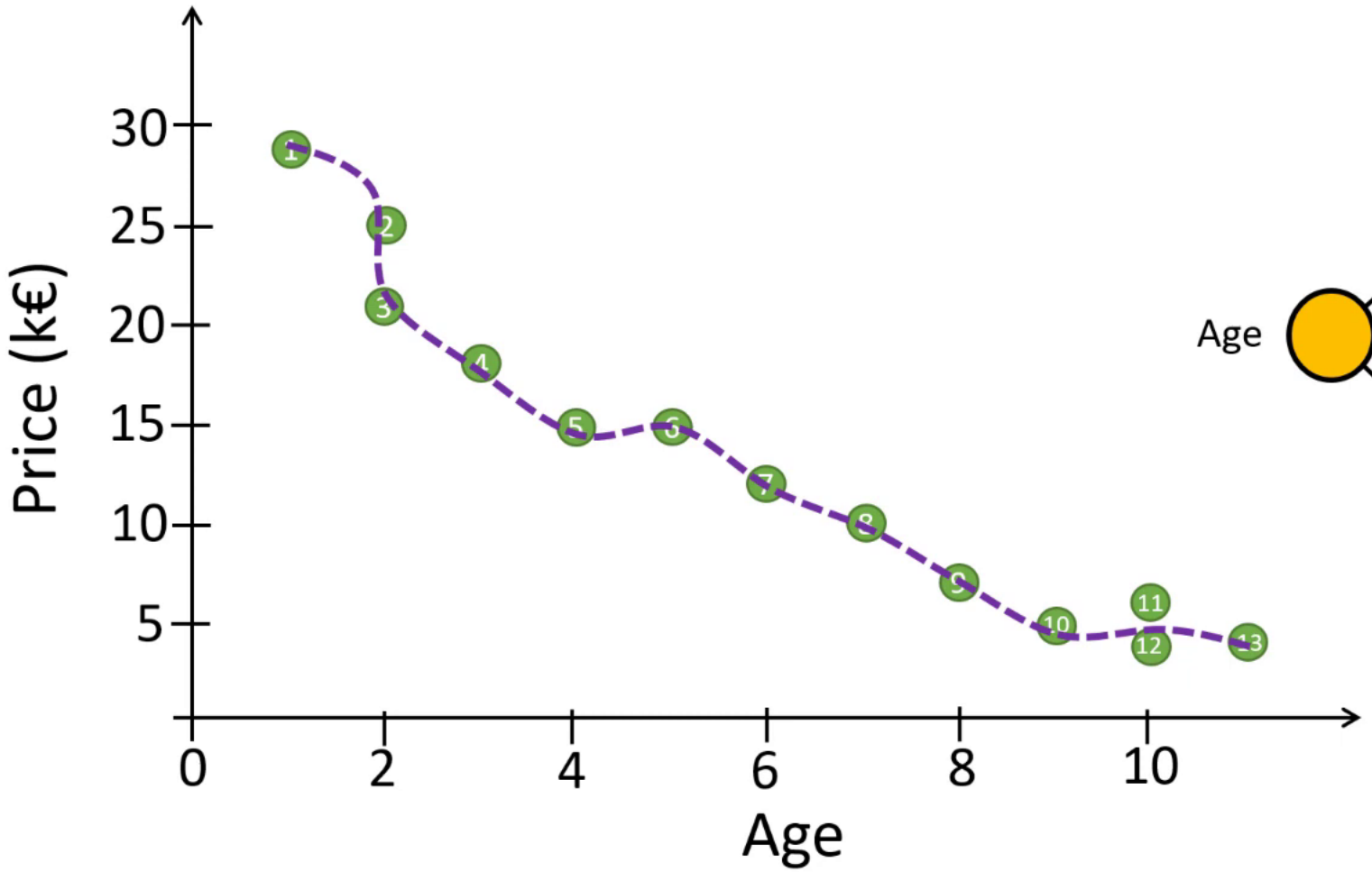
Overfitting

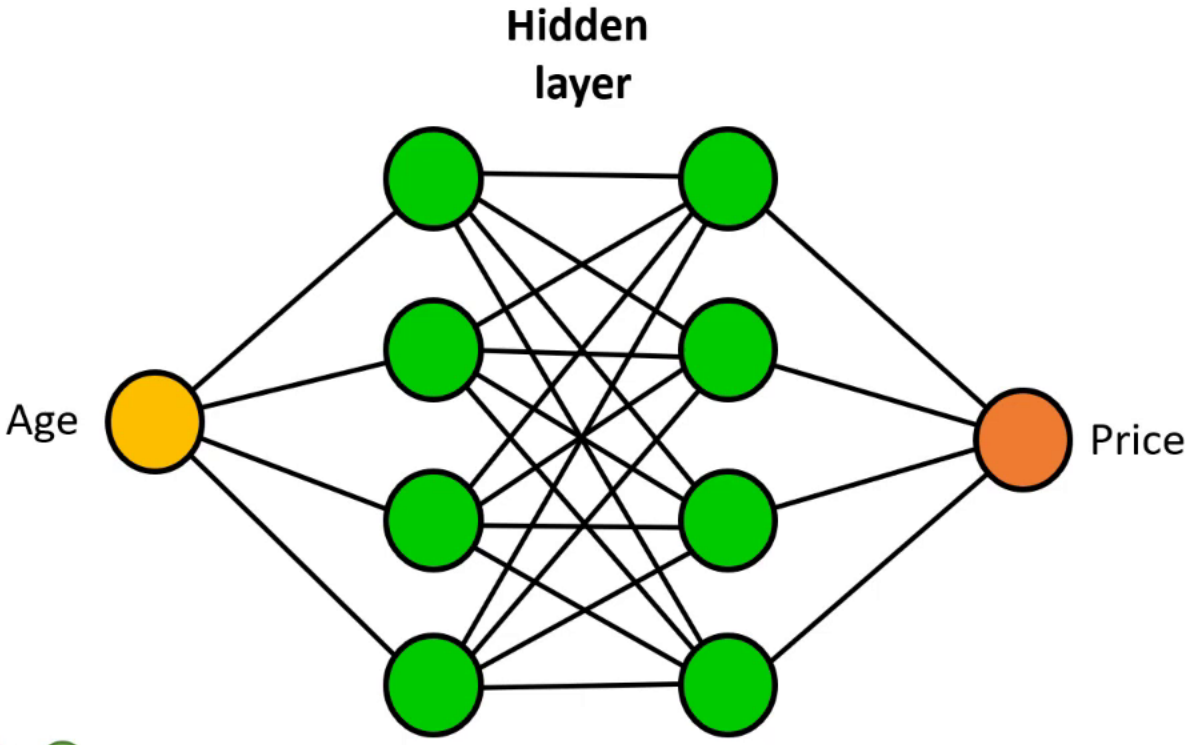




If we use a network with many hidden nodes and layers,

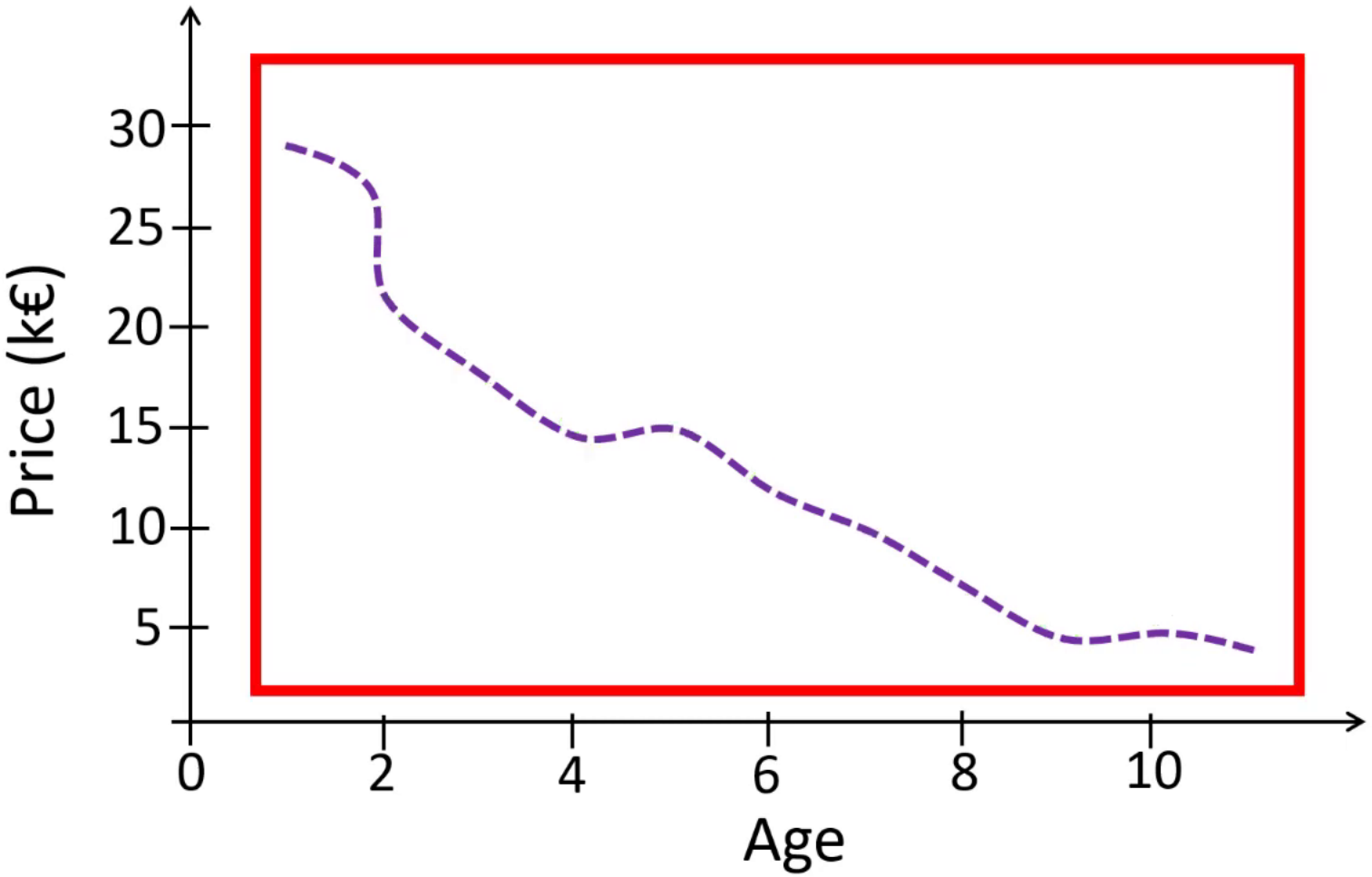
Overfitting





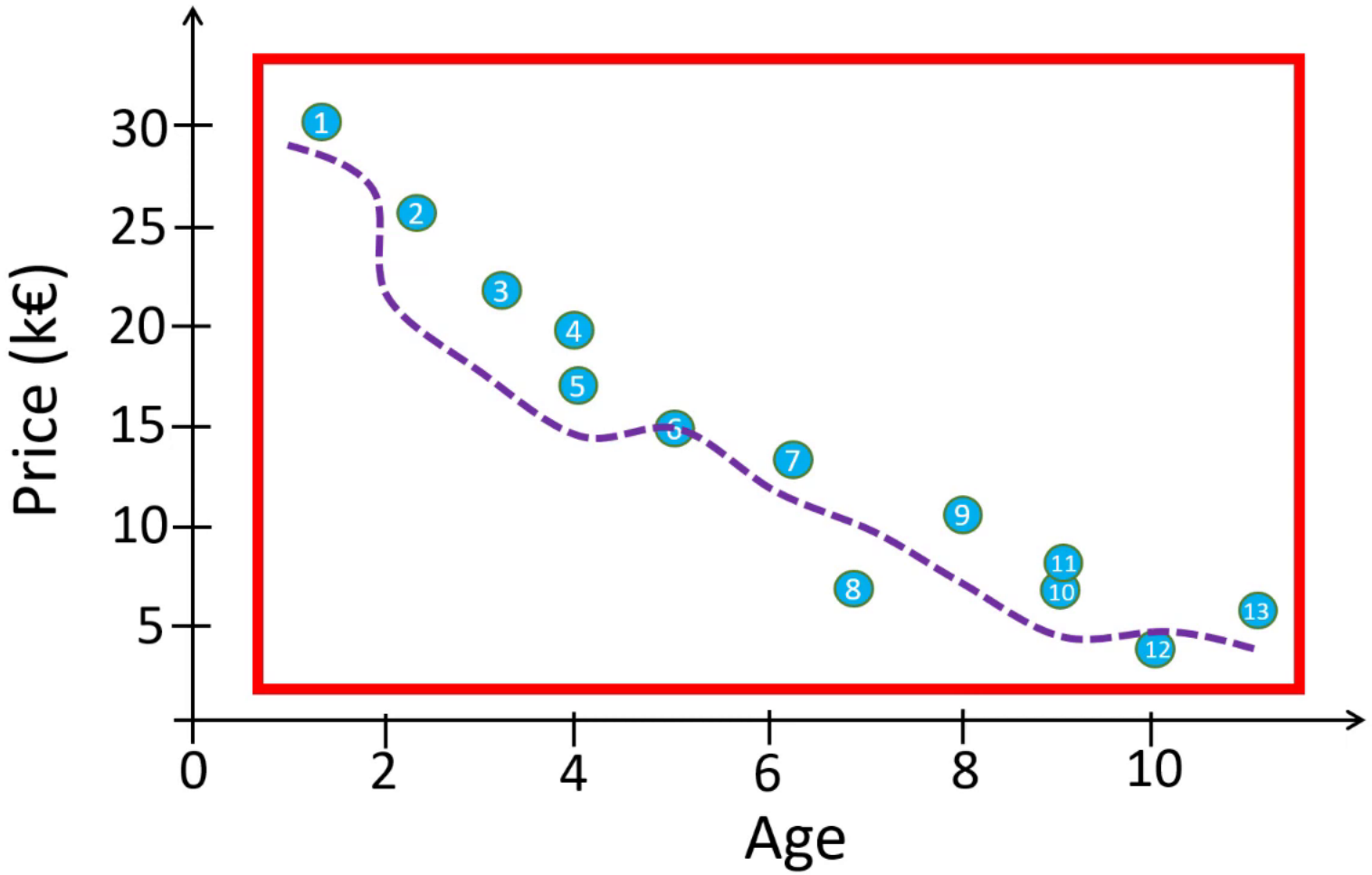
the network can fit the data a bit too well like in this case because the curve intercepts through almost every data point.

Overfitting



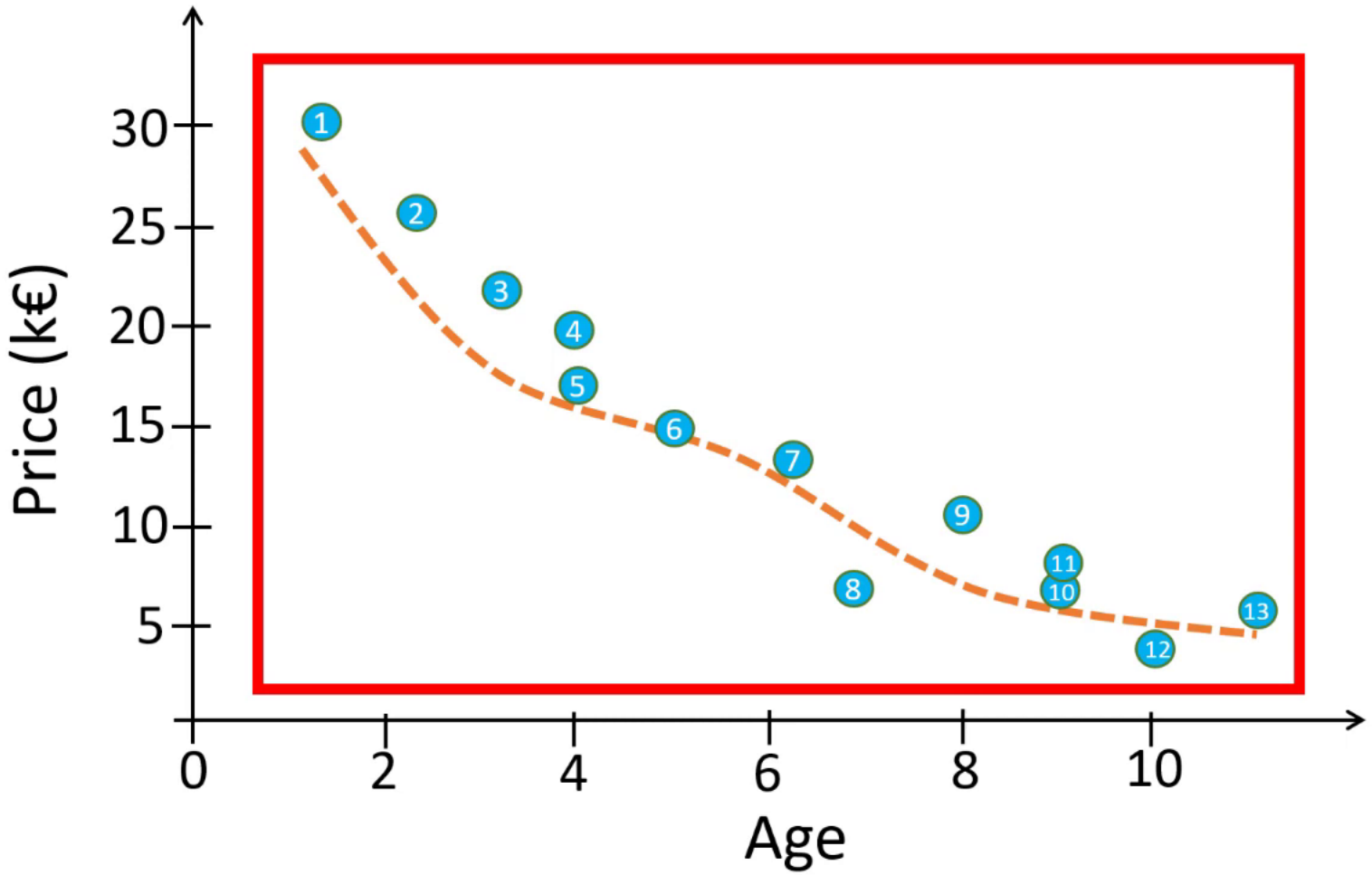
In this example, the network learns too well because if we would remove the training data,

Overfitting



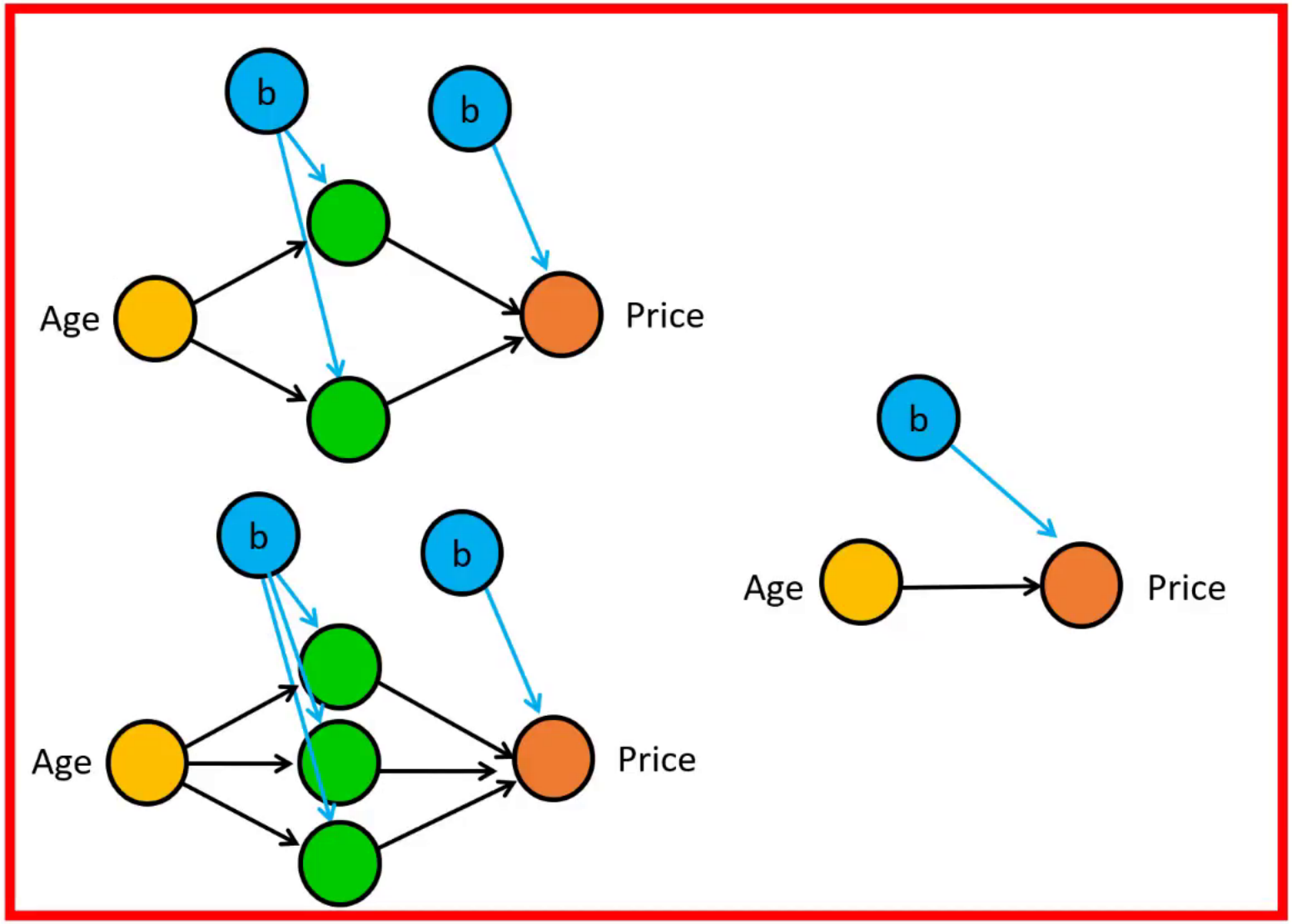
and collect a new data set on similar cars, then we see that the curve does no longer fit well with the data.

Overfitting



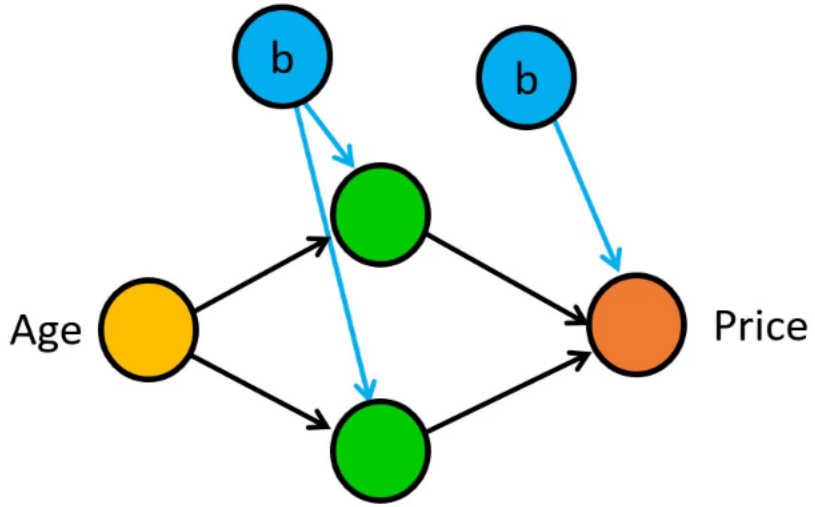
Our previous curve based on a simpler network with just two hidden nodes makes more sense because it tries to find the general trend rather than fitting all training data perfectly.

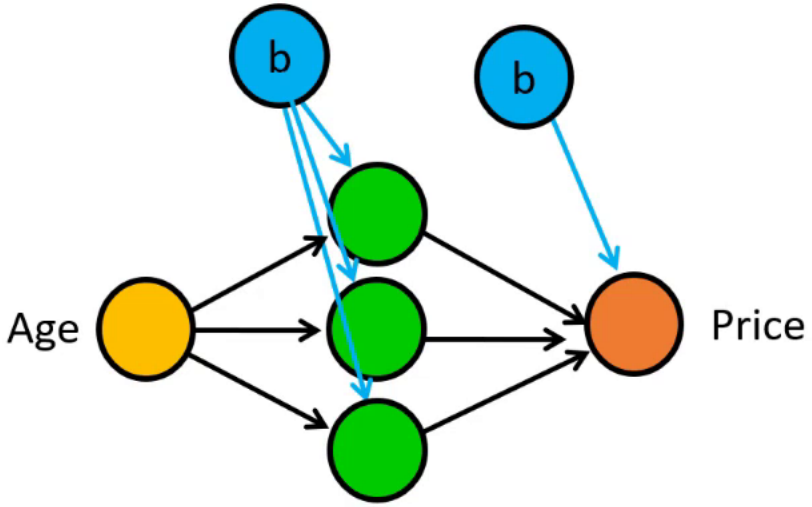
Validate

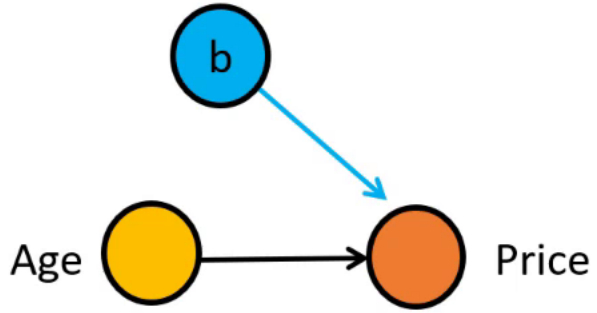


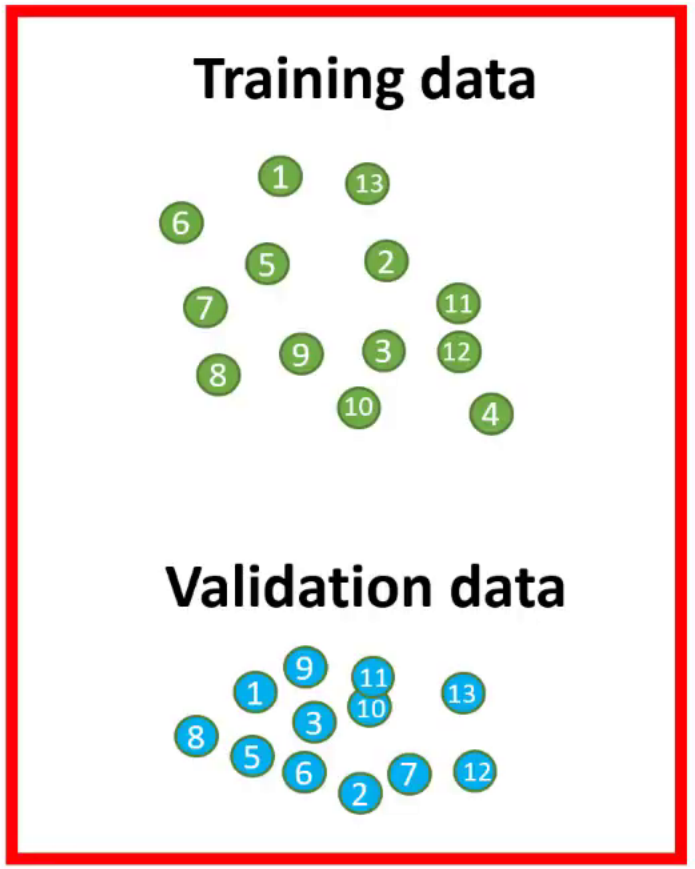
To compare a number of different network models and activation functions,

Validate



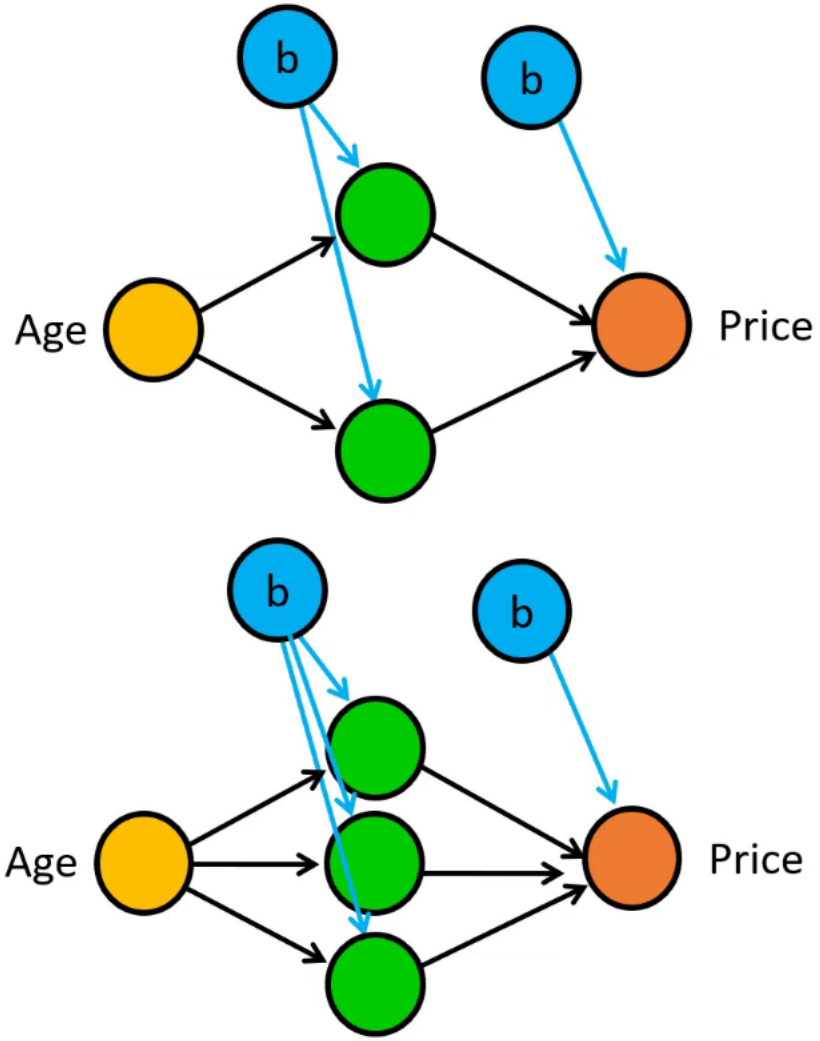


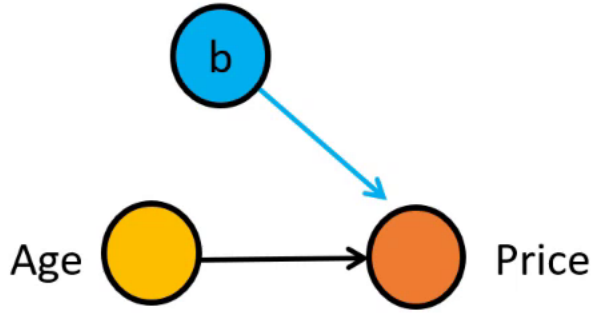


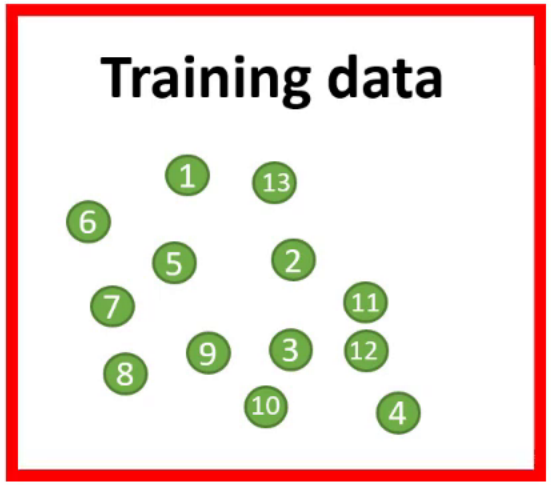


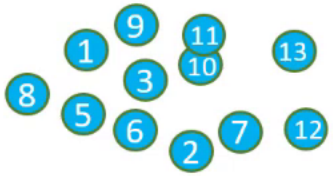
we could, for example, collect a data set and split it into a training data set and a validation data set.

Validate



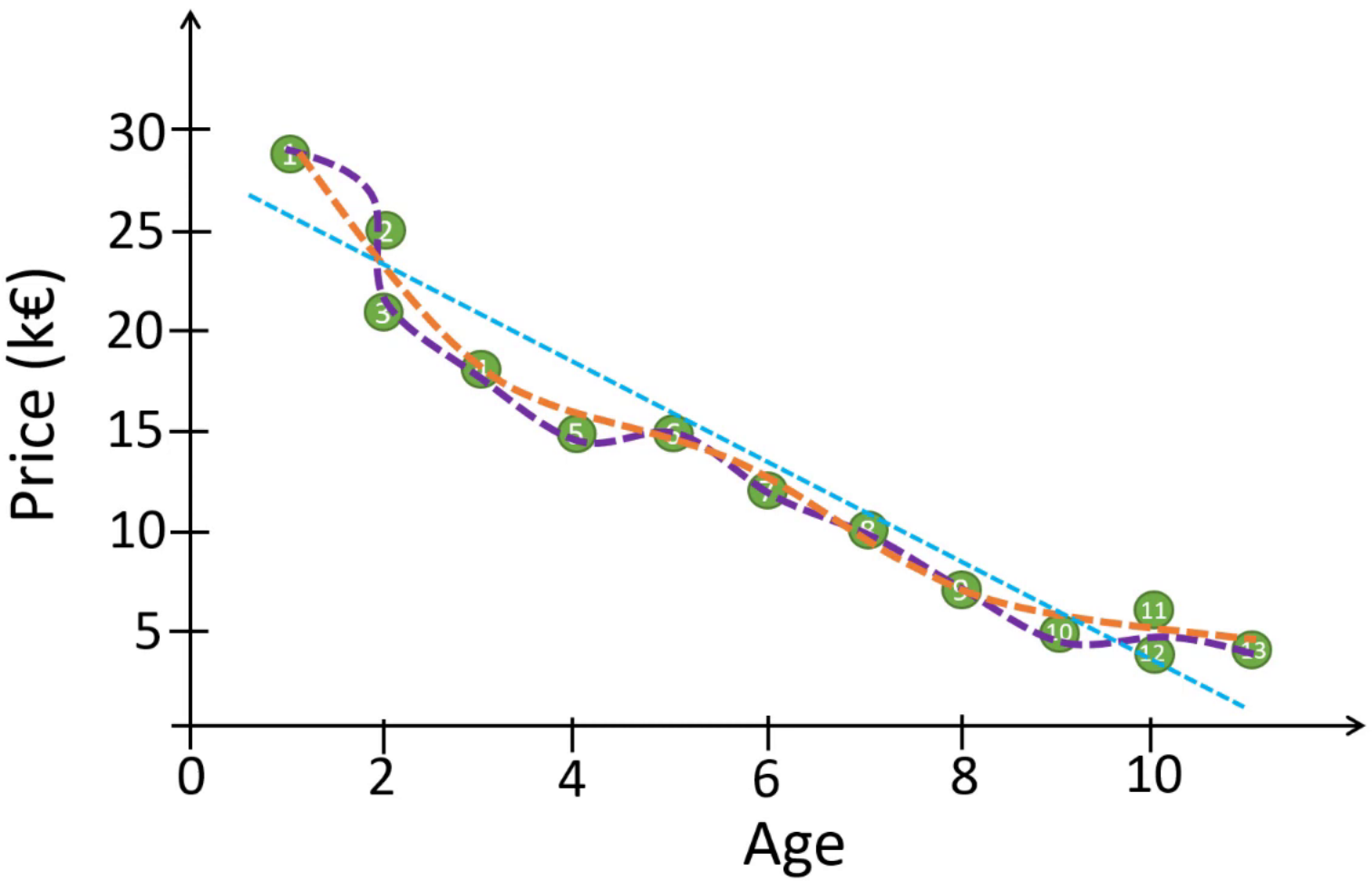




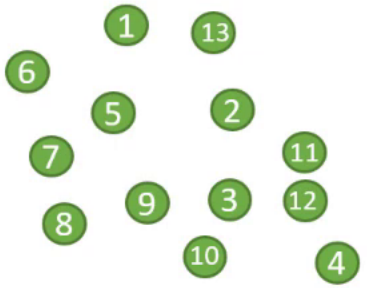


Then we use the training data, where we train all the different networks on this data.

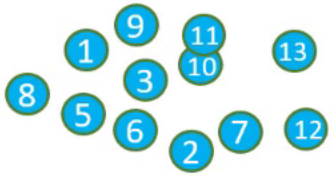
Validate



Training data

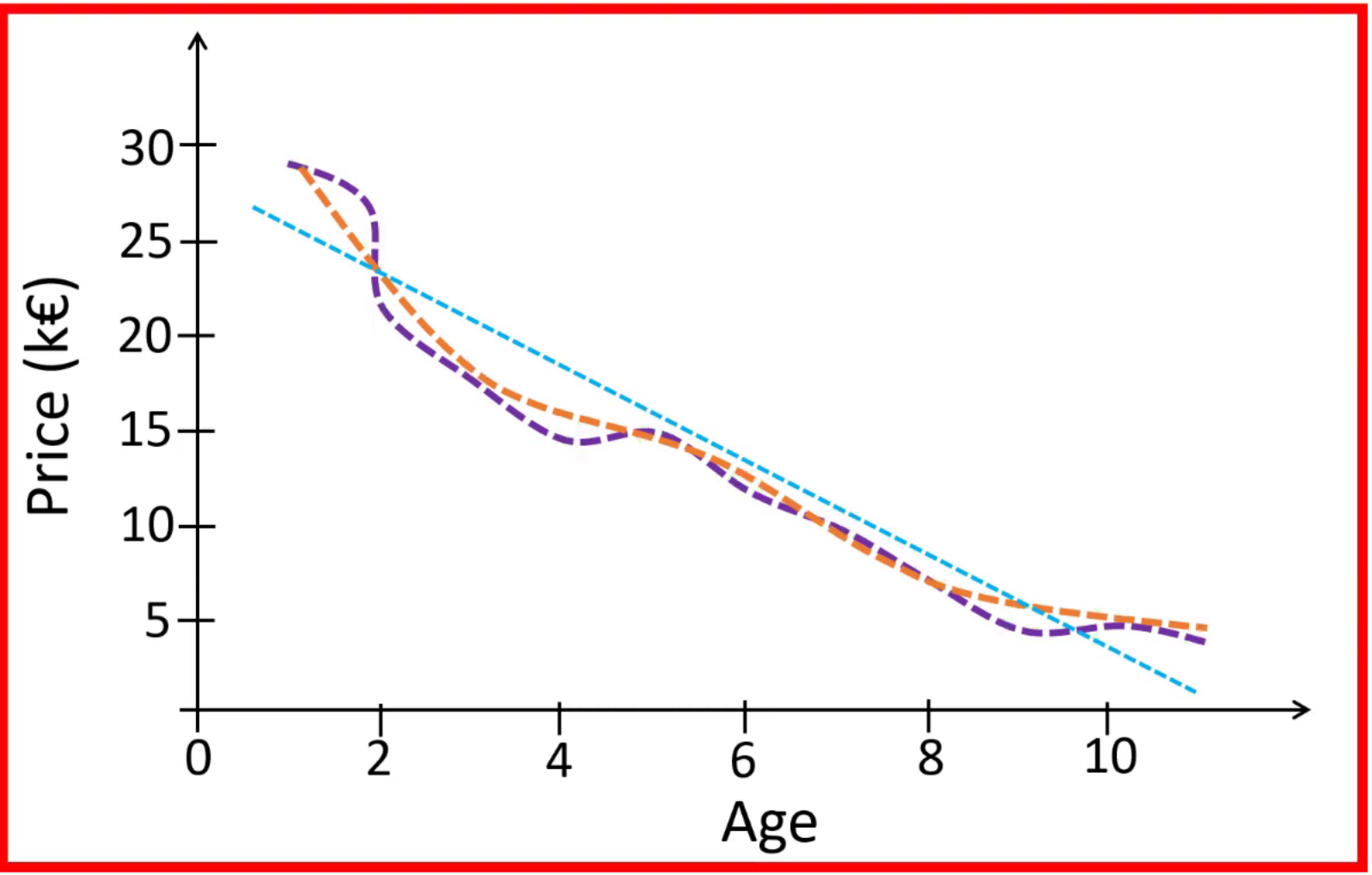


Validation data



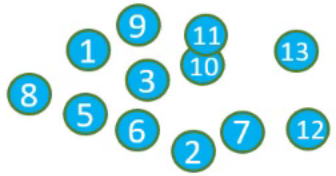
Suppose that we have fitted three different network models to the training data that are represented by the three different curves.

Validate



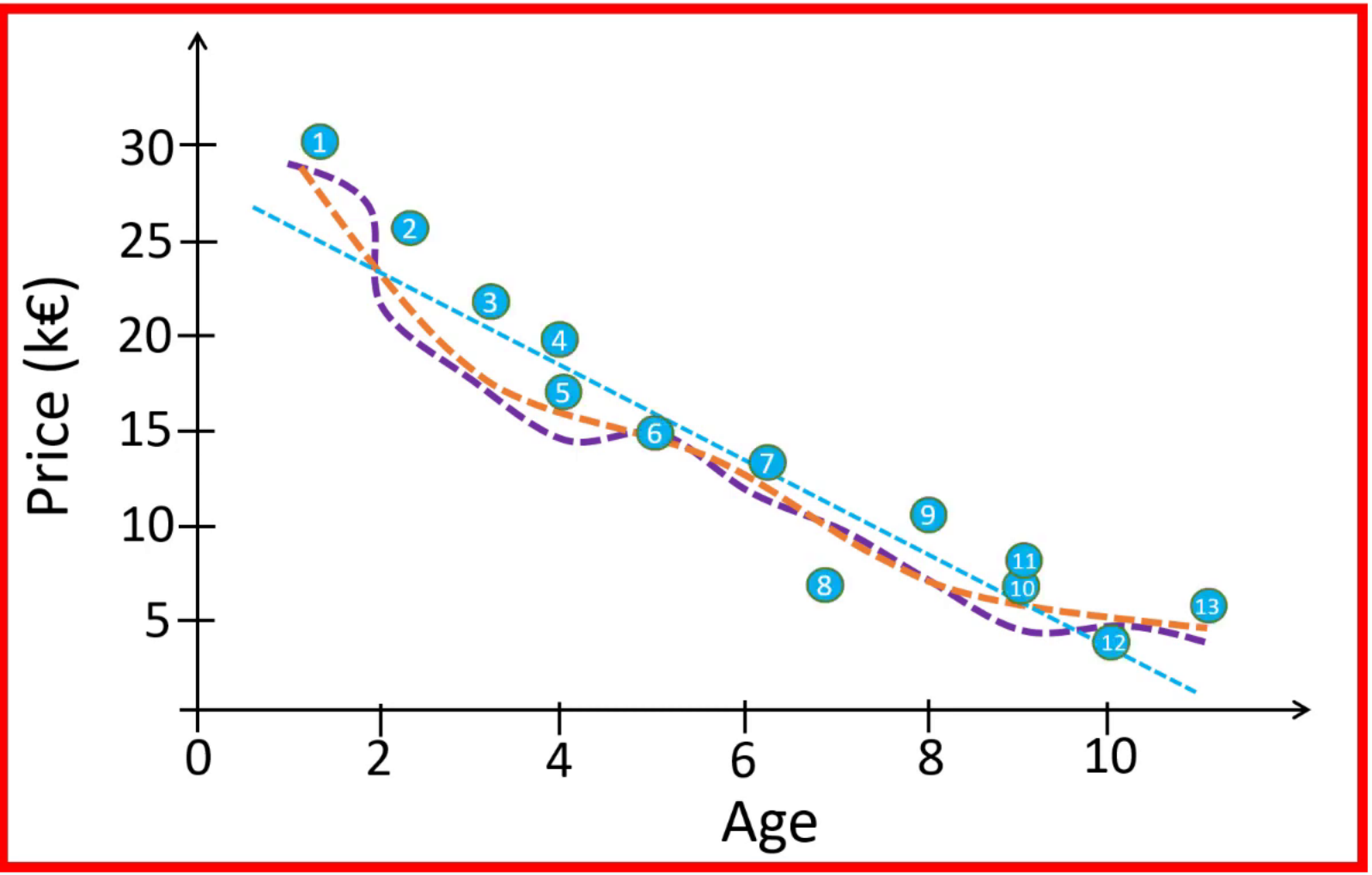
Training data

Validation data

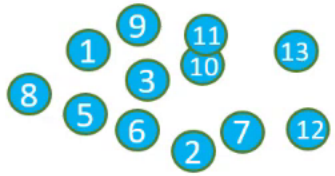


Then we remove the training data,

Validate

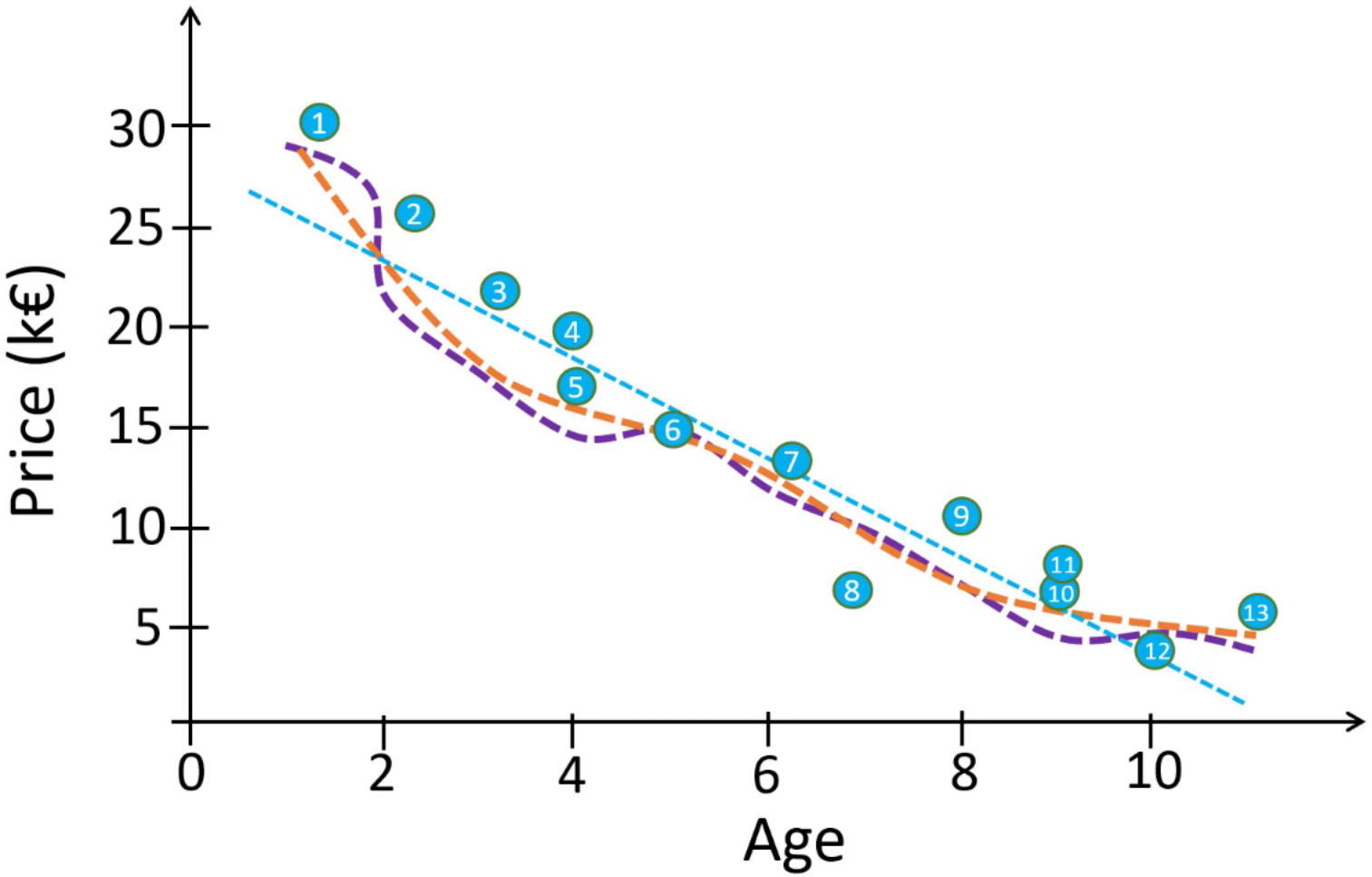


Validation data



and plug in the validation data,

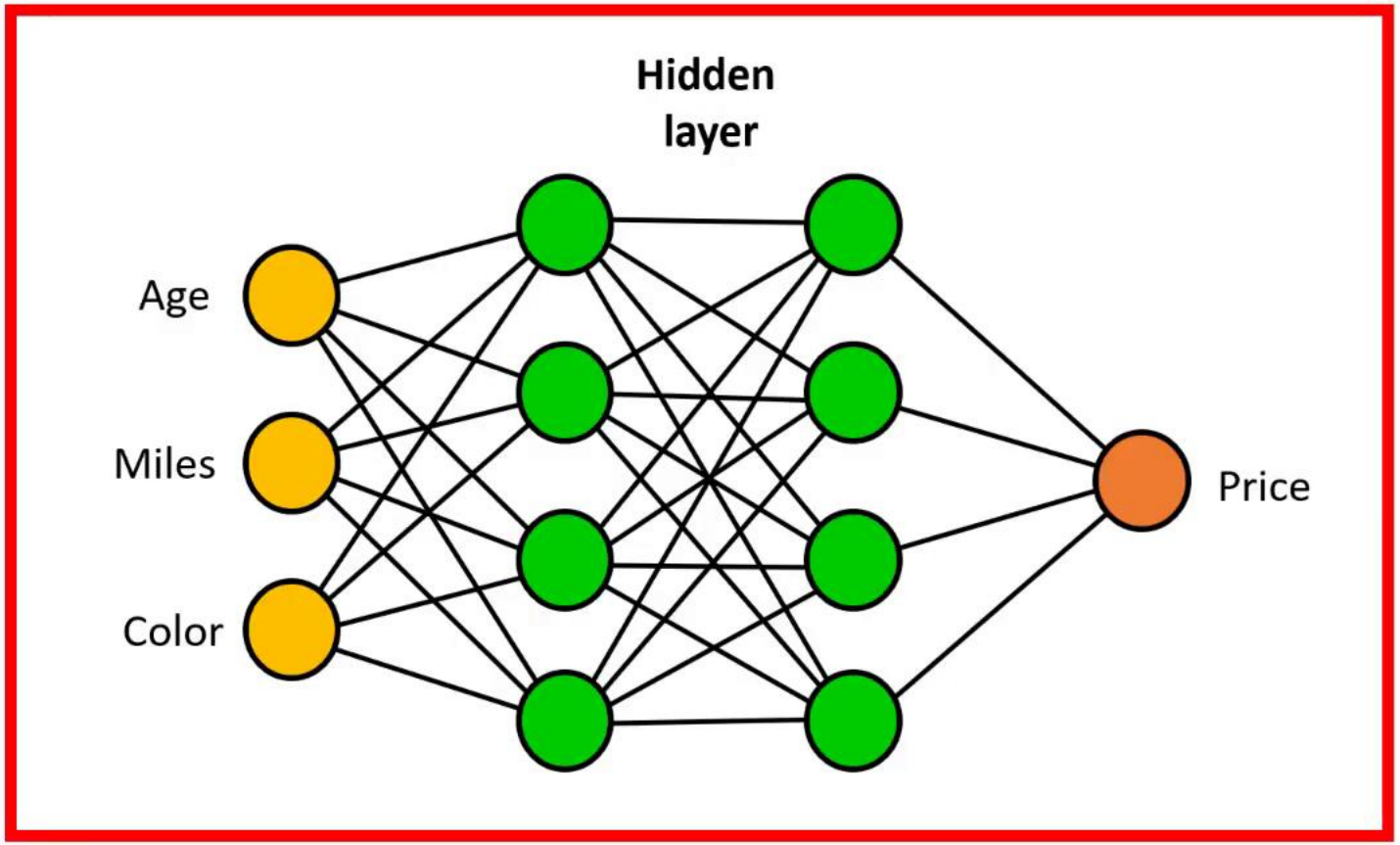
Validate



$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

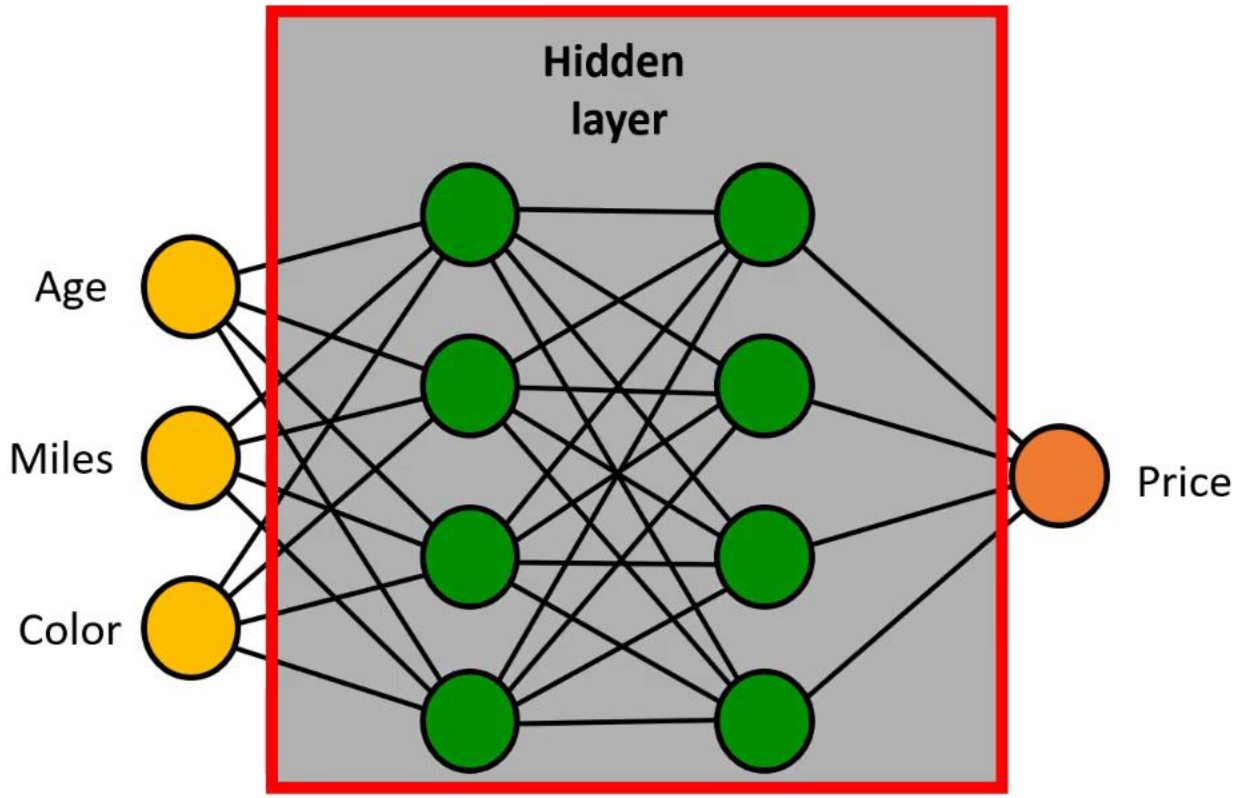
and calculate the sum of squared errors for each model based on the validation data. We can then select the model with the lowest SSE because such a model is likely to perform best on new data.

ANN



We can of course extend the network to include more input nodes like this but the math behind such a network would be way too complicated to show in this video.

ANN



This is why this part of the network is usually considered as a black box.